When Banks Retreat: The Demand for Public Safe Assets*

Jialu Sun† Northwestern University

Yuan Tian (JMP)‡ London School of Economics

This version: 8th November 2026

Click here for the latest version

Abstract

This paper provides the first causal estimate of how private safe-asset supply affects public safe-asset prices. An increase in money market fund (MMF) demand for Treasury bills equal to 1 percent of the bill market's value lowers bill yield spreads by 4.5 basis points. We use banks' quarter-end window dressing as a laboratory. Around regulatory reporting dates, non-U.S. banks reduce borrowing from U.S. MMFs, effectively reducing private safe-asset supply. Constrained to hold short-term, high-quality assets, the funds redeploy the freed-up cash into Treasury bills. As a result, short-maturity U.S. Treasury yields fall by 3–9 basis points at quarter-ends, reflecting an increase in the convenience yield. We identify the causal effect through a network shift-share design that exploits predetermined fund–bank relationships and cross-sectional variation in funds' preferred habitats along the Treasury bill maturity spectrum. We then examine why the Fed's reverse repo facility (RRP) does not fully absorb this demand and whether flexible term operations could improve its ability to stabilize quarter-end Treasury yields.

Keywords: Safe Asset; Treasury Bills; Money Market Funds; Bank Regulation; Market Macrostructure.

JEL Codes: E43, E58, G12, G23, G28, H63.

*We thank Ulf Axelson, Svetlana Bryzgalova, Manav Chaudhary, Nicolas Crouzet, Janice Eberly, Antoine Hubert de Fraisse, Zhengyang Jiang, Ian Martin, Martin Oehmke, Dimitris Papanikolaou, Daniel Paravisini, Cameron Peng, Tarun Ramadorai, Sergio Rebelo, Dimitri Vayanos, Kathy Yuan, Linyan Zhu, and participants at the LSE and Kellogg Finance Student Seminars for helpful comments.

†Kellogg School of Management, Northwestern University. Email: jialu.sun@kellogg.northwestern.edu.

‡London School of Economics. Email: y.tian11@lse.ac.uk.

1 Introduction

Safe assets—including government securities and private claims such as bank deposits and repurchase agreements—play a central role in the financial system by serving as stores of value (Caballero et al., 2017), providing collateral (Gorton, 2017; Ozdenoren et al., 2023), and facilitating liquidity management (Nagel, 2016). Because public and private safe assets provide overlapping safety and liquidity services, a contraction in the supply of private safe assets can shift investor demand toward public safe assets, raising their prices (Holmström and Tirole, 1998; Krishnamurthy and Vissing-Jorgensen, 2012). Yet in deep and liquid public safe-asset markets such as the Treasury bill market, other investors may readily accommodate this additional demand by adjusting their holdings, limiting the price response. Moreover, changes in private safe asset supply often coincide with shifts in investors' demand for safety and liquidity, making supply-driven price effects difficult to isolate. Direct causal evidence on this substitution channel and its magnitude therefore remains limited.

This paper provides the first causal estimate of this substitution channel, showing that bank capital regulation can raise Treasury bill prices by restricting banks' provision of private safe assets. An increase in money market fund (MMF) demand for Treasury bills equal to 1 percent of the bill market's value lowers bill yield spreads by 4.5 basis points. This channel also highlights the asset-pricing implications of banks' role as issuers of safe claims, complementing existing research on how bank capital requirements affect prices by constraining intermediaries' balance sheet capacity.[1]

We use banks' quarter-end window dressing as a laboratory. Banks' short-term liabilities, including repurchase agreements and commercial paper, serve as private safe assets for institutional cash investors such as MMFs. Regulatory reporting based on quarter-end snapshots creates incentives for non-U.S. banks to temporarily contract their balance sheets. Around these reporting dates, non-U.S. banks reduce borrowing from U.S. money market funds to shrink their reported balance sheets (Munyan, 2017), effectively reducing available private safe asset supply. Constrained to hold short-term, high-quality assets, money mar-

[1] See evidence from repo markets (Munyan, 2017; Correa et al., 2020; Copeland et al., 2025), deviations from covered interest parity (Du et al., 2018b), FX swap markets (Wallen, 2020; Kloks et al., 2024; Du et al., 2025), and the long-term Treasury futures market (Fleckenstein and Longstaff, 2020a).

ket funds redeploy some of the cash into public safe assets: Treasury securities with short maturities. Treasury supply, however, does not expand to absorb this temporary substitution. With dealers' intermediation capacity constrained around reporting dates (Du et al., 2018b), arbitrageurs' ability to trade against MMFs' additional demand is limited, allowing the resulting price pressure to persist. Consistent with this mechanism, we find that short-maturity Treasury yields fall by 3–9 basis points at quarter-ends.

This identified channel matters because even when investors' underlying demand for safety remains unchanged, a contraction in banks' short-term liabilities can make an unchanged stock of public safe assets scarce relative to demand. Microprudential policies such as limits on bank leverage aim to make the financial system safer. Yet they can also reduce private safe-asset creation and raise the price of public safe assets. The Federal Reserve's RRP facility could mitigate these pressures by providing an alternative destination for temporary cash. However, investments through the facility may be imperfect substitutes for short-maturity Treasuries, allowing price pressure on Treasuries to persist.

In this paper, we first document the pricing pattern using daily security-level data on short-term U.S. Treasury securities from 2012 through 2024. Comparing the last trading day of each calendar quarter to other trading days, and controlling for security characteristics, monetary policy, auction, and issuance-type fixed effects, we find that quarter-end yields are 3–9 basis points lower, with the effect declining monotonically across maturity buckets below one year. In addition, we discover that yields begin declining roughly two weeks before the quarter-end date, reach their trough on the quarter-end day itself, and take another two weeks to fully recover, with the sharpest movement concentrated at the shortest maturities. The pattern is absent for Treasury notes and bonds with more than one year to maturity, pointing to the short-maturity investor base rather than a generic market-wide effect.

To explain the price pattern, our story combines MMFs' portfolio constraints with limited arbitrage capacity. At quarter-end, banks reduce borrowing from money funds to economize on balance sheet capacity. Funds' investment mandates direct some of the freed-up cash toward Treasury securities, increasing demand and lowering yields. Other investors' maturity preferences limit their willingness to accommodate this reallocation, while arbitrageurs must commit costly balance sheet capacity to offset the price pressure (Du et al., 2018b). These

frictions allow predictable demand shifts to generate persistent quarter-end price effects. The theory yields three testable predictions: short-maturity Treasury yields should fall at quarter-end; money funds should reduce bank-facing exposures and substitute into public safe assets; and Treasury yield declines should be larger where predicted money-fund demand pressure is stronger. A simple model in the appendix formalizes this story and its predictions.

We verify the story using monthly portfolio holdings from Form N-MFP filings, which report the complete security-level portfolios of all U.S. money market funds at month-end. The data reveal a sawtooth pattern in funds' bank-debt exposures: holdings contract sharply at each quarter-end and rebound the following month. The decline is concentrated among non-U.S. banks, especially those headquartered in the euro area, the United Kingdom, and Switzerland, whose leverage ratios are measured from quarter-end snapshots rather than daily averages. In aggregate, the quarter-end contraction amounts to roughly $100 billion, split between secured and unsecured instruments. The cash released by reductions in repo and commercial paper holdings is not accompanied by a statistically significant increase in aggregate net redemptions; instead, it flows into government money-like assets, predominantly Treasury bills and the Federal Reserve's overnight reverse repurchase facility. A within-fund analysis confirms the substitution directly: funds that reduce bank-funding exposures at quarter-end increase their Treasury holdings within the same reporting month. In total, MMF Treasury holdings rise by roughly $30 billion at quarter-ends—large enough to generate the yield movements we document. We also discuss why arbitrage does not eliminate the price effects of predictable demand shifts. Dealer balance-sheet constraints limit arbitrage capacity at quarter-end, while evidence from a companion paper shows that transaction costs eliminate the gross gains from an interest-rate-hedged bill-timing trading strategy (Tian and Yuan, 2026).

To quantify this substitution channel, we develop a network-composed shift-share design that traces quarter-end bank liability shocks through money market funds into maturity-specific Treasury demand and prices. The design combines predetermined fund–bank lending relationships with predetermined fund-level Treasury allocations across maturities. The first layer maps bank-specific liability contractions into predicted reallocations by individual funds, while the second maps those reallocations into demand pressure at each Treasury

tenor. We also document that individual MMFs specialize in different maturity segments even within the Treasury bill market, a pattern consistent with preferred-habitat behavior (Vayanos and Vila, 2021). These predetermined links in the bank–fund–Treasury network provide plausibly exogenous variation in Treasury demand across maturities, allowing us to estimate its causal effect on yields. We estimate that additional money-fund demand equal to 1% of outstanding market value lowers Treasury bill yield spreads by 4.5 basis points. This estimate is robust to alternative lag structures, subsample definitions, and weaker identification assumptions, and is supported by placebo tests. This effect is larger than comparable estimates for short-maturity Treasuries in the literature, which imply yield declines of around 1 basis point for an equivalent increase in demand.

Finally, we examine whether the existing RRP facility can mitigate quarter-end pressure on Treasury-bill prices. We find that funds continue to invest in bills despite having unused overnight RRP capacity, even initiating holding positions in bills whose yields are below the RRP rate. This evidence suggests that capacity constraints and return differences alone do not fully explain funds' allocations. One possible explanation is that bills allow funds to obtain longer maturity exposure, which limits substitution toward overnight investments. We discuss whether a flexible term RRP facility could complement the existing overnight facility and alleviate quarter-end Treasury-bill price pressure.

Literature Our paper contributes to the literature on safe and near-money assets. Treasury bills and other short-term government liabilities serve as public safe assets, while deposits, repo, commercial paper, certificates of deposit, and other intermediary liabilities serve as private safe assets. Complementing evidence on how public supply affects private money creation and prices, we identify and provide the first causal quantification of the reverse transmission channel at high frequency. The literature starts from the understanding that these instruments are imperfect substitutes (Gorton and Pennacchi, 1990; Krishnamurthy and Vissing-Jorgensen, 2012; Sunderam, 2015; Nagel, 2016; Krishnamurthy and Li, 2022). Private safe asset provision can be insufficient (Holmström and Tirole, 1998) or excessive because of run and fire-sale externalities (Stein, 2012), and public short-term debt can crowd out socially costly private money creation (Greenwood et al., 2015). Banks' production of

private safe assets responds to public safe asset supply (Kacperczyk et al., 2021), while the spread between bills and other short-term rates reflects the liquidity services of public money-like assets (Lenel et al., 2019). Recent work shows that Treasury bill scarcity affects the pricing of near-money assets (d'Avernas and Vandeweyer, 2023) and Treasury bill yields become more sensitive to changes in bill supply as money funds exhaust their bill holdings and less elastic investors become marginal (Stein and Wallen, 2025).

The intermediary asset-pricing literature links fluctuations in risk-bearing capacity and balance-sheet constraints to risk premia and arbitrage wedges (He and Krishnamurthy, 2013; Adrian et al., 2014; He et al., 2017; Haddad and Muir, 2021; Baron and Muir, 2022). In the quarter-end setting, existing research emphasizes how post-crisis leverage regulation and reporting incentives constrain intermediaries' arbitrage capacity, contributing to covered-interest-parity deviations, negative swap spreads, and related dislocations (Du et al., 2018b; Jermann, 2020; Boyarchenko et al., 2018; Du et al., 2023; Fleckenstein and Longstaff, 2020a). [2] We identify a complementary channel operating through banks' role as issuers of safe claims: regulation-induced balance-sheet contractions reduce private safe asset supply, shifting money market fund demand toward Treasury bills and raising their prices.

The paper also relates to Treasury-market anomalies. Research on auction cycles documents profits from trading old against newly issued bonds (Krishnamurthy, 2002), on-the-run premia (Lou et al., 2013), related patterns in TIPS (Christensen et al., 2020), and search-based explanations (Vayanos and Weill, 2008). Replication studies find persistent gaps between actual and synthetic TIPS, floating-rate notes, and Treasuries (Fleckenstein et al., 2014; Fleckenstein and Longstaff, 2020b; Du et al., 2018a). We instead study a repeated calendar-end effect in highly liquid short maturities, where duration risk is negligible, rather than pricing around issuance.

Our paper also contributes to the emerging literature on market macrostructure, which studies how large institutions' identities, mandates, and constraints shape asset prices (Haddad and Muir, 2025). Related work emphasizes institutional demand in segmented and

[2] Quarter-end balance-sheet contractions are also directly documented in U.S. repo and reserve markets (Munyan, 2017; Anbil and Senyuz, 2018; Correa et al., 2020; Copeland et al., 2025) and euro repo markets (Bassi et al., 2024), while granular studies trace associated spillovers into FX swaps (Aldasoro et al., 2022; Kloks et al., 2024; Wallen, 2020; Du et al., 2025).

preferred-habitat bond markets (Vayanos and Vila, 2021; Greenwood et al., 2023), heterogeneous demand elasticities (Koijen and Yogo, 2019; Gabaix and Koijen, 2021), and the maturity-specific effects of investor composition in Treasury markets (Chaudhary et al., 2024; Jansen et al., 2024). Our empirical contribution is a network shift-share design built on Adao et al. (2019); Goldsmith-Pinkham et al. (2020); Borusyak et al. (2022) that identifies institutional demand shocks in the highly liquid Treasury bill market, where relevant demand instruments are difficult to obtain.

Roadmap Section 2 provides institutional background and describes the data. Section 3 documents the quarter-end yield drop. Section 4 analyzes the portfolio holdings of money market funds around quarter-end. Section 5 constructs a proxy for window-dressing intensity and quantifies this channel. Section 6 discusses the implications of our findings for the design of the ON RRP facility. Section 7 concludes.

2 Institutional Background and Data

This section describes the institutional details of the short-term Treasury market. We first discuss its main participants and the mandates that shape their behavior, and then the data we use to measure short-term Treasury prices and money market fund portfolio reallocation.

Short-term U.S. Treasury Securities We define the short-term Treasury universe as marketable nominal Treasury securities with one year or less remaining to maturity. This universe includes Treasury bills as well as originally longer-maturity notes and bonds that have rolled down to the front end of the curve.[3] Treasury bills are central to our analysis because they are issued directly at short maturities, sold at a discount or at par, and redeemed at face value at maturity.[4] Regular bills are issued at 4-, 8-, 13-, 26-, and 52-week maturities

[3]We exclude TIPS and FRNs from the price analysis. FRNs carry a floating coupon that resets to the 13-week bill rate, so the fixed-rate spread we use—the yield over the federal funds lower bound—is not directly comparable across FRNs and conventional bills and notes; TIPS are inflation-indexed and trade in an institutionally distinct market from the nominal Treasuries we study.

[4]Figure B.27 plots the composition of Treasury securities with remaining maturity below one year and shows that bills are the dominant component of this short-maturity universe. Additional details on

(a 17-week tenor was added in 2022), while cash-management bills are issued irregularly to meet short-run financing needs.

The Treasury bill market is large and central to government financing. By the end of 2025, the amount of bills outstanding exceeded $6.6 trillion. Its investor base is distinct from that of longer-maturity Treasuries. According to Jansen et al. (2024), short-term Treasuries are held primarily by U.S. money market funds (MMFs), U.S. nonfinancial investors, and official institutions. MMFs have held the largest average share since 2012, accounting for roughly one-third of all outstanding short-term Treasury securities. In contrast, dealers and hedge funds play a substantially smaller role in this segment than in intermediate- and long-maturity Treasury markets. Jansen et al. (2024) further show that MMFs are the key marginal investors absorbing changes in Treasury bill supply. Figure B.25 shows that U.S. MMFs' footprint is most pronounced in Treasury bills, especially in the 0–6 month maturity range.

Money Market Funds Money market funds (MMFs) are regulated cash-management vehicles that invest in short-term, high-quality money-market instruments. As of 2024Q4, U.S. MMFs managed approximately $7.4 trillion in assets. MMFs are commonly classified into government, prime, and tax-exempt funds, as well as retail and institutional share classes. Government money market funds invest primarily in Treasury securities, government agency securities, cash, and repos collateralized by government securities, whereas prime funds may also invest in private money-market instruments such as commercial paper and certificates of deposit.

MMF portfolio choices are shaped by SEC Rule 2a-7, which imposes maturity, liquidity, and diversification requirements. In particular, MMFs generally may not acquire securities with remaining maturity greater than 397 days, must maintain a weighted average maturity (WAM) of no more than 60 days, and a weighted average life (WAL) of no more than 120 days. The rule also imposes issuer-concentration limits and requires funds to maintain minimum levels of daily and weekly liquid assets. These restrictions concentrate MMF portfolios in short-term government securities and money-market instruments and make MMFs natural

Treasury auction mechanisms are provided in Appendix B.6.3.

investors in Treasury bills. Figure B.1 provides a snapshot of MMF asset allocations at the end of 2024.

Data Our empirical analysis combines daily security-level Treasury prices with monthly security-level MMF portfolio holdings. For Treasury prices, we use the CRSP daily U.S. Treasury security file from January 2012 to December 2024. The file covers outstanding marketable Treasury securities and reports, for each security and trading day, end-of-day yields, prices, duration, amount outstanding, coupon rate, CUSIP, issue date, maturity date, and security type. We merge the CRSP Treasury file with TreasuryDirect auction data, from which we construct issuance amounts by tenor bucket to control for new-supply effects, and with a daily series on the lower bound of the federal funds target range to control for changes in the policy-rate environment. We restrict the sample to Treasury securities with remaining maturity of one year or less—Treasury bills plus originally longer-maturity notes and bonds that have rolled down to the front end.

For MMF holdings, we use monthly Form N-MFP filings submitted to the Securities and Exchange Commission by U.S. money market funds. These filings report each fund's complete portfolio holdings as of the final business day of the month. We obtain all N-MFP filings from EDGAR for 2012 through 2024 and classify each position into a major asset category: Treasury securities, agency securities, repurchase agreements, commercial paper, certificates of deposit, time deposits, municipal securities, and Federal Reserve overnight reverse repurchase agreements. For private money-market instruments, we standardize issuer names and map each issuer to its ultimate parent, country of domicile, and bank-or-nonbank status. Because N-MFP filings lack standardized issuer identifiers and report principal amounts and credit ratings unevenly, we supplement them with proprietary security-level MMF holdings data from Crane Data, our primary auxiliary source for issuer identity, principal, and short-term ratings. For repo positions, we observe the counterparty identity and collateral information.

The CRSP Treasury data allow us to measure daily yield dynamics around quarter-end reporting dates, while the N-MFP data allow us to observe month-end portfolio reallocations across asset classes, issuers, and counterparties. Because the two datasets differ in frequency,

we use the daily Treasury data to identify the timing and magnitude of price effects, and the monthly holdings data to trace the underlying portfolio substitution mechanism. Our main price-impact tests use the common January 2012–December 2024 sample. Additional details on issuer-name cleaning, parent mapping, and asset-class classification are provided in Appendix C.

3 The Quarter-End Short-Term Yield Pattern

We begin with the central pricing fact: short-maturity Treasury yields fall systematically at quarter-ends. The result is robust—it holds conditional on a rich set of security- and market-level controls and survives a battery of robustness checks. We develop the story linking it to MMF portfolio reallocation in Section 4.

3.1 Empirical Specification

For each Treasury security i and trading day t, we define the security-level Treasury yield spread as the difference between security i's yield and the contemporaneous lower bound of the federal funds target range:

$$\text{Spread}_{i,t} \;=\; \text{Yield}_{i,t} \;-\; \text{FedFundsLB}_t. \tag{1}$$

This construction removes the level of the contemporaneous policy-rate benchmark; the residual spread reflects other pricing components at the front end of the Treasury curve, which can represent a proxy for the convenience yield (Acharya and Laarits, 2023).[5]

We then define an indicator QEnd_t equal to one on the last trading day of each calendar quarter and zero otherwise, and estimate

$$\text{Spread}_{i,t} \;=\; \beta\,\text{QEnd}_t \;+\; \gamma' X_{i,t} \;+\; \delta' Z_t \;+\; \lambda_{k(i)} \times q(t) \;+\; \varepsilon_{i,t}, \tag{2}$$

[5]The time series of the cross-sectional spread distribution, partitioned into three remaining-maturity buckets (0–3, 3–6, and 6–12 months), is reported in Appendix B.2.1 (Figure B.4).

where $X_{i,t}$ collects security-level controls,[6] Z_t collects market-level controls, and $\lambda_{k(i)} \times q(t)$ denotes issuance-type-by-year-quarter fixed effects, with $k(i) \in \{\text{bill}, \text{note}, \text{bond}\}$ and $q(t)$ the calendar year-quarter containing day t. Standard errors are clustered at the issuance-type-by-year-quarter level. The coefficient of interest is β, which measures the average quarter-end change in the spread relative to other trading days; a negative β indicates a quarter-end yield drop.

Table 1 reports the estimates for three remaining-maturity buckets and two complementary samples. Panel A uses Treasury bills only—the universe most directly relevant to MMF demand and our focus throughout—while Panel B adds rolled-down notes and bonds with one year or less remaining maturity. Two patterns stand out. First, the quarter-end coefficient is negative in every cell and the absolute magnitude declines with remaining maturity: in the bills-only sample it is roughly 5–6 basis points at 0–3 months, 3–4 at 3–6 months, and 3 at 6–12 months (insignificant at the longest tenor), with comparable but larger magnitudes (9, 4, and 3 basis points) in the pooled sample. Second, the effect at the shortest tenor is larger in Panel B than in Panel A, indicating that rolled-down coupon securities—thinner in secondary-market liquidity than freshly issued bills—contribute disproportionately to the 0–3 month drop. The effect is strongest at short maturities, where MMF demand is concentrated.

The estimates are robust to a range of additional checks reported in Appendix B.2. Appendix B.2.2 examines the results before and after the 2015 phase-in of foreign-bank Basel III leverage-ratio disclosure and discusses the 2016 MMF reform.[7] We also restrict the comparison group to month-end trading days (Appendix B.2.3) and allow the effect to vary across calendar quarters (Appendix B.2.4). Further analyses decompose the pooled effect by security type and original tenor (Appendix B.2.8) and compare zero-lower-bound and positive-rate periods (Appendix B.2.10). The yield decline is concentrated in positive-rate periods.

[6] We omit the coupon rate from $X_{i,t}$ because it is absorbed by the issuance-type-by-year-quarter fixed effects: within a given issuance-type-quarter cell, T-bill coupons are zero and Treasury-note coupons are fixed at issuance.

[7] The SEC adopted the reform in 2014, with its principal provisions taking effect in October 2016. We refer to the reform throughout by its effective date.

Table 1: Quarter-End Effect on Short-Term Treasury Spreads

Dependent Variable:	Spread (pp)					
Sample:	Panel A. T-bills only			Panel B. Notes + Bonds + Bills		
Remaining maturity:	0–3 m	3–6 m	6–12 m	0–3 m	3–6 m	6–12 m
QEnd_t	-0.0565^{***}	-0.0359^{**}	-0.0344	-0.0912^{***}	-0.0412^{***}	-0.0303^{*}
	(0.0182)	(0.0160)	(0.0218)	(0.0184)	(0.0116)	(0.0172)
Issuance × Year-Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Security controls	Yes	Yes	Yes	Yes	Yes	Yes
Market-level controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	60,467	48,742	24,899	99,526	88,176	106,261
R^2	0.244	0.752	0.827	0.187	0.747	0.845
Within R^2	0.065	0.037	0.021	0.017	0.032	0.027

Notes: OLS estimates of equation (2) on the full daily sample, January 2012 through December 2024. Security-level controls $X_{i,t}$ are duration and the bid–ask spread; market-level controls Z_t are the dollar volume of newly auctioned Treasury bills and the face amount of Treasury debt maturing on day t. Standard errors, clustered at the issuance-type-by-year-quarter level, are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

3.2 Event-Window Analysis

To shed more light on the dynamics around quarter-ends, we construct an event-window panel in which each quarter-end week is compared to adjacent non-quarter-end weeks in the same, previous, or next quarter, indexed by offsets $k \in \{-4, \ldots, +4\}$ weeks. For each k and each remaining-maturity bucket, we estimate

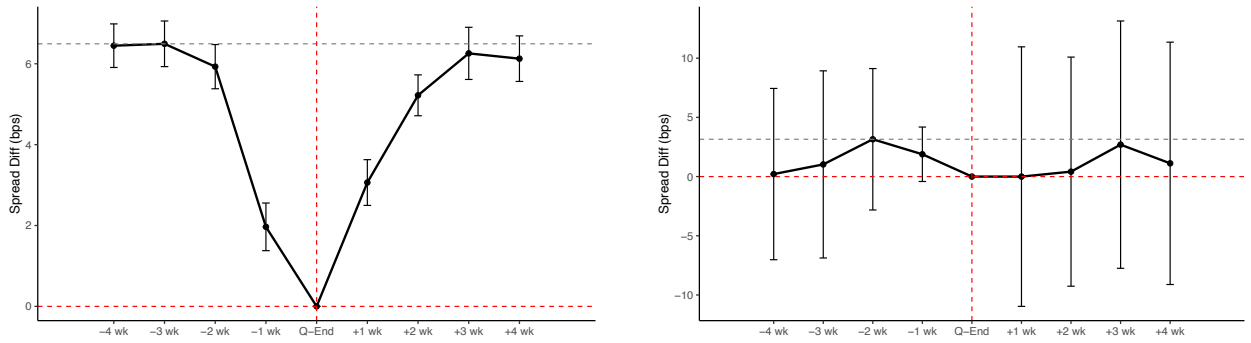
$$\text{Spread}_{i,t} = \beta_k \, \text{QEnd}_t + \gamma' X_{i,t} + \delta' Z_t + q(t) + \varepsilon_{i,t}, \tag{3}$$

where QEnd_t equals one for observations in the quarter-end week and zero for those in the offset-k week, $X_{i,t}$ and Z_t are the security- and market-level controls of equation (2), and $q(t)$ denotes year-quarter fixed effects. By construction, this design identifies β_k from a short-window contrast that holds the monetary policy regime, macroeconomic conditions, and aggregate Treasury supply approximately fixed.

Figure 1 traces out a pronounced V-shape in the entire Treasury-bill sample over the full 2012–2024 window: spreads are lowest in the quarter-end week and rise on either side, with a trough of roughly 6 basis points in absolute magnitude and tight confidence bands,

in line with the overall bills-only point estimate in Panel A of Table 1. Notably, the trough does not appear abruptly at the quarter-end: spreads begin declining roughly two weeks in advance, consistent with slow-moving capital widely documented in the literature.[8] The right panel reports a placebo estimated on Treasury notes with more than one year remaining maturity, which shows no comparable quarter-end V-shape—consistent with the effect being concentrated in the short-maturity bills held by MMFs rather than a generic quarter-end phenomenon across the Treasury curve.

Figure 1: Event-Study around Quarter-Ends: T-bills and Notes Placebo



T-bills

Placebo: Notes (more than one year), pooled

Notes: This figure plots the event-window coefficients $-\hat{\beta}_k$ from equation (3) for $k \in \{-4, \ldots, +4\}$ weeks relative to quarter-end, with the quarter-end week normalized to zero. The left panel uses the entire Treasury bills sample; the right panel is a placebo estimated on Treasury notes with more than one year remaining maturity (pooled). Error bars are 95% confidence intervals. The sample spans January 2012 through December 2024.

Separate event-window plots by remaining-maturity bucket, together with the notes-only

[8] In practice, MMFs reduce their bank exposure ahead of quarter-end not by selling positions—the secondary markets for banks' short-term liabilities are over-the-counter and relatively illiquid—but by letting maturing instruments run off without rolling them over, so the timing of the withdrawal tracks the tenor of each funding leg. Unsecured bank debt is comparatively long-dated: measured at month-end, the median remaining maturity is about 66 days for certificates of deposit and time deposits (CD/TD) and 67 days for financial commercial paper, with a 25th percentile of 18 and 27 days, respectively. Because this market is large, even the shortest quartile is a substantial dollar amount maturing in the weeks before quarter-end, and the freed cash flows into Treasuries over that window, producing the gradual pre-trend. The post-quarter-end reversal can likewise be gradual: banks rebuild their unsecured funding through new issuance spread over subsequent days, and MMFs progressively reallocate from Treasuries back into bank debt. The temporary demand pressure on Treasury bills therefore dissipates over time, consistent with yields taking roughly two weeks to recover rather than rebounding immediately. Repo, by contrast, is overwhelmingly short-dated—a median remaining maturity of roughly three days for Treasury and Agency repo and a 25th percentile of one day—so positions are unwound and re-established quickly, around and just after quarter-end. For further duration breakdown, see Table B.25.

and bonds-only placebo tests, are reported in Appendix B.2.9. The V-shape is present in every Treasury-bill maturity bucket but flattens and loses precision as remaining maturity rises, while neither placebo exhibits a comparable quarter-end pattern.

Taken together, the evidence in this section establishes the central empirical price fact of the paper: short-maturity Treasury yields fall by roughly 3–9 basis points on quarter-end days, with the largest effect at 0–3 months and a monotonic decline across remaining maturities. The pattern shows up both in the broader sample of all noncallable short-maturity Treasuries and in the bills-only sample, and is robust to a wide range of alternative samples and specifications.

4 MMF Portfolio Reallocation around Quarter-Ends

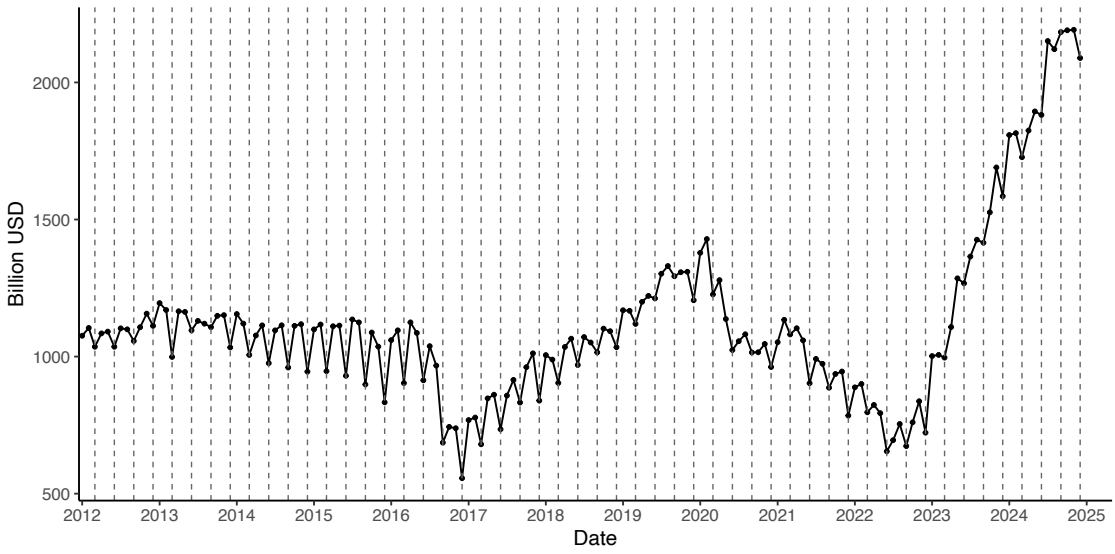
This section examines the portfolio reallocation channel linking banks' quarter-end window dressing to Treasury demand. As banks reduce their short-term borrowing around regulatory reporting dates, MMFs reallocate freed-up funds toward public safe assets, particularly Treasuries and the Federal Reserve's overnight reverse repurchase facility. We first document the magnitude of the contraction in MMFs' bank-debt holdings and its concentration among non-U.S. banks. We then decompose this contraction into changes in market participation and reductions in exposures among continuing funds. Next, we use fund-level data to trace the destinations of the released funding, examining whether funds that reduce their bank exposures increase their Treasury holdings. Finally, we consider whether trading against these predictable portfolio shifts offers arbitrageurs an opportunity to earn risk-free profits. Together, these analyses clarify the portfolio adjustments underlying quarter-end Treasury demand pressure and the limits to arbitraging its price effects.

4.1 Magnitude of the Window-Dressing Contraction

Non-U.S. banks have incentives to compress balance-sheet-intensive activities around regulatory reporting dates to appear less leveraged, because leverage ratios and related metrics are measured from quarter-end snapshots of their balance sheets. Figure 2 plots aggregate

MMF holdings of bank debt[9] with dashed vertical lines marking quarter-end dates. The series traces a pronounced *sawtooth* pattern: MMF holdings of bank debt fall sharply at each quarter-end and rebound the following month, quarter after quarter throughout the sample.

Figure 2: Aggregate Bank Debt Holdings by MMFs



Notes: The figure plots aggregate MMF holdings of bank debt at month-ends; dashed vertical lines mark quarter-end months. The sample spans 2012–2024.

The magnitudes are economically large. In non-quarter-end months, MMFs hold approximately $1.15 trillion of bank debt in aggregate, of which about $0.89 trillion is issued by non-U.S. banks. Table 2 reports the year-month regression estimates of aggregate holdings on a quarter-end indicator. At quarter-end, these holdings decline on average by roughly $96.6 billion, equivalent to 8.4% of total bank-debt holdings and 10.9% of foreign-bank holdings. The reduction is split almost evenly between repo and unsecured instruments. Appendix B.3.1 presents complementary event-study evidence, confirming that the decline is concentrated around quarter-end dates for both instruments.

[9]Bank debt here includes both secured instruments (repurchase agreements collateralized by bank-pledged collateral) and unsecured instruments (commercial paper, certificates of deposit, and time deposits) issued by the bank.

Table 2: Aggregate Quarter-End Shrinkage in MMF Bank-Debt Holdings

Dependent Variable:	Aggregate market value ($ bn)		
Sample:	Repo	Unsecured	Total
QEnd_t	-52.54^{***}	-42.84^{***}	-96.57^{***}
	(7.925)	(7.290)	(10.85)
Year-Quarter FE	Yes	Yes	Yes
Observations	156	156	156
R^2	0.990	0.975	0.982

Notes: Year-month panel, one observation per month-end ($N = 156$). The dependent variable is the aggregate market value (in USD billions) of MMF holdings of bank-issued debt (Repo / Unsecured / Total), summed across all funds and bank issuers. Each column regresses the corresponding aggregate on the quarter-end indicator with Year-Quarter fixed effects, so β is identified from within-quarter variation. Standard errors, clustered by year-quarter, in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

This window-dressing incentive is stronger for banks in jurisdictions that rely on quarter-end snapshots rather than period averages. Appendix B.3.2 decomposes the contraction by issuer region and by secured-versus-unsecured instrument and reports the corresponding event-window plots. These confirm that the quarter-end pattern is concentrated in non-U.S. banks—especially those headquartered in the euro area, the UK, and Switzerland—and operates symmetrically across repo and unsecured funding.

4.2 Intensive- and Extensive-Margin Decomposition

To further investigate these patterns, we decompose the quarter-end contraction into two margins. The *extensive* margin is the number of funds that leave the bank-funding market entirely at quarter-end; the *intensive* margin is the amount by which a fund that remains active reduces its position, conditional on continuing to invest in at least one bank through the quarter-end month. By an approximate decomposition, the change in total exposure is the sum of (i) the average change in position per fund, weighted by the number of active funds (the intensive margin), and (ii) the change in the number of active funds, weighted by

the average position size (the extensive margin):

$$\Delta \text{Value}_t \approx \underbrace{\Delta\big(\text{average position per fund}\big)}_{\text{intensive}} \times N_{\text{fund}} + \underbrace{\Delta N_{\text{fund}}}_{\text{extensive}} \times \big(\text{average position per fund}\big). \tag{4}$$

We begin with the extensive margin. Table 3 estimates the quarter-end reduction in the number of active funds at quarter-end, regressing the count of active funds per month on the quarter-end indicator. The number of participating funds falls by about 1.56 for repo, 0.96 for unsecured debt, and 1.80 with both asset classes pooled, all statistically significant. At an average investment of about $7.6 billion per fund-month, the pooled estimate implies a market-level extensive-margin reduction of roughly $13.7 billion on average.

Table 3: Magnitude of Window Dressing: Extensive Margin (Active Funds per Month)

Dependent Variable (count of):	Active Funds per Month		
Panel:	Year-Month		
Asset Class:	Repo	Unsecured	Both
QEnd_t	-1.563^{***}	-0.9615^{***}	-1.798^{***}
	(0.5759)	(0.3423)	(0.6048)
Year-Quarter FE	Yes	Yes	Yes
Observations	156	156	156
R^2	0.987	0.998	0.994
Within R^2	0.314	0.139	0.163

Notes: This table aggregates the data to a year-month panel and regresses the number of unique money funds with a positive position in the indicated asset class at month-end t on QEnd_t, capturing aggregate fund participation in the bank-funding market. Standard errors, clustered by year-quarter, in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

We next quantify the intensive margin, the reduction in position for a fund that remains active through the quarter-end month. We estimate a fund-month-level panel regression:

$$\text{Value}_{i,t} = \beta\, \text{QEnd}_t + \text{Controls}_{i,t} + \alpha_{i,q(t)} + \varepsilon_{i,t} \tag{5}$$

Table 4 reports the estimates and shows that MMFs significantly reduce their bank-funding exposures at quarter-end. The estimated coefficient on QEnd_t is negative and statisti-

cally significant across all specifications, implying that, within the same fund-year-quarter, quarter-end months are associated with reductions of about $0.35 billion in repo holdings, $0.57 billion in unsecured debt holdings, and $0.62 billion when both asset classes are pooled. The estimates are nearly identical whether holdings are measured using book value or market value, suggesting that the results are not driven by valuation differences. Across the roughly 140 unique funds with observations in both quarter-end and non-quarter-end months, aggregating the fund-level estimates implies an intensive-margin contraction on the order of $87 billion. Combining the two margins reproduces the order of magnitude of the directly measured aggregate quarter-end shrinkage of $96.6 billion.

Table 4: Magnitude of Window Dressing: Intensive Margin

Dependent Variables:	Book Value	Market Value	Book Value	Market Value	Book Value	Market Value
Asset Class:	Repo		Unsecured Debt		Both	
QEnd_t	-0.3449^{***}	-0.3448^{***}	-0.5734^{***}	-0.5718^{***}	-0.6248^{***}	-0.6240^{***}
	(0.0245)	(0.0245)	(0.0389)	(0.0389)	(0.0291)	(0.0291)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fund × Year-Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	21,909	21,909	11,173	11,173	23,215	23,215
R^2	0.987	0.987	0.990	0.990	0.991	0.991
Within R^2	0.033	0.033	0.054	0.054	0.048	0.048

Notes: Fund-level OLS estimates of $\text{Value}_{i,t} = \beta\,\text{QEnd}_t + \text{Controls}_{i,t} + \alpha_{i,q(t)} + \varepsilon_{i,t}$, where i indexes a money fund, t indexes a month-end, and $\alpha_{i,q(t)}$ are Fund × Year-Quarter fixed effects. The dependent variable is the dollar amount (book or market value, in USD billions) that fund i holds in the indicated asset class at month-end t. Controls include security characteristics, weighted average maturity (WAM), credit ratings, fund size, fund age, and (for repo) collateral type, collateral days-to-maturity, and haircut. Standard errors, clustered by fund × year-quarter, in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

As a robustness check, we repeat the analysis using the fund–bank-level dataset in Appendix B.3.3. At this level, the extensive margin decomposes further into declines in the number of funds per bank and the number of banks per fund, while the intensive margin captures the within-pair contraction in a fund's exposure to a given bank. Results for both margins confirm the patterns documented above.

4.3 Where Does the Withdrawn Bank Funding Go?

By an accounting identity, the bank funding that MMFs withdraw must be reallocated somewhere: fund shareholders may redeem their shares; the money may shift to other private issuers that do not window-dress, such as U.S. banks or non-financial corporations; it may flow into other asset classes, such as municipal bonds, Treasuries, or the Federal Reserve's overnight reverse repurchase facility (RRP); or it may remain as cash. Table 5 estimates this reallocation across all candidate destinations at the fund-month level on a balanced panel. The quarter-end inflow is concentrated in Treasuries and the RRP: the typical fund shifts about \$0.14 billion into Treasuries and \$0.56 billion into the RRP at quarter-end.

Table 5: Where Does the Money Go? Quarter-End Reallocation Across Asset Classes

Account	Cash	Netflow	Muni	Other	Treasury	RRP	Agency bonds	Other MMF shares
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
QEnd_t	0.0076**	0.0362	-0.0027***	-0.0344***	0.1392***	0.5603***	0.0286**	0.0152*
	(0.0037)	(0.0321)	(0.0010)	(0.0096)	(0.0265)	(0.0286)	(0.0137)	(0.0087)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fund × Year-Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	31,445	31,445	31,445	31,445	31,445	31,445	31,445	31,445
R^2	0.91792	0.36047	0.96890	0.96888	0.98871	0.97296	0.99171	0.99458
Within R^2	0.00024	6.5×10^{-5}	0.00050	0.00076	0.00174	0.02661	0.00031	0.00020

Notes: OLS estimates of $\text{Quantity}_{i,t} = \beta\,\text{QEnd}_t + \text{Controls}_{i,t} + \alpha_{i,q(t)} + \varepsilon_{i,t}$ at the fund-month level, on a balanced panel. Each column uses a different dependent variable: the dollar position in the named asset class (in USD billions). The control is fund netflow except for specification (2), and all specifications include fund × year-quarter fixed effects. Standard errors, clustered by fund × year-quarter, in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

To identify the destination of the freed-up cash *within the same fund*, we relate the quarter-end change in a fund's holdings of each candidate asset class to the contemporaneous change in its bank funding, controlling for net investor flows:

$$\Delta\text{Asset}_{i,t} = \beta_{Asset}\left(-\Delta\text{BankFunding}_{i,t}\right) + \delta\,\text{Netflow}_{i,t} + \alpha_{i,y(t)} + \varepsilon_{i,t}, \tag{6}$$

We restrict the sample to quarter-end months, where Asset includes Treasury Debt, RRP Facility, Agency Debt, Municipal Debt, Holdings of other MMFs, Other private debt, Cash, and Others, $-\Delta\text{BankFunding}_{i,t}$ is the within-fund reduction in fund i's holdings of bank-

issued repo and unsecured debt, $Netflow_{i,t}$ is net subscriptions to the fund (subscriptions less redemptions), and $\alpha_{i,y(t)}$ are fund-by-year fixed effects. A positive β means that the dollars a fund pulls from bank funding flow into the named asset. We report the regression in both portfolio-share (percentage points of AUM) and dollar units.

The fund-level accounting identity implies $\sum_{\text{Asset}} \Delta\text{Asset}_{i,t} + (\text{Netflow}_{i,t} - \Delta\text{AUM}_{i,t}) = -\Delta\text{BankFunding}_{i,t} + \text{Netflow}_{i,t}$, where the sum covers all nonbank accounts. We therefore include $\text{Netflow}_{i,t} - \Delta\text{AUM}_{i,t}$ as an additional outcome to reconcile investor flows with changes in fund size. Because all regressions use the same sample, controls, and fixed effects, the bank-funding coefficients across asset accounts and this adjustment sum to one. This adding-up property holds in both dollar units and units scaled by lagged AUM.

Table 6 reports the estimates. Holding net investor flows fixed, each \$1 reduction in bank funding is associated with approximately \$0.12 more in Treasuries and \$0.55 more in RRP within the same fund. In the lagged-AUM specification, a reduction in bank funding equal to one percentage point of the previous month's net assets is associated with increases of 0.09 and 0.18 percentage points in Treasury and RRP holdings, respectively. The larger RRP loading is consistent with its role as a temporary destination for cash, while Treasuries absorb an economically meaningful share of the reallocation. Table 5 shows no statistically significant quarter-end increase in fund net inflows. Together, these findings are consistent with reallocations within MMFs contributing to quarter-end Treasury demand.

Table 6: Where Does the Bank-Funding Dollar Go?

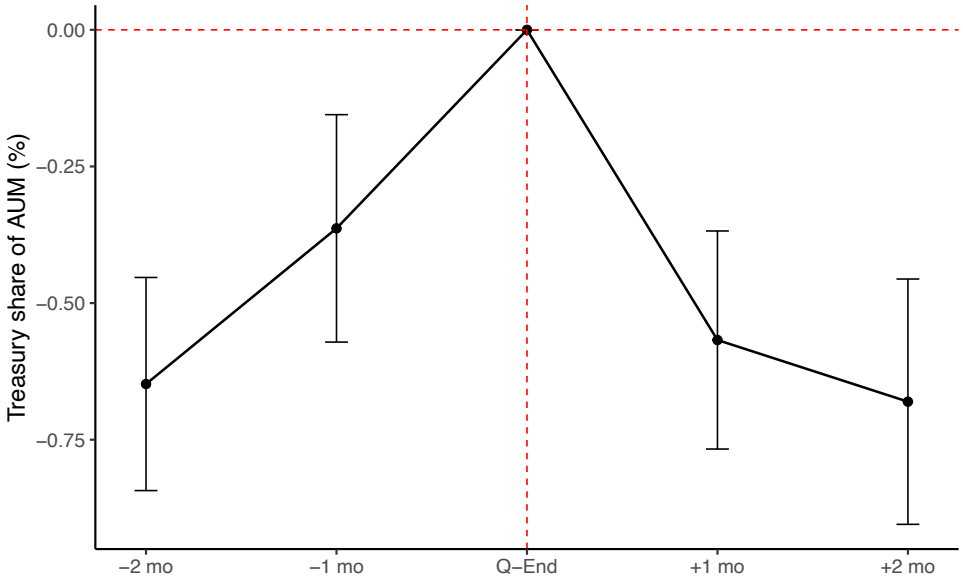
Outcome	Dollars ($)	Lagged AUM (pp)
Treasury debt	0.11774***	0.09065***
	(0.01449)	(0.00834)
RRP	0.55128***	0.17537***
	(0.01447)	(0.00867)
Agency debt	0.08063***	0.10069***
	(0.00912)	(0.00887)
Foreign Sovereign Debt	−0.00087***	−0.00032
	(0.00032)	(0.00071)
Municipal Debt	0.00138*	0.00894***
	(0.00076)	(0.00216)
Holdings of other MMFs	0.05397***	0.47818***
	(0.00602)	(0.01334)
Other Private Debt	0.03017***	0.04674***
	(0.00714)	(0.00785)
Others	0.00274*	0.00356*
	(0.00161)	(0.00181)
Cash	0.02062***	0.04887***
	(0.00566)	(0.00482)
Netflow - Δ AUM	0.14234***	0.04734***
	(0.00638)	(0.00599)
Sum of displayed coefficients	1	1
Observations	9,285	9,285
Funds	301	301
Fund × Year Fixed effects	Yes	Yes

Notes: Each row reports the coefficient on a reduction in bank funding from a separate regression for the indicated account. All regressions control for net investor flows and include fund-by-year fixed effects. Changes are measured from the previous month. Column (1) uses dollar amounts; column (2) expresses changes and flows as percentages of the previous month's net assets. Other Private Debt includes asset-backed, financial, non-financial, and other commercial paper; Treasury, agency, and other repo; and CDs/time deposits. Holdings already included in bank funding are excluded to avoid double counting. The dataset's Other category is reported separately. Cash and other net assets equals net assets less total reported security values. Net inflow minus AUM change is an accounting adjustment rather than a security holding. These accounts complete the accounting identity, so the reported coefficients sum to one in each column. All regressions use the same 9,285 quarter-end fund-month observations from 2012 to 2024. Conventional standard errors are in parentheses; ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Finally, we conduct an event-study analysis of Treasury holdings following the approach in Section 3.2. Figure 3 plots changes in the fund-level Treasury share of AUM relative to quarter-end over a two-month window on either side. Each horizon is estimated separately

against the quarter-end month, controlling for Treasury duration and including fund-by-event-quarter fixed effects. The estimates show a quarter-end peak: relative to an average quarter-end Treasury share of 34.0%, the share is 0.65 pp (1.90%) lower two months before and 0.36 pp (1.07%) lower one month before, and 0.57 pp (1.67%) lower one month after and 0.68 pp (2.00%) lower two months after. This pattern is consistent with portfolio reallocation toward Treasuries at quarter-end. Normalizing by AUM removes the mechanical effect of fund size.

Figure 3: Treasury Share of MMF AUM Event Window



Notes: The figure plots the fund-level MMF Treasury share (Treasury holdings as a share of the fund's AUM, in percentage points) in event time, over a window of ±2 months around the quarter-end ($t = 0$, normalized to zero). The specification includes within-fund × event-quarter fixed effects. The sample spans 2012–2024. We exclude the March–May 2020 dash-for-cash months due to the COVID crisis, and error bars denote 95% confidence intervals.

Taken together, the same-fund reallocation estimates and the Treasury-share event study support substitution from bank funding into Treasuries at quarter-end. The estimates in Table 5, which control for net investor flows, imply an aggregate quarter-end increase in MMF Treasury holdings of roughly $30 billion, consistent with the aggregate increase observed in the holdings data. Appendix B.3.5 shows that the same-fund substitution is robust across sub-periods and reports Treasury-share event windows separately for government and prime

MMFs.

4.4 Limits to Arbitrage at Quarter-End

Having established the quarter-end price pattern and the predictable shifts in demand for Treasury bills, we now ask why arbitrage does not eliminate these effects. We consider two explanations.

First, quarter-ends are associated with tighter dealer balance-sheet constraints and reduced intermediation capacity. Related price patterns have therefore been documented across several markets. For example, Du et al. (2018b) document arbitrage opportunities arising from deviations from covered interest parity, while Fleckenstein and Longstaff (2020a) examine related long–short strategies in the longer-term Treasury market. From this perspective, the quarter-end premium reflects an access rent: the institutions best positioned to trade it away face their highest balance-sheet costs precisely when the opportunity arises.

Second, even an investor with ample balance-sheet capacity may find it difficult to profit from the pattern. Consider a strategy that purchases Treasury bills one week before quarter-end and sells them on the quarter-end date, when the price premium peaks. In a companion paper, Tian and Yuan (2026) implement this strategy four times per year. Over a sample spanning more than a decade beginning in 2012, the strategy generates a cumulative loss of \$0.15 per \$100 invested, a Sharpe ratio of -0.77, and no profitable quarters. Broader interest-rate movements dominate the quarter-end component of the outright bill return. A predictable quarter-end price premium therefore does not, by itself, constitute an arbitrage opportunity.

Hedging interest-rate exposure with an overnight index swap improves performance, producing a modestly positive gross return and a Sharpe ratio of approximately 0.32, comparable to the long-run U.S. equity-market benchmark of 0.33. However, this performance does not survive transaction costs. Once bill bid–ask costs are included, the Sharpe ratio falls to approximately -0.42, eliminating the strategy's gross advantage. Implementation also requires access to interest-rate derivatives, creating additional barriers for investors without established derivatives-trading arrangements, for example, an ISDA membership. Thus, even with sufficient balance-sheet capacity, exploiting the quarter-end premium requires managing

interest-rate risk, covering transaction costs, and securing access to the necessary hedging instruments.

5 Estimating the Treasury Price Multiplier

We now quantify this substitution channel and estimate the corresponding price multiplier.

Money market funds (MMFs) allocate freed-up cash across Treasuries with different remaining maturities. In our empirical panel, let τ index one-month buckets of remaining maturity, ranging from one to twelve months; let t index month-end observations; and let $q(t)$ denote the corresponding calendar quarter. The tenor-by-quarter fixed effects $\gamma_{\tau,q(t)}$ correspond to the "year-quarter $\times$ maturity bucket" fixed effects reported in the tables. Let $\Delta spread_{t,\tau}$ denote the quarter-end spread change for tenor τ. We write:

$$\Delta spread_{t,\tau} = \beta \Delta \theta^{MMF}_{t,\tau} + \mathcal{A}_{t,\tau}, \tag{7}$$

where $\Delta \theta^{MMF}_{t,\tau}$ measures MMF Treasury demand pressure in tenor τ, normalized by the amount outstanding in that tenor bucket, and $\mathcal{A}_{t,\tau}$ collects other non-MMF pricing forces. We decompose the latter as

$$\mathcal{A}_{t,\tau} = \alpha_t + \gamma_{\tau,q(t)} + X'_{t,\tau}\delta + \nu_{t,\tau}, \tag{8}$$

where the date fixed effects α_t absorb pricing shocks common to all tenors at a given month-end, including the common effects of monetary-policy news, liquidity conditions, quarter-end dynamics, and arbitrage capacity. The tenor-by-quarter fixed effects $\gamma_{\tau,q(t)}$ absorb tenor-specific differences in spread behavior and pricing conditions that persist within a quarter. The vector $X_{t,\tau}$ contains observed controls for other pricing forces such as new auction amount and tenor-specific liquidity (proxied by bid-ask spread), and $\nu_{t,\tau}$ captures the remaining tenor-month pricing variation.

A regression on realized $\Delta \theta^{MMF}_{t,\tau}$ would be difficult to interpret causally, because realized Treasury holdings are equilibrium outcomes: yields affect MMF allocations, and unobserved

Treasury-market shocks move both prices and holdings. For example, if bill yields rise for reasons unrelated to MMF demand, MMFs may respond by buying more bills. The resulting positive relationship between MMF holdings and bill yields would therefore reflect MMFs responding to yields, rather than the causal effect of MMF demand on yields. Our strategy therefore uses predicted window-dressing-induced Treasury demand pressure directly as the regressor of interest. Specifically, we estimate

$$\Delta spread_{t,\tau} = \beta WDIT^{QE}_{t,\tau} + \alpha_t + \gamma_{\tau,q(t)} + X'_{t,\tau}\delta + \epsilon_{t,\tau}, \tag{9}$$

where $WDIT^{QE}_{t,\tau}$ denotes predicted quarter-end window-dressing-induced Treasury demand pressure in tenor τ, constructed below. The coefficient of interest, β, measures the reduced-form equilibrium price multiplier from predicted MMF demand pressure to short-maturity Treasury spread movements. It captures the average price response across tenors at quarter-ends, net of all other pricing forces absorbed by fixed effects and controls.

We construct $WDIT^{QE}_{t,\tau}$ in two steps. We first define a fund-level exposure to bank window-dressing shocks:

$$B_{i,t} = \sum_j \theta^{Bank}_{i,j,t-1} \times \Delta Debt_{j,t}, \tag{10}$$

where i indexes money market funds, j indexes banks, and $t-1$ denotes the month preceding the quarter-end. The term $\theta^{Bank}_{i,j,t-1} = Value_{i,j,t-1} / \sum_{i'} Value_{i',j,t-1}$ is the predetermined share of bank j's MMF funding provided by fund i before the quarter-end window, where $Value_{i,j,t-1}$ is the market value of fund i's holdings of bank j's repo and unsecured debt and j ranges over non-U.S., non-Canadian banks. The bank-level shift, $\Delta Debt_{j,t}$, is the quarter-end change in bank j's debt held by MMFs. We then map the fund-level bank-side shocks into Treasury-tenor-level pressure using funds' lagged Treasury portfolio weights, and normalize by the tenor's market size for interpretability:

$$WDIT^{QE}_{t,\tau} = \text{QEnd}_t \frac{100}{Size_{t-1,\tau}} \sum_i \theta^{Bill}_{i,\tau,t-1} B_{i,t}, \tag{11}$$

where $\theta^{Bill}_{i,\tau,t-1}$ denotes fund i's lagged portfolio weight on Treasury securities of tenor τ,

$Size_{t-1,\tau}$ is the lagged outstanding market value of tenor τ, and QEnd_t is an indicator variable equal to one for quarter-end months and zero otherwise. Because a quarter-end contraction makes $\Delta Debt_{j,t} < 0$, the constructed pressure $WDIT^{QE}_{t,\tau}$ is negative for the tenors into which money funds channel the freed-up cash. A positive β estimate means a larger predicted demand inflow (a more negative $WDIT^{QE}_{t,\tau}$) is associated with a larger spread decline.

For consistency of the OLS estimator, we assume that residual tenor-month pricing shocks are orthogonal to predicted window-dressing pressure after conditioning on fixed effects and controls:

$$\mathbb{E}\left[\epsilon_{t,\tau} WDIT^{QE}_{t,\tau} \mid \alpha_t, \gamma_{\tau,q(t)}, X_{t,\tau}\right] = 0. \tag{12}$$

In Appendix B.4, we show that Assumption 1, if satisfied, implies the orthogonality condition (12) and therefore allows us to recover β consistently.

Assumption 1. *(Share Exogeneity) For any Treasury tenor τ and bank j in each period t, the MMF's lagged holding share in tenor τ, $\theta^{Bill}_{i,\tau,t-1}$, times the lagged bank–fund exposure $\theta^{Bank}_{i,j,t-1}$ is orthogonal to the unobserved tenor-level price shock $\epsilon_{t,\tau}$:*

$$\mathbb{E}_{\tau,j,t}\left[\epsilon_{t,\tau}\theta^{Bill}_{i,\tau,t-1}\theta^{Bank}_{i,j,t-1}\middle|\ \alpha_t,\ \gamma_{\tau,q(t)},\ X_{t,\tau}\right] = 0, \quad \forall i, j, \tau, t$$

This identification strategy builds on the insight of Goldsmith-Pinkham et al. (2020) that Bartik instruments can be interpreted as instruments constructed from predetermined exposure shares. In this sense, our approach resembles an exposure design: Treasury bill tenors are exposed to common window-dressing shocks, and the magnitude of that exposure is jointly determined by lagged MMF portfolio allocations and lagged bank–fund relationships (Chaudhary et al., 2023).

To illustrate the intuition, consider a quarter-end at which a bank reduces its borrowing from MMFs by $100 billion. The size of this window-dressing shock may itself be correlated with Treasury bill prices.[10] Suppose the bank has historically obtained 20% of its MMF

[10]Some dealer banks may use capital borrowed from MMFs to engage in arbitrage across Treasury securities. Through term-structure linkages, these activities may be correlated with Treasury bill prices.

funding from Fund A and 80% from Fund B, reflecting persistent bank–fund relationships. Under our construction, the bank's reduction in borrowing generates an additional $20 billion of liquidity for Fund A and $80 billion for Fund B that must be reallocated.

Next, suppose each fund mechanically reallocates this liquidity according to its predetermined portfolio weights. For simplicity, assume Fund A allocates half of its assets to Treasury securities and invests exclusively in one-month bills, while Fund B also allocates half of its assets to Treasury securities but invests exclusively in three-month bills. The bank's borrowing reduction therefore generates a $10 billion demand shock to one-month bills and a $40 billion demand shock to three-month bills.

If lagged MMF portfolio weights and lagged bank–fund relationships are orthogonal to contemporaneous shocks to Treasury bill prices, then the relative demand pressure across bill tenors induced by the common bank shock is as good as randomly assigned. This variation provides the identifying source of cross-sectional demand shocks used in our empirical design.

Table 7 reports the baseline estimates of equation (9). The sample includes nominal Treasury bills, notes, and bonds with less than one year to maturity.[11] All columns include report-month and year-quarter-by-maturity-bucket fixed effects and control for the bid–ask spread and auction amount. Column (1) uses the full sample, column (2) uses observations before 2015, and column (3) uses observations from 2015 onward. Across the three samples, the coefficient on $WDIT^{QE}_{t,\tau}$ is positive and statistically significant. In the full sample the coefficient is 4.5, so a predicted demand inflow of one percent of a tenor's market size lowers its spread by about 4.5 basis points.

Moreover, the estimated multiplier is larger before 2015 (6.4) than from 2015 onward (4.2), although the pre- and post-2015 coefficients are not statistically distinguishable from each other ($p = 0.46$). Two forces may potentially explain this pattern. First, the Federal Reserve's overnight reverse repo (ON RRP) facility expanded over the sample period. When its capacity was limited before 2015, MMFs allocated cash displaced from bank funding primarily to Treasury bills. As capacity increased (Appendix B.4.8), the facility absorbed more of this reallocation, dampening its impact on Treasury prices. Second, the substantial growth in the outstanding stock of Treasury bills may have been accompanied by greater

[11] We exclude securities within 15 days of maturity. Near maturity, small price differences imply large annualized yield swings and trading thins out, making their measured spreads unreliable.

market depth and more elastic demand from other investors, reducing the price response to MMF demand pressure.

Table 7: Window-Dressing Spillovers to Short-Maturity Treasury Spreads

Dependent Variable:	$\Delta spread$		
	Full Sample	2012–2014	2015–2024
Specification:	(1)	(2)	(3)
$WDIT^{QE}$	4.476***	6.434***	4.187**
	(1.725)	(2.232)	(1.946)
Controls	Yes	Yes	Yes
YearQuarter × Maturity bucket	Yes	Yes	Yes
Report month	Yes	Yes	Yes
Observations	1,840	432	1,408
R^2	0.831	0.585	0.833
Within R^2	0.006	0.048	0.005

Notes: OLS estimates of equation (9). The dependent variable is the month-end change in the spread to the federal-funds lower bound, in basis points. The same short-maturity Treasury universe—nominal bills, notes, and bonds with under one year to maturity—is used to construct both $WDIT^{QE}_{t,\tau}$ and $\Delta spread_{t,\tau}$, and securities within 15 days of maturity are excluded. All specifications include report-month and year-quarter-by-maturity-bucket fixed effects and control for the bid–ask spread. Standard errors, clustered by year-quarter-by-maturity bucket, are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

5.1 Exogenous Shares

To assess the plausibility of joint exogeneity of the exposure shares with respect to unobserved pricing shocks, we examine whether our constructed regressors are systematically related to observable financial conditions or business models of the underlying funds or banks. We conduct these checks using the same tenor–month observations and estimation sample as the specification reported in Table 7, mapping fund- and bank-level characteristics to the tenor–month level.[12]

[12] Let $X_{i,t-1}$ denote a fund-level characteristic and $Z_{j,t-1}$ a bank-level characteristic. We aggregate fund characteristics using each fund's lagged Treasury portfolio share in tenor τ:

$$\bar{X}_{t-1,\tau} = \frac{\sum_i \theta^{Bill}_{i,\tau,t-1} X_{i,t-1}}{\sum_i \theta^{Bill}_{i,\tau,t-1}}. \tag{13}$$

For bank characteristics, define the lagged bank–tenor exposure share as

$$s_{j,\tau,t-1} = \sum_i \theta^{Bill}_{i,\tau,t-1} \theta^{Bank}_{i,j,t-1}.$$

Tables 8 and 9 report the corresponding results.[13] Conditional on the included controls and fixed effects, neither fund nor bank characteristics are statistically significantly associated with $WDIT^{QE}_{t,\tau}$ in these specifications. These tests therefore provide no evidence of a systematic linear relationship between the regressor and the observed characteristics of funds or their bank counterparties. We also add these characteristics as direct controls to the baseline regression. Table B.16 shows that the estimated coefficient on $WDIT^{QE}_{t,\tau}$ remains close to the baseline estimate, changing from 4.48 to 4.77 when both sets of controls are included.

Table 8: Bank Characteristics and Aggregate WDIT

Dependent Variable:	$WDIT^{QE}$				
	(1)	(2)	(3)	(4)	(5)
Log CDS spread	0.0616				0.0653
	(0.0654)				(0.0689)
Tier 1 capital ratio (pp)		0.0019			-0.0030
		(0.0110)			(0.0117)
Short-term borrowing / total funding (pp)			-0.0009		-0.0011
			(0.0017)		(0.0018)
ROA (pp)				-0.0222	0.0005
				(0.0676)	(0.0748)
Report Month	Yes	Yes	Yes	Yes	Yes
YearQuarter × Maturity bucket	Yes	Yes	Yes	Yes	Yes
Observations	1,840	1,840	1,840	1,840	1,840
R^2	0.77180	0.77151	0.77157	0.77154	0.77188
Within R^2	0.00412	0.00286	0.00309	0.00298	0.00447

Notes: Columns (1)–(4) regress predicted quarter-end window-dressing-induced Treasury demand pressure, $WDIT^{QE}_{t,\tau}$, on each bank characteristic separately. Column (5) includes all four jointly. All columns use the same 1,840 cells. Log CDS spread measures the market's assessment of bank credit risk. The Tier 1 capital ratio is regulatory Tier 1 capital divided by risk-weighted assets and measures capital strength. The short-term borrowing ratio is short-term borrowings divided by total funding, as reported by Bloomberg, and measures reliance on short-term funding. ROA is return on assets and measures bank profitability. Variables labeled "pp" are expressed in percentage points. Each bank characteristic is mapped to the cell with the lagged bank–tenor exposure shares of equation (14), built for every month from holdings in the prior calendar month. Following Table B.15, the 45 cells without usable lagged bank exposure are retained by filling the characteristic with its observed-cell mean; every column carries the four coverage fractions and the missing-value indicator, and standard errors are clustered by year-quarter-by-maturity bucket as in Table 7. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

We then construct the corresponding tenor-level bank characteristic as

$$\bar{Z}_{t-1,\tau} = \frac{\sum_j s_{j,\tau,t-1} Z_{j,t-1}}{\sum_j s_{j,\tau,t-1}}. \tag{14}$$

[13]Bank-level characteristics are unavailable for some bank–month observations. Their coverage is reported in Appendix Table B.15.

Table 9: Fund Characteristics and Aggregate WDIT

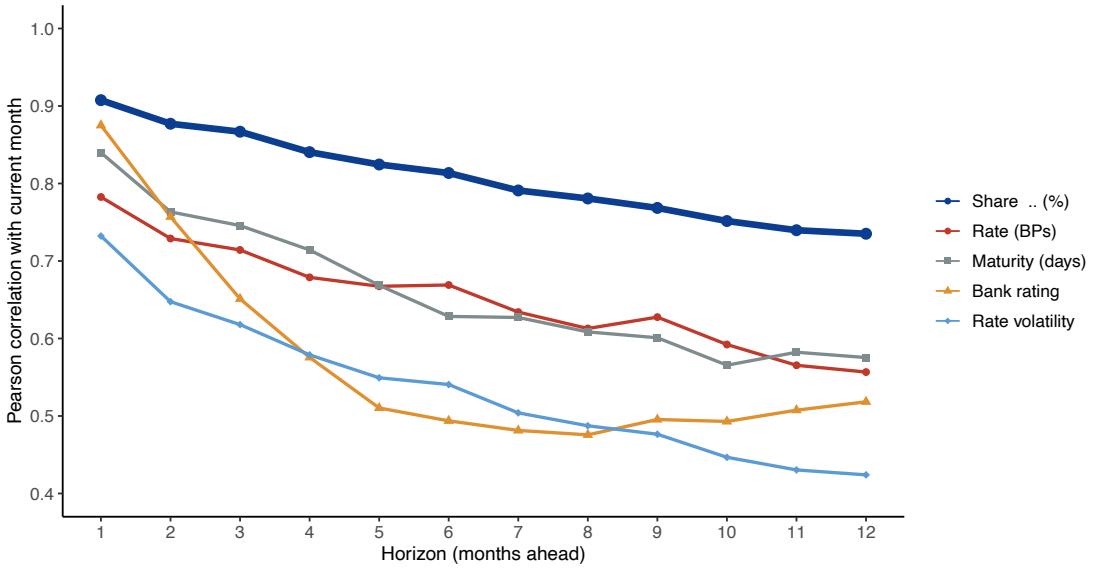
Dependent Variable:	$WDIT^{QE}$						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Cash / AUM (pp)	0.0036	0.0038	0.0038	0.0042	0.0044	0.0044	0.0040
	(0.0027)	(0.0028)	(0.0028)	(0.0029)	(0.0029)	(0.0028)	(0.0029)
Government-fund share (pp)	0.0007	0.0007	0.0007	0.0007	0.0007	0.0007	0.0007
	(0.0005)	(0.0005)	(0.0005)	(0.0005)	(0.0005)	(0.0005)	(0.0005)
Seven-day gross yield (pp)		0.0047	0.0054	0.0063	0.0070	0.0071	0.0067
		(0.0237)	(0.0233)	(0.0230)	(0.0228)	(0.0229)	(0.0232)
WAM (days)			0.0003	0.0004	0.0005	0.0005	0.0005
			(0.0013)	(0.0013)	(0.0013)	(0.0013)	(0.0013)
Daily liquid assets / AUM (pp)				-0.0004	-0.0005	-0.0005	-0.0005
				(0.0009)	(0.0009)	(0.0009)	(0.0009)
Log(1 + minimum investment)					0.0021	0.0021	0.0022
					(0.0026)	(0.0026)	(0.0026)
Log fund AUM						0.0011	0.0022
						(0.0075)	(0.0077)
Net flow / AUM (pp)							-0.0014
							(0.0015)
Report Month	Yes	Yes	Yes	Yes	Yes	Yes	Yes
YearQuarter × Maturity bucket	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,840	1,840	1,840	1,840	1,840	1,840	1,840
R^2	0.77215	0.77217	0.77218	0.77222	0.77239	0.77240	0.77261
Within R^2	0.00566	0.00572	0.00577	0.00596	0.00670	0.00672	0.00764

Notes: Each column reports a regression of predicted quarter-end window-dressing-induced Treasury demand pressure, $WDIT^{QE}_{t,\tau}$, on lagged fund characteristics. The dependent variable is measured in percentage points of the outstanding Treasury stock in the corresponding tenor bucket. Fund characteristics are mapped to the tenor–month level using lagged Treasury portfolio shares as weights, following equation (13) and (14). All specifications include month-end date and year-quarter × maturity-bucket fixed effects. Standard errors are reported in parentheses. AUM denotes assets under management. Cash denotes total fund cash holdings. Government-fund share is the weighted average of an indicator for government MMFs. Seven-day gross yield is the annualized seven-day gross yield offered to shareholders. WAM is the weighted-average maturity of the fund's overall portfolio. Daily liquid assets are defined according to SEC Rule 2a-7. Minimum investment is the minimum initial investment required by the fund. Net flow is the monthly net capital flow into the fund. Variables labeled "pp" are expressed in percentage points. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

We next analyze $\theta^{Bank}_{i,j,t}$ and $\theta^{Bill}_{i,\tau,t}$ separately, beginning with evidence that the exposure shares $\theta^{Bank}_{i,j,t}$ capture stable fund–bank relationships. Figure 4 shows that these shares are highly persistent: their autocorrelation is approximately 0.91 at one month and remains approximately 0.81 at six months. Moreover, exposure shares are substantially more persistent than observable fundamentals that might motivate portfolio reallocation: the autocorrelations of the funding rate, its volatility, position maturity, and the bank's credit rating all decline more rapidly. These patterns support interpreting $\theta^{Bank}_{i,j,t}$ as a predetermined relationship exposure rather than a tactical portfolio adjustment around quarter-end. Evidence at

the extensive margin reinforces this interpretation: Figure B.19 in the appendix shows that 85% of fund–bank relationships remain active one year after they are established.

Figure 4: Autocorrelation of $\theta_{i,j,t}^{Bank}$ and Its Drivers

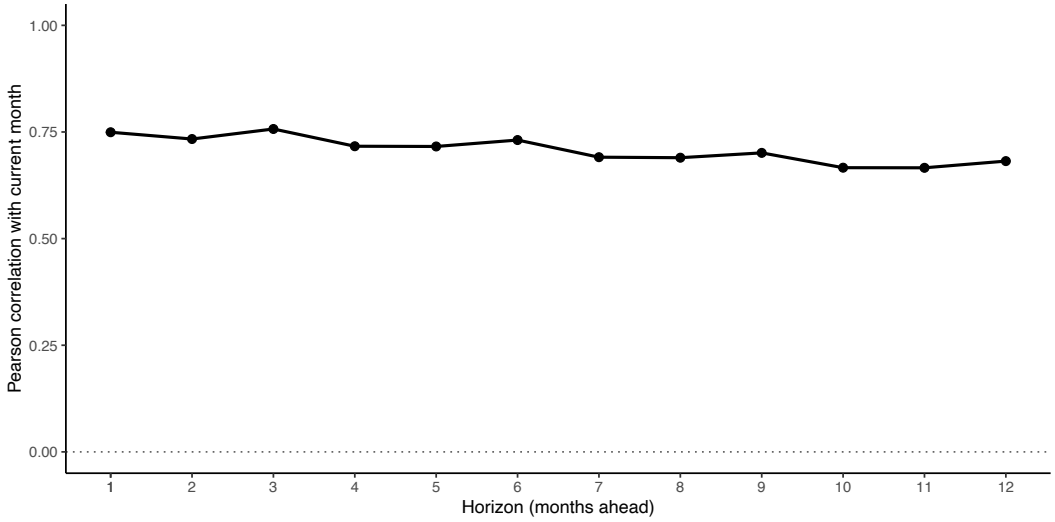


Notes: This figure plots the pooled Pearson autocorrelation of the fund–bank funding share $\theta_{i,j,t}^{Bank}$ (dark blue) at horizons of $x = 1, \ldots, 12$ months, together with the autocorrelation of the observable fundamentals of the same fund–bank position: the value-weighted funding rate over EFFR (Rate, in bps), the within-position dispersion of security spreads (Rate volatility), the value-weighted days-to-maturity (Maturity), and the value-weighted numeric Moody's rating (Bank rating). Each series is the pooled correlation of the pair-month value with its value x months later. The share is markedly more persistent than any of its drivers at every horizon. The x-axis measures the horizon in months.

The persistence of these relationships is also consistent with the institutional features of the MMF industry. Since the Global Financial Crisis, MMFs have increasingly prioritized capital preservation, liquidity, and regulatory compliance over yield maximization. Successive reforms have tightened liquidity, credit quality, and maturity requirements, reinforcing incentives for funds to maintain stable funding and trading relationships rather than continuously search for marginal yield differentials. In this environment, relationship-based intermediation is a natural outcome. Our findings are consistent with the evidence in Huber (2023), who shows that MMFs often transact simultaneously at different repo rates rather than exclusively supplying funds to the highest-yielding counterparty, even after conditioning on issuer risk.

What about MMFs' Treasury investment decisions? The lagged portfolio weights $\theta^{Bill}_{i,\tau,t-1}$ are the second key component of our identification strategy. These weights are measured prior to the quarter-end window and therefore predate the Treasury-market price movements that we seek to explain. Repeating the same persistence exercise reveals that Treasury-tenor allocations are also highly stable over time, as shown in Figure 5. This evidence supports the interpretation of $\theta^{Bill}_{i,\tau,t-1}$ as predetermined portfolio exposures rather than active responses to quarter-end price fluctuations.

Figure 5: Autocorrelation of $\theta^{Bill}_{i,\tau,t}$



Notes: This figure plots the autocorrelation of $\theta^{Bill}_{i,\tau,t}$ measured in percentage points. The x-axis measures the time lags.

A natural follow-up question is what determines a fund's allocation across Treasury bill tenors. Beyond idiosyncratic fund preferences, a primary determinant is duration management under the prevailing interest-rate environment. As illustrated in Figure B.20, when interest rates are rising, MMFs have incentives to shorten portfolio maturity in order to preserve reinvestment flexibility and reduce interest-rate risk. Conversely, when rates are expected to decline, funds may modestly extend maturity to lock in prevailing yields. Beyond the first moment—that is, whether interest rates are rising or falling—Abdullah and Tase (2025) find that the second moment of the policy rate also matters. More specifically, they use survey data to measure policy rate uncertainty and find MMFs tend to lower their overall

portfolio duration by investing in shorter Treasuries when facing higher uncertainty. For our analysis, these duration-management considerations are driven by aggregate interest-rate conditions that are common across funds and bill tenors. Our empirical specification absorbs such common forces through month-end fixed effects and tenor-by-quarter fixed effects. Consequently, identification does not rely on variation generated by changing interest-rate expectations. Instead, conditional on these fixed effects and other controls, the lagged tenor weights provide a predetermined mapping that translates fund-level bank-funding shocks into predicted Treasury-bill demand pressure across tenors.

For completeness, Figures B.17 and B.18 in the appendix plot average monthly changes in bill and bank shares, respectively. These averages remain close to zero over time, and 80% of monthly share changes fall within ±2 percentage points.

The baseline specification measures both exposure shares one month before the quarter-end window. Appendix Table B.19 re-estimates the full-sample spillover coefficient using fund–bank exposure shares lagged $k_1 \in \{1, \ldots, 6\}$ months and Treasury tenor weights lagged $k_2 \in \{1, \ldots, 6\}$ months, as well as their respective averages over the preceding twelve months. The ΔDebt shocks remain measured over the one-month quarter-end window. Consistent with the persistence of exposure shares documented above, the coefficient on $WDIT^{QE}$ remains positive and statistically significant at conventional levels across all specifications. Table B.24 complements this analysis by comparing OLS and IV estimates of the relationship between Treasury spread changes and actual changes in MMF Treasury holdings, scaled by lagged market size in the corresponding maturity bucket. The OLS coefficients are statistically indistinguishable from zero while the IV coefficient remains similar to our baseline estimate. In addition, in Appendix B.4, we further compare the predicted demand pressure, $WDIT^{QE}_{t,\tau}$, with the realized quarter-end portfolio reallocation, $\Delta Bill^{QE}_{t,\tau}$, which incorporates the impact of fund flows, and document a close correspondence between the two measures.

Appendix B.4.10 shows that our results are robust to a weaker identification assumption, which requires exogeneity only of one layer of the shares. The resulting estimates are quantitatively very similar to our benchmark specification. The results are also similar when the post-quarter-end reversal period is included, with the coefficient on $WDIT$ remaining positive and statistically significant (Appendix Table B.23). As a timing placebo, Appendix

Table B.22 shows that current induced trading is not significantly associated with the previous month's spread change, both in the full sample and in the pre- and post-2015 subsamples. Table B.21 provides an alternative construction by netting out fund i's own contribution to the fund-level shock. Finally, Figure B.23 benchmarks our estimated price impact against comparable estimates in the existing literature, placing our results in the context of the broader evidence on institutional demand elasticities.

5.2 Unpacking the $WDIT^{QE}$ Regressor

Having presented our benchmark estimates of the price multiplier and discussed the plausibility of our identifying assumptions, we further assess the validity of our identification strategy using the diagnostic tests and visualization procedures proposed by Goldsmith-Pinkham et al. (2020).

In our setting, the conventional "industry share" corresponds to the exposure of Treasury bill tenor τ to bank issuer j through fund i, given by

$$\sum_i \theta^{Bill}_{i,\tau,t-1}\theta^{Bank}_{i,j,t-1}.$$

Our sample comprises funds that reallocate capital from the liabilities of 79 banks into Treasury securities, implying that the shift-share regressor is a weighted average of $J = 79$ underlying bank-specific shocks, one for each issuer bank j. The Rotemberg decomposition below operates at the issuer-by-quarter-end-month level, so the underlying components are these issuer-month cells, and Panel D later aggregates them back to the issuer level to rank the most influential banks. Since these banks differ substantially in their contribution to the aggregate instrument, the overall validity of the identification strategy depends primarily on the plausibility of the identifying assumptions for the most influential banks.

5.2.1 Rotemberg Weights

To formally rank the importance of individual banks, we perform a Rotemberg decomposition of the regressor outlined in Goldsmith-Pinkham et al. (2020). If any particular instrument is misspecified, the Rotemberg weight tells us how sensitive the overall estimator

is to the misspecification of the individual component. To be specific, we can decompose our estimator in the following way[14]:

$$\hat{\beta} = \sum_{j,t} \hat{\alpha}_{j,t} \hat{\beta}_{j,t} \tag{15}$$

where $\sum_{j,t} \hat{\alpha}_{j,t} = 1$ and $\hat{\alpha}_{j,t}$ is the Rotemberg weight. Each $\hat{\beta}_{j,t}$ is the just-identified estimate of the price multiplier, obtained using only bank j's month-t component as the instrument.

We illustrate the Rotemberg decomposition using our benchmark specification, reported in Column (1) of Table 7. Appendix Table B.17 summarizes the distribution of the resulting Rotemberg weights. Panel A shows that the Rotemberg weights are predominantly positive across banks. Panel B further shows that these weights are strongly negatively correlated (-0.83) with the signed window-dressing shock. Thus, the banks experiencing the largest quarter-end liability contractions contribute most to identification, making the validity of the identifying assumptions for these banks particularly important. Reassuringly, the component-level estimates $\hat{\beta}_{jt}$ are essentially uncorrelated with both the Rotemberg weights and the size of the underlying shock (Panel B), so the aggregate estimate is not mechanically driven by up-weighting components with a particular $\hat{\beta}$. To assess the sensitivity of our results, we identify the five banks with the largest Rotemberg weights and re-estimate the benchmark specification using each of these banks individually. The corresponding estimates, reported in Panel D, range from 0.3 to 6.6. Because each is identified off a single bank's variation alone, the individual confidence intervals are wide, but the estimates are uniformly positive, and four of the five confidence intervals contain the benchmark estimate of 4.5, consistent with the identifying assumptions. Collectively, these five banks account for approximately 65.3% of the total Rotemberg weight. Panel C shows that observations from 2016–2018 contribute the most to identification. Finally, Panel E demonstrates that the overall estimator is driven primarily by observations with positive α values, while observations with negative α values contribute relatively little. Appendix Figure B.16 visualizes the bank-level weights, F-statistics, and just-identified estimates.

Following Goldsmith-Pinkham et al. (2020), we then regress the bank–tenor exposure

[14]The way to compute the individual weight and the just-identified estimator is shown in the appendix.

share for each of the five banks with the largest Rotemberg weights on these mapped characteristics. Table B.18 reports the results. Overall, we find little evidence that observable fund characteristics systematically explain the bank–tenor exposure shares. While a small number of coefficients are statistically significant, these estimates are economically small, do not exhibit a consistent sign across banks, and are not robust across specifications. A joint test that the coefficients on all mapped characteristics are zero fails to reject the null in every column (p between 0.21 and 0.91), and aggregate $WDIT_{t,\tau}^{QE}$ in column (6) loads on no characteristic individually or jointly. These findings suggest that the identifying variation in our shift-share instrument is not driven by observable fund characteristics, providing additional support for the exogeneity of the predetermined exposure shares underlying our empirical design.

6 ON RRP and Quarter-End Treasury Market Stability

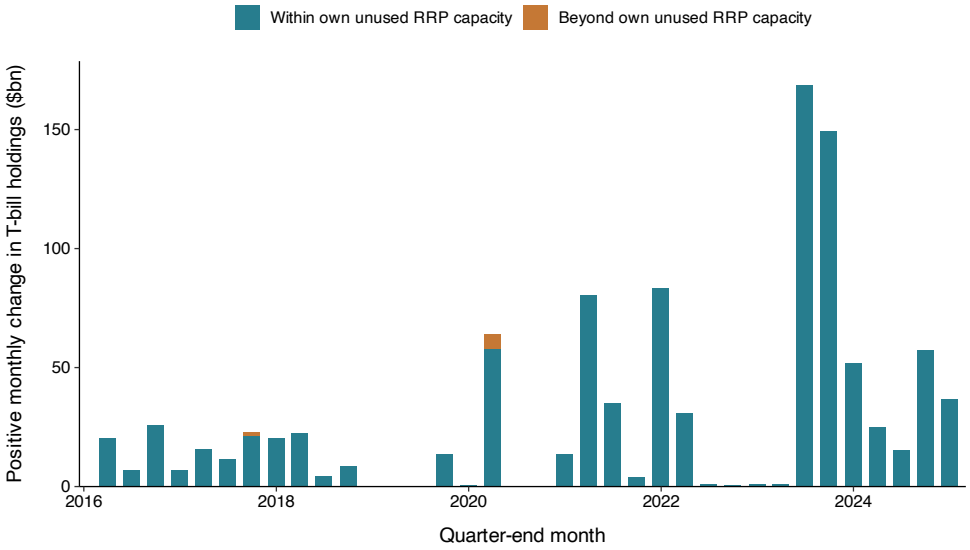
We have documented that quarter-end contractions in the supply of private safe assets induce money market funds (MMFs) to reallocate toward Treasury bills. Yet eligible MMFs also have access to the Federal Reserve's overnight reverse repurchase (ON RRP) facility, which began operating on a trial basis in September 2013. Why, then, does this facility not fully absorb the resulting demand shift and cushion its impact on the Treasury bill market? In this section, we examine the limits to substitution between Treasury bills and ON RRP.

First, MMFs exhibit substantial heterogeneity in their portfolio management strategies, including their preferred maturity profiles. The three monthly snapshots in Appendix B.5.1 show wide cross-sectional dispersion in the weighted-average maturity (WAM) of funds' Treasury holdings. Because each panel compares funds within the same month, this dispersion arises within a common interest-rate and macroeconomic environment. The pattern is consistent with funds pursuing distinct maturity targets, potentially reflecting differences in beliefs, liquidity needs, or specialized portfolio management strategies. Related evidence from the broader mutual fund industry documents persistent demand for particular factor exposures (Peng and Wang, 2021). For funds seeking to maintain exposure to longer ma-

turities, an overnight investment in ON RRP may therefore be an imperfect substitute for Treasury bills.

While differences in maturity preferences provide one possible explanation for limited substitution, binding ON RRP limits could also induce funds to turn to Treasury bills. We therefore examine whether funds had sufficient unused ON RRP capacity to accommodate their observed increases in bill holdings at quarter-end. Figure 6 shows that, in most sampled quarter-end months, the positive monthly increases in Treasury-bill holdings could have been accommodated almost entirely within the same funds' unused ON RRP capacity. This suggests that individual ON RRP limits are unlikely to be the main explanation for the quarter-end reallocation toward Treasury bills. Instead, the continued accumulation of bills despite this available headroom is consistent with imperfect substitutability between Treasury bills and ON RRP. The availability of a public safe asset therefore need not fully insulate the Treasury bill market from shifts in demand.

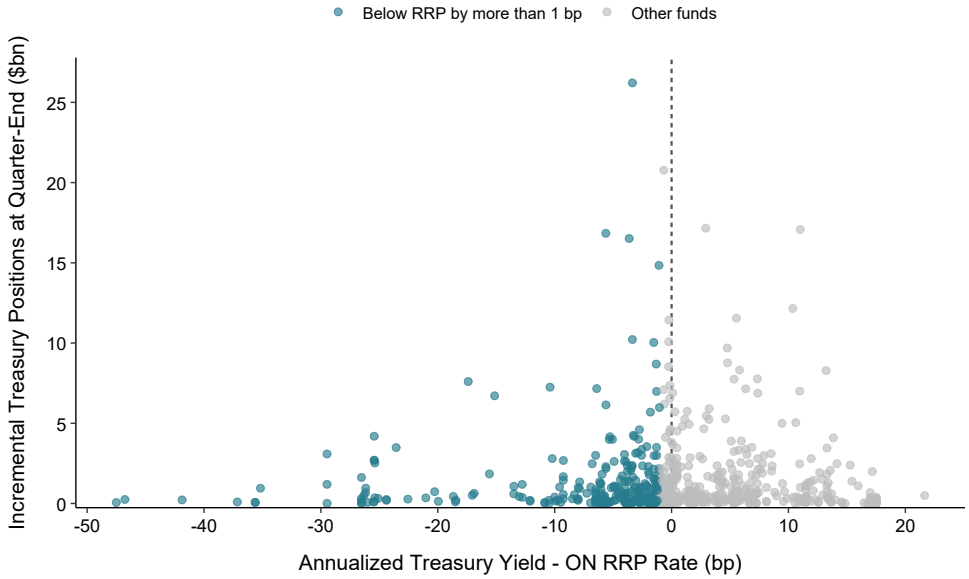
Figure 6: Quarter-End Treasury-Bill Increases and Unused ON RRP Capacity



Notes: Each bar reports the sum of positive changes in Treasury-bill holdings between the preceding month-end and the indicated quarter-end, measured in billions of dollars. The sample covers 2016–2024 and includes prime and government money market funds with positive reported ON RRP investments and an available previous-month filing. For each fund, unused ON RRP capacity equals the applicable per-counterparty limit less the principal invested in ON RRP at quarter-end. The teal portion represents bill increases that could fit within the same funds' unused capacity; the orange portion represents increases exceeding that capacity. This decomposition is performed separately for each fund before aggregation. Declines in holdings are not netted against increases. Treasury bills are identified using CRSP security classifications, and holdings are measured at reported market values. The comparison measures headroom under individual limits rather than aggregate facility availability.

Finally, the availability of unused ON RRP capacity leaves open a rate-based explanation: funds may favor Treasury bills simply because the RRP facility offers a lower rate. Figure 7 shows that this explanation is incomplete. The figure compares the annualized yields on newly appearing Treasury-bill positions with the contemporaneous ON RRP rate. In 41.6% of the sampled fund-quarter observations, the market-value-weighted yield on these positions is more than one basis point below the ON RRP rate, even though the funds have demonstrated access to the facility through contemporaneous use. Thus, the continued allocation to Treasury bills cannot be fully explained by their offering higher current yields. This pattern is consistent with funds valuing maturity exposure and other portfolio attributes alongside contemporaneous yield.

Figure 7: Treasury-Bill Additions and Yield Spreads Relative to ON RRP



Notes: Each point represents one fund at one quarter-end. The sample contains 697 fund-quarter observations from 76 prime and government money market funds during 2016–2024, restricted to funds with positive verified ON RRP use. New bill positions are identified by CUSIPs held at quarter-end but absent from the same fund's preceding month-end filing. Eligible bills have 16–90 days remaining to maturity, and their combined new-position value must exceed $1 million. The horizontal axis reports the market-value-weighted annualized yield on these positions minus the contemporaneous ON RRP rate, in basis points. Bill yields are calculated from CRSP midpoint prices using an ACT/360 convention. The vertical axis reports the quarter-end market value of these positions, in billions of dollars. Price coverage must reach at least 99% by value for both the eligible bill portfolio and its new positions. Teal points indicate an average yield more than one basis point below ON RRP; the dashed line marks equal yields. New-position values are ending holdings rather than observed purchase expenditures. Yields are measured at quarter-end, not at execution, and the comparison does not adjust for expected future overnight rates.

These findings motivate complementing the ON RRP facility with a flexible term RRP

program that offers eligible MMFs a menu of short maturities spanning quarter-end. Allowing funds to select maturities consistent with their portfolio objectives could make the facility a closer substitute for Treasury bills and enable it to absorb more of the reallocation triggered by contractions in private safe-asset supply. Such a program could reduce concentrated demand for particular bill tenors and dampen quarter-end price distortions.

The Federal Reserve previously conducted term RRP operations spanning quarter-ends in 2014–2015. Following the expansion of ON RRP capacity in December 2015, policymakers judged that additional term operations were unlikely to materially improve control of the federal funds rate.[15] That assessment, however, does not establish whether term operations could mitigate maturity-specific price distortions in the Treasury bill market. Moreover, most of the testing occurred near the zero lower bound. Although the final year-end 2015 operations took place after liftoff, this limited experience leaves open how a flexible term RRP program would perform across interest-rate regimes.

7 Conclusion

This paper identifies and quantifies a safe-asset substitution channel through which bank regulation affects short-term Treasury prices, complementing explanations centered on limited intermediation capacity. At quarter-end, non-U.S. banks reduce short-term borrowing from money market funds, contracting the supply of private safe assets through repo and unsecured funding markets. Money funds redirect part of the displaced cash toward Treasury bills. A reduction in private safe-asset supply can therefore increase demand for public safe assets even without a change in investors' underlying demand for safety.

Using daily security-level Treasury prices from 2012 to 2024, we document systematic quarter-end yield declines of approximately 3–9 basis points across maturity buckets below one year, with the largest effects at the shortest maturities. These patterns remain after controlling for security characteristics, market conditions, and monetary policy. To quantify the reallocation channel, we develop a network shift-share design that maps bank funding contractions into predicted Treasury demand using predetermined fund–bank relationships

[15] See Simon Potter, "Money Markets after Liftoff: Assessment to Date and the Road Ahead," February 22, 2016, https://www.newyorkfed.org/newsevents/speeches/2016/pot160222.

and fund-level maturity allocations. Tenors exposed to greater predicted demand pressure experience larger yield declines. The estimated local price multiplier implies that a predicted demand inflow of 1% of a tenor's outstanding market value lowers its yield spread by approximately 4.5 basis points.

Our findings suggest that ON RRP investments and Treasury bills are imperfect substitutes. Among sampled funds already using the facility, unused capacity could accommodate most bill additions, yet funds hold newly appearing bill positions whose average quarter-end yields fall below the contemporaneous ON RRP rate. Together with differences in funds' Treasury maturity profiles, this evidence suggests that capacity limits and current yield differences alone cannot fully explain bill demand. A flexible term RRP program offering maturities spanning quarter-end could better match funds' portfolio preferences, absorb more of the demand displaced by bank window dressing, and mitigate Treasury price pressure. Its effectiveness would depend on its maturity, pricing, and liquidity terms.

More broadly, our findings shed light on the effects of microprudential regulation beyond the banking sector. Limits on leverage can strengthen bank resilience while restricting the supply of private safe assets. The resulting substitution toward public safe assets can raise Treasury prices even when investors' underlying demand for safety is unchanged. Policies intended to make banks safer can therefore generate price pressures in public safe asset markets. A broader assessment of prudential policy must account for how constraints on individual institutions reshape the supply and pricing of safe assets across markets.

References

Abdullah, S. and Tase, M. (2025). Policy rate uncertainty and money market funds (MMF) portfolio allocations. Finance and Economics Discussion Series 2025-063, Board of Governors of the Federal Reserve System.

Acharya, V. V. and Laarits, T. (2023). When do treasuries earn the convenience yield?: A hedging perspective.

Adao, R., Kolesár, M., and Morales, E. (2019). Shift-share designs: Theory and inference. *The Quarterly Journal of Economics*, 134(4):1949–2010.

Adrian, T., Etula, E., and Muir, T. (2014). Financial intermediaries and the cross-section of asset returns. *The Journal of Finance*, 69(6):2557–2596.

Aldasoro, I., Ehlers, T., and Eren, E. (2022). Global banks, dollar funding, and regulation. *Journal of International Economics*, 137:103609.

Anbil, S. and Senyuz, Z. (2018). The regulatory and monetary policy nexus in the repo market. Finance and Economics Discussion Series 2018-027, Board of Governors of the Federal Reserve System.

Baron, M. and Muir, T. (2022). Intermediaries and asset prices: International evidence since 1870. *The Review of Financial Studies*, 35(5):2144–2189.

Bassi, C., Behn, M., Grill, M., and Waibel, M. (2024). Window dressing of regulatory metrics: Evidence from repo markets. *Journal of Financial Intermediation*, 57:101071.

Borusyak, K., Hull, P., and Jaravel, X. (2022). Quasi-experimental shift-share research designs. *The Review of Economic Studies*, 89(1):181–213.

Boyarchenko, N., Eisenbach, T. M., Gupta, P., Shachar, O., and Van Tassel, P. (2018). Bank-intermediated arbitrage. *FRB of New York Staff Report*, (858).

Caballero, R. J., Farhi, E., and Gourinchas, P.-O. (2017). The safe assets shortage conundrum. *Journal of Economic Perspectives*, 31(3):29–46.

Chaudhary, M., Fu, J. Z., and Li, J. (2026). Which intermediary costs matter for asset prices? *Working Paper*.

Chaudhary, M., Fu, J. Z., and Zhou, H. (2024). Anatomy of the treasury market: Who moves yields? *Olin Business School Center for Finance & Accounting Research Paper*, (2024/14).

Chaudhary, M., Fu, Z., and Li, J. (2023). Corporate bond multipliers: Substitutes matter. *Working Paper*.

Christensen, J. H. E., Fischer, J. A. L., and Shultz, P. L. (2020). Is there an on-the-run premium in TIPS? *Federal Reserve Bank of San Francisco Working Paper*.

Copeland, A., Duffie, D., and Yang, Y. (2025). Reserves were not so ample after all. *The Quarterly Journal of Economics*, 140(1):239–281.

Correa, R., Du, W., and Liao, G. Y. (2020). US banks and global liquidity. Technical report, National Bureau of Economic Research.

d'Avernas, A. and Vandeweyer, Q. (2023). Treasury bill shortages and the pricing of short-term assets. *The Journal of Finance*.

Doerr, S. and Eren, E. (2025). Money market funds and the pricing of near-money assets. *The Journal of Finance*. Forthcoming; previously circulated as SFI Research Paper 23-04.

Du, W., Hébert, B., and Huber, A. W. (2023). Are intermediary constraints priced? *The Review of Financial Studies*, 36(4):1464–1507.

Du, W., Im, J., and Schreger, J. (2018a). The U.S. Treasury premium. *Journal of International Economics*, 112:167–181.

Du, W., Strasser, G., and Verdelhan, A. (2025). Repo and fx swap: A tale of two markets. Technical report, Working Paper.

Du, W., Tepper, A., and Verdelhan, A. (2018b). Deviations from covered interest rate parity. *The Journal of Finance*, 73(3):915–957.

Fleckenstein, M. and Longstaff, F. A. (2020a). Renting balance sheet space: Intermediary balance sheet rental costs and the valuation of derivatives. *The Review of Financial Studies*, 33(11):5051–5091.

Fleckenstein, M. and Longstaff, F. A. (2020b). The US Treasury floating rate note puzzle: Is there a premium for mark-to-market stability? *Journal of Financial Economics*, 137(3):637–658.

Fleckenstein, M., Longstaff, F. A., and Lustig, H. (2014). The TIPS–Treasury bond puzzle. *The Journal of Finance*, 69(5):2151–2197.

Gabaix, X. and Koijen, R. S. J. (2021). In search of the origins of financial fluctuations: The inelastic markets hypothesis. NBER Working Paper 28967, National Bureau of Economic Research.

Goldsmith-Pinkham, P., Sorkin, I., and Swift, H. (2020). Bartik instruments: What, when, why, and how. *American Economic Review*, 110(8):2586–2624.

Gorton, G. (2017). The history and economics of safe assets. *Annual Review of Economics*, 9:547–586.

Gorton, G. and Pennacchi, G. (1990). Financial intermediaries and liquidity creation. *The Journal of Finance*, 45(1):49–71.

Greenwood, R., Hanson, S., Stein, J. C., and Sunderam, A. (2023). A quantity-driven theory of term premia and exchange rates. *The Quarterly Journal of Economics*, 138(4):2327–2389.

Greenwood, R., Hanson, S. G., and Stein, J. C. (2015). A comparative-advantage approach to government debt maturity. *The Journal of Finance*, 70(4):1683–1722.

Haddad, V. and Muir, T. (2021). Do intermediaries matter for aggregate asset prices? *The Journal of Finance*, 76(6):2719–2761.

Haddad, V. and Muir, T. (2025). Market macrostructure: Institutions and asset prices. *Annual Review of Financial Economics*, 17.

He, Z., Kelly, B., and Manela, A. (2017). Intermediary asset pricing: New evidence from many asset classes. *Journal of Financial Economics*, 126(1):1–35.

He, Z. and Krishnamurthy, A. (2013). Intermediary asset pricing. *American Economic Review*, 103(2):732–70.

Holmström, B. and Tirole, J. (1998). Private and public supply of liquidity. *Journal of Political Economy*, 106(1):1–40.

Huber, A. W. (2023). Market power in wholesale funding: A structural perspective from the triparty repo market. *Journal of Financial Economics*, 149(2):235–259.

Jansen, K. A., Li, W., and Schmid, L. (2024). Granular treasury demand with arbitrageurs. Technical report, National Bureau of Economic Research.

Jermann, U. J. (2020). Negative swap spreads and limited arbitrage. *The Review of Financial Studies*, 33(1):212–238.

Kacperczyk, M., Pérignon, C., and Vuillemey, G. (2021). The private production of safe assets. *The Journal of Finance*, 76(2):495–535.

Kloks, P., Mattille, E., and Ranaldo, A. (2024). Hunting for dollars. *Swiss Finance Institute Research Paper*, (24-10).

Koijen, R. S. and Yogo, M. (2019). A demand system approach to asset pricing. *Journal of Political Economy*, 127(4):1475–1515.

Krishnamurthy, A. (2002). The bond/old-bond spread. *Journal of Financial Economics*, 66(2-3):463–506.

Krishnamurthy, A. and Li, W. (2022). The Demand for Money, Near-Money, and Treasury Bonds. *The Review of Financial Studies*.

Krishnamurthy, A. and Vissing-Jorgensen, A. (2012). The aggregate demand for Treasury debt. *Journal of Political Economy*, 120(2):233–267.

Lenel, M., Piazzesi, M., and Schneider, M. (2019). The short rate disconnect in a monetary economy. *Journal of Monetary Economics*, 106:59–77.

Lou, D., Yan, H., and Zhang, J. (2013). Anticipated and repeated shocks in liquid markets. *The Review of Financial Studies*, 26(8):1891–1912.

Munyan, B. (2017). Regulatory arbitrage in repo markets. *Working Paper*.

Nagel, S. (2016). The liquidity premium of near-money assets. *The Quarterly Journal of Economics*, 131(4):1927–1971.

Ozdenoren, E., Yuan, K., and Zhang, S. (2023). Dynamic asset-backed security design. *The Review of Economic Studies*, 90(6):3282–3314.

Peng, C. and Wang, C. (2021). Factor demand and factor returns. Technical report, London School of Economics and Political Science, LSE Library.

Siriwardane, E. N., Sunderam, A., and Wallen, J. (2025). Segmented arbitrage. *The Journal of Finance*, 80(5):2543–2590.

Stein, J. C. (2012). Monetary policy as financial stability regulation. *The Quarterly Journal of Economics*, 127(1):57–95.

Stein, J. C. and Wallen, J. (2025). The imperfect intermediation of money-like assets. *The Journal of Finance*. Forthcoming.

Sunderam, A. (2015). Money creation and the shadow banking system. *The Review of Financial Studies*, 28(4):939–977.

Tian, Y. and Yuan, K. (2026). The price of balance-sheet warehousing: Quarter-end dislocations in u.s. money markets and the limits to arbitrage. *LSE Working Paper*.

Vayanos, D. and Vila, J.-L. (2021). A preferred-habitat model of the term structure of interest rates. *Econometrica*, 89(1):77–112.

Vayanos, D. and Weill, P.-O. (2008). A search-based theory of the on-the-run phenomenon. *The Journal of Finance*, 63(3):1361–1398.

Wallen, J. (2020). Markups to financial intermediation in foreign exchange markets. *Working Paper*.

Appendix Contents

A A Theoretical Model **49**

A.1 Discussion of Modeling Choices 53

B Empirical Appendix **55**

B.1 Empirical Support for the Model Assumptions 55

B.2 Robustness and Heterogeneity of the Results in Section 3 57

B.2.1 Time Series of the Cross-Sectional Spread Distribution 57

B.2.2 Regime Shifts: Basel III Phase-In and the 2016 MMF Reform 58

B.2.3 Month-End Comparison-Group Robustness 59

B.2.4 Heterogeneity by Calendar Quarter 60

B.2.5 FOMC Meeting Robustness 61

B.2.6 Leave-One-Quarter-Out Analysis 62

B.2.7 Index-Level Regression 62

B.2.8 Decomposition by Security Type and Original Tenor 63

B.2.9 Event-Window Plots by Maturity Bucket and Placebo Tests 65

B.2.10 Zero-Lower-Bound Regime Heterogeneity 66

B.3 Robustness and Heterogeneity of the Results in Section 4 70

B.3.1 Aggregate Bank-Funding Event Study 70

B.3.2 Decomposition by Issuer Region and Instrument 71

B.3.3 Bank–Fund-Level Evidence on the Intensive and Extensive Margins . 75

B.3.4 Window-Dressing Magnitude: Pre vs. Post Basel III 76

B.3.5 Same-Fund Substitution by Sub-Period and by Fund Type 79

B.4 Derivations and Robustness for Section 5 81

B.4.1 Proof of Orthogonality under the Strong Orthogonality Assumption . 81

B.4.2 Bank-side Characteristics Coverage 81

B.4.3 Baseline Regression with Additional Controls 82

B.4.4 Rotemberg Weights: Algebra and Full Results 82

B.4.5 $\Delta\theta^{Bill}_{i,\tau,t}$ and $\Delta\theta^{Bank}_{i,j,t}$ over Time 86

B.4.6 Persistence of Fund–Bank Relationships 87

B.4.7 MMF Portfolio Duration and the Policy Rate . . . 88
B.4.8 The Overnight Reverse Repurchase (ON RRP) Facility . . . 88
B.4.9 Further Share Lags . . . 89
B.4.10 Weaker Identification Assumption and Estimates . . . 90
B.4.11 Leave-One-Out Bank Shift . . . 92
B.4.12 Prior-Month Placebo . . . 94
B.4.13 Additional Results Incorporating the Reversal Period . . . 95
B.4.14 $WDIT^{QE}_{t,\tau}$ versus $\Delta Bill^{QE}_{t,\tau}$ . . . 95
B.5 Additional Results for Section 6 . . . 101
B.5.1 Heterogeneous Duration Targets . . . 101
B.6 Additional Empirical Results . . . 101
B.6.1 MMF Holdings of Treasury Securities . . . 101
B.6.2 Composition of the Short-Term Treasury Market . . . 103
B.6.3 Treasury Auctions and Reopenings . . . 104

C Data Preparation and Cleaning **107**

A A Theoretical Model

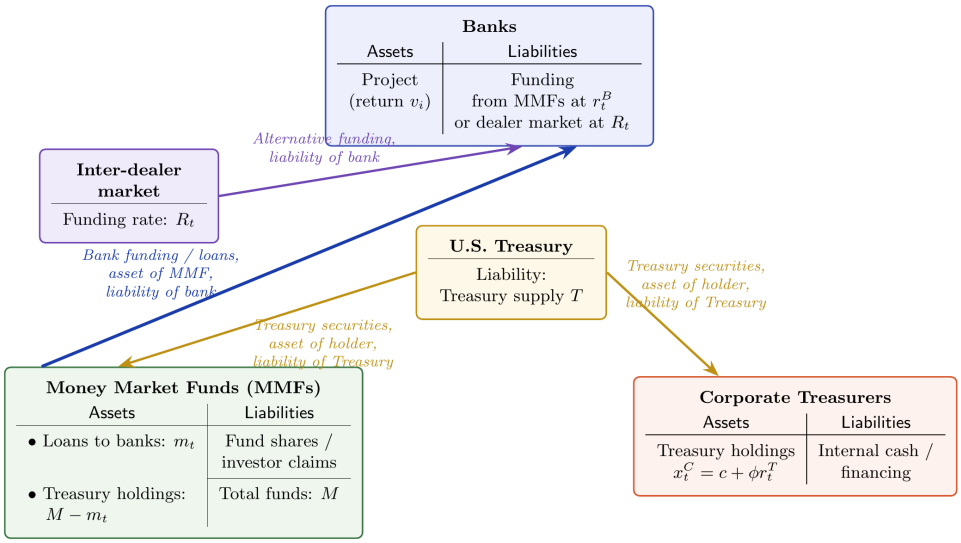
We present a simple model that jointly rationalizes our findings on prices in Section 3 and money fund portfolio allocations in Section 4, as well as price patterns in repo and unsecured funding markets.

Time is discrete with two periods, $t \in \{0, 1\}$. We interpret $t = 0$ as the inter-quarter period and $t = 1$ as the quarter-end. There are three types of agents: (i) banks with total mass B, (ii) money market funds with total mass 1, and (iii) corporate treasurers with total mass 1. For simplicity, we assume a unit discount factor across periods for all agents.

Banks are short-lived agents that seek funding to finance an indivisible project that requires a one-dollar investment. They are heterogeneous in their project returns: each bank $i \in I$ generates payoff v_i, where v_i is drawn from an exogenous distribution $F(\cdot)$. We assume $F(\cdot)$ is a continuous and strictly increasing cumulative distribution function; existence and uniqueness of equilibrium are established below. After observing v_i, a bank chooses whether to finance its project by borrowing in the inter-dealer market at rate R_t, or from money market funds at rate $r^B_{i,t}$. The rate $r^B_{i,t}$ is determined via bilateral bargaining.

Money market funds are homogeneous, short-lived agents endowed with M dollars at the beginning of each period. To preserve capital and generate returns for shareholders, they allocate funds between Treasury securities with yield r^T_t and loans to banks at the negotiated rate $r^B_{i,t}$. Corporate treasurers are preferred-habitat investors, modeled as in Vayanos and Vila (2021), with an exogenous demand for Treasury securities. The model setup is summarized in Figure A.1.

Figure A.1: Model Illustration



Bank–Fund Bargaining We first solve for the equilibrium rate between banks and money funds. The negotiated rate is determined via Nash bargaining:

$$\max_{r_{i,t}^B} \left[(v_i - r_{i,t}^B) - (v_i - R_t)\right]^{\alpha} \left[r_{i,t}^B - r_t^T\right]^{1-\alpha}. \tag{A.1}$$

The first term captures the bank's surplus from borrowing from the fund rather than the dealer market, while the second term captures the fund's surplus from lending to the bank instead of investing in Treasuries. Since the disagreement payoffs do not depend on i, the equilibrium rate is identical across banks:

$$r_t^B = (1-\alpha)R_t + \alpha r_t^T. \tag{A.2}$$

Bank Borrowing Not all banks choose to borrow. The participation constraint is $v_i - r_t^B > 0$, implying that in equilibrium the total borrowing is:

$$m_t = B\big(1 - F(r_t^B)\big). \tag{A.3}$$

Treasury Market Let total Treasury supply be T. Corporate treasurer demand is given by $x_t^c = c + \phi r_t^T$. Market clearing implies:

$$T = \underbrace{c + \phi r_t^T}_{\text{corporate demand}} + \underbrace{M - m_t}_{\text{money fund demand}} . \tag{A.4}$$

Lemma 1. *For each $t \in \{0, 1\}$, there exists a unique equilibrium $\{r_t^B, r_t^T, m_t\}$.*

Combining (A.2)–(A.4), we obtain:

$$m_t = B\left(1 - F\left((1-\alpha)R_t + \alpha\left(\frac{T - c - M}{\phi} + \frac{m_t}{\phi}\right)\right)\right). \tag{A.5}$$

Because the right-hand side is continuous and nonincreasing in m_t while the left-hand side is strictly increasing in m_t, a unique fixed point m_t exists.

Quarter-End Shock We now introduce a quarter-end funding shock by assuming $R_1 > R_0$, reflecting tighter conditions in the inter-dealer market due to window dressing.[16] This yields:

Proposition 1. $r_1^B > r_0^B, \quad m_1 < m_0, \quad$ *and* $\quad r_1^T < r_0^T$.

The intuition is as follows. At quarter-end, balance-sheet constraints reduce inter-dealer lending, raising R_t. This worsens the banks' outside borrowing option and raises the negotiated borrowing rate r_t^B. Higher funding costs reduce bank borrowing ($m_1 < m_0$), leaving money funds with excess liquidity. This liquidity is reallocated into Treasuries, increasing demand and lowering equilibrium yields ($r_1^T < r_0^T$).

Arbitrageurs We now introduce arbitrageurs, who operate across both periods and exploit predictable price movements. Anticipating higher Treasury prices (lower yields) at quarter-end, arbitrageurs take offsetting positions against money funds.

[16] $R_1 > R_0$ can be interpreted as a higher borrowing rate in interdealer markets, either collateralized or uncollateralized. In practice, collateralized interdealer borrowing can occur in several markets, including (i) the GCF repo market, (ii) the FICC DVP repo market, and (iii) private bilateral repo markets. Due to data limitations, Figure B.3 plots the GCF repo market rate.

Let s denote the arbitrageur's Treasury position and W its initial wealth. Due to frictions[17], arbitrageurs incur a convex cost $c(s; R_1)$ when building positions. The cost depends on the funding rate R_1, which reflects funding-market conditions at quarter-end (Du et al., 2018b). Their problem is:

$$\max_s (p_1 - p_0)s + [W - sp_0]R_0 - c(s; R_1), \tag{A.6}$$

where, consistent with the net rates used above, Treasury prices and yields satisfy:

$$p_t = \frac{1}{1 + r_t^T}.$$

To solve the model analytically, we assume the functional form $c(s; R_1) = \frac{1}{2\eta(R_1)}s^2$, where $\eta(R_1) > 0$ and $\eta'(R_1) < 0$. The first-order condition then yields:

$$s = \eta(R_1)\left(\frac{1}{1 + r_1^T} - \frac{1 + R_0}{1 + r_0^T}\right). \tag{A.7}$$

Proposition 2. *A lower R_1 reduces the equilibrium yield spread $r_0^T - r_1^T$.*

Intuitively, a lower R_1 means a larger η, which implies more elastic arbitrage demand. Arbitrageurs take larger positions for a given intertemporal price difference, increasing demand in period 0 and supply in period 1. This compresses the cross-period yield differential in equilibrium. Note that R_1 enters twice: it sets the size of the quarter-end funding shock and, through $\eta(R_1)$, the elasticity of arbitrage. The proposition combines both effects; the arbitrage channel on its own is the comparative static in η holding the shock fixed.

Interpretation The model captures quarter-end window dressing in reduced form through an exogenous increase in the inter-dealer rate ($R_1 > R_0$).[18] Banks reduce borrowing through the price margin: a higher R raises the negotiated rate r_t^B, pricing marginal banks out of the market. This abstracts from a direct balance-sheet quantity channel, whereby banks curtail quarter-end borrowing to improve reported leverage ratios even at unchanged rates.

[17] We discuss the interpretation of these frictions in Appendix A.1.

[18] In practice, foreign banks are important providers of short-term dollar funding to U.S. banks. At quarter-end, they reduce the supply of dollar funding, putting upward pressure on R.

In practice, both channels may operate. Introducing an explicit quarter-end balance-sheet cost for banks would add this quantity channel and reinforce the model's predictions that $m_1 < m_0$ and $r_1^T < r_0^T$. We discuss our modeling choices further in Appendix A.1.

A.1 Discussion of Modeling Choices

First, we do not distinguish between the different types of bank debt traded between banks and money market funds. In practice, both unsecured instruments—such as commercial paper, certificates of deposit, and time deposits—and secured instruments, such as repos, are widely used. We provide a detailed breakdown in the empirical section. Since the equilibrium implications are qualitatively identical across these instruments, we abstract from this distinction for simplicity.[19]

Second, we do not explicitly model the single-issuer exposure constraints for bank debt under SEC Rule 2a-7. A simple extension would introduce a penalty term into equation (A.2), such that money funds effectively receive $r_{i,t}^B - r^c$, where r^c represents the shadow cost of the exposure constraint. However, since all propositions and conclusions remain unchanged relative to the benchmark, we omit this feature for better tractability.

Third, in reality, money market funds allocate a substantial share of their portfolios to the Federal Reserve's reverse repo facility (RRP), which offers an exogenously set rate. The model can be extended by modifying the fund's outside option in (A.2) to $\max(r_t^T, r_t^{RRP})$, where r_t^{RRP} tracks the lower bound of the policy rate. However, this modification alone would imply corner solutions—full allocation to either Treasuries or RRP—which is inconsistent with the data. Empirically, allocations vary with the monetary policy regime (see Figure B.20). The intuition is straightforward: the RRP is an overnight instrument with effectively zero duration, whereas Treasuries have longer duration. In a rising rate environment, funds prefer RRP to benefit from higher reinvestment rates; in a declining rate environment, longer-duration Treasuries become more attractive. Extending the model to include RRP would require specifying the extent of substitution between overnight investments and Treasuries. The Treasury-price spillover would persist under conditions that prevent RRP from fully absorbing displaced funding.

[19] Empirical equilibrium rates can be found in Figure B.2.

Fourth, the model does not distinguish between U.S. and non-U.S. banks. While the model matches aggregate empirical patterns—namely $R_1 > R_0$, $r_1^B > r_0^B$, and $m_1 < m_0$—the data show that most of the reduction in bank borrowing at quarter-end is driven by non-U.S. banks. A detailed decomposition in the data is provided in Section 4.

Fifth, the friction parameter η can be interpreted from two perspectives. (i) Even within Treasury markets, arbitrageurs face multiple trading strategies—such as covered interest parity (CIP), TIPS–Treasury arbitrage, Treasury spot–futures basis trades, and Treasury–swap spreads (see Siriwardane et al., 2025). Substitution across these opportunities is imperfect, limiting arbitrage capacity in any single strategy. (ii) Quarter-end arbitrage resembles a timing strategy rather than a pure long–short trade, and thus requires balance sheet usage, as emphasized in recent work (Chaudhary et al., 2026). Therefore, η can also be interpreted as capturing the cost of balance sheet capacity. Both perspectives are correlated with conditions in the inter-dealer funding market, hence η is a function of R_1.

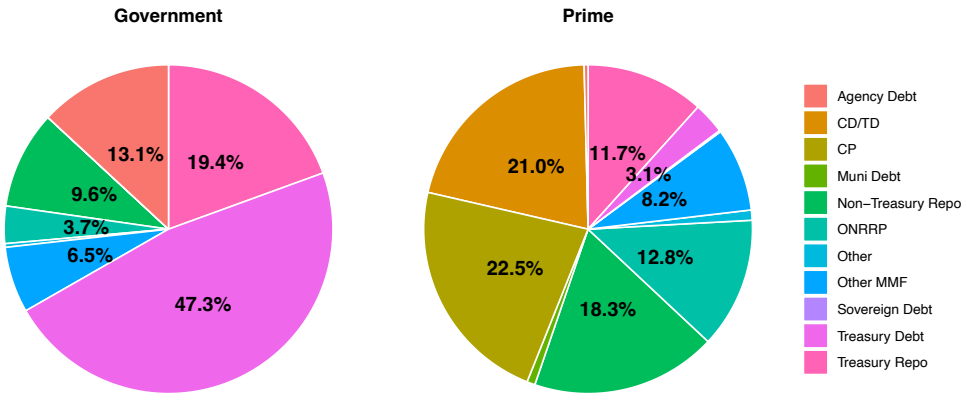
Finally, we abstract from the precise market structure governing bargaining between money funds and banks. The literature has not reached consensus on which side holds market power. For example, Huber (2023) argues that primary dealers price-discriminate across funds, suggesting borrower-side market power, while Doerr and Eren (2025) find evidence consistent with lender-side power. Our model is agnostic about pricing power in normal periods, but it departs from perfect competition at quarter-end. Under a Walrasian benchmark, window dressing would reduce borrowing demand, implying a lower equilibrium repo rate. In contrast, repo rates increase at quarter-end (see Figure B.2), suggesting that a contraction in banks' borrowing demand alone cannot explain the observed rate movements.

B Empirical Appendix

B.1 Empirical Support for the Model Assumptions

Consistent with the model's portfolio choice between lending to banks and holding Treasuries, Figure B.1 shows the composition of holdings by fund type. Government funds hold predominantly Treasury debt (about 47%) and Treasury repo (19%), alongside agency debt (13%) and non-Treasury repo (10%). Prime funds tilt toward bank claims—CD/TD (21%), commercial paper (23%), and Treasury repo (12%)—with only about 3% in Treasury debt.

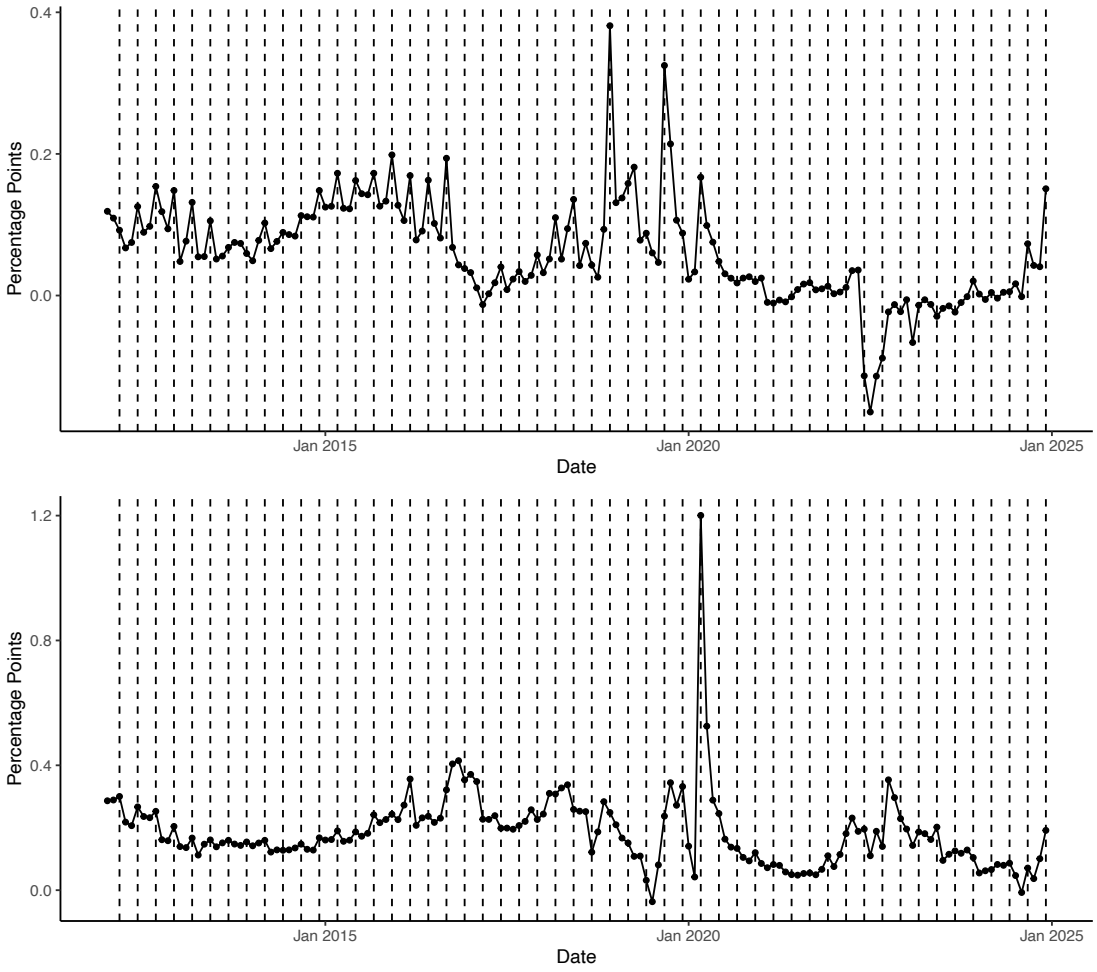
Figure B.1: MMF Holdings by Fund Type



Notes: This figure shows the composition of MMF holdings by fund type as of December 2024. Agency debt refers to debt issued by U.S. agency institutions and government-sponsored enterprises, such as Freddie Mac and Fannie Mae. CD/TD denotes certificates of deposit and time deposits. CP denotes commercial paper. Municipal debt refers to bonds and notes issued by municipalities or related public-sector entities. Sovereign debt refers to U.S. dollar-denominated government debt issued by non-U.S. governments. Treasury repo refers to repurchase agreements collateralized by Treasury securities. ONRRP denotes the Federal Reserve's overnight reverse repurchase agreement facility.

As the empirical counterpart to the quarter-end funding shock ($R_1 > R_0$, and hence $r_1^B > r_0^B$), Figure B.2 shows both repo and unsecured bank borrowing rates rising at quarter-ends.

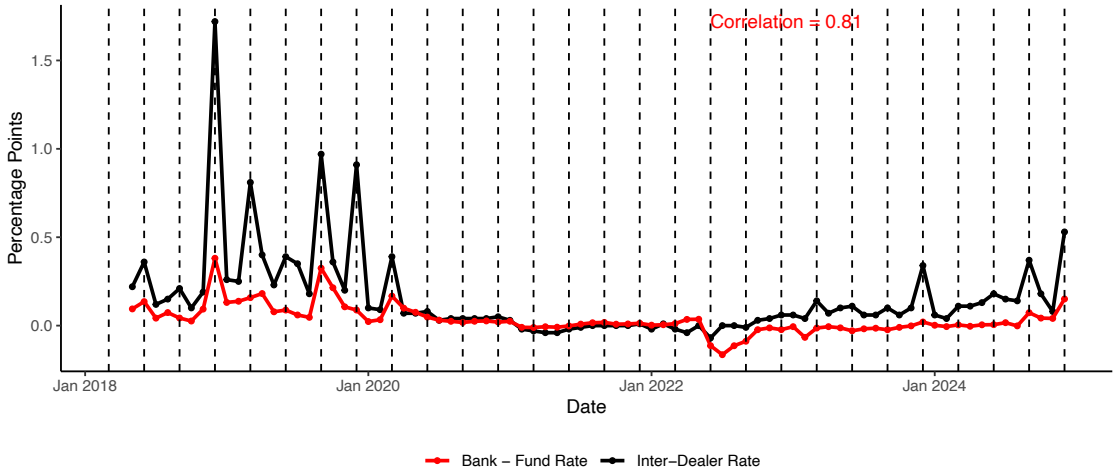
Figure B.2: Bank Debt Borrowing Rates at Quarter-Ends



Notes: This figure plots the market-value-weighted repo rate spreads, defined as in equation (1), in the top panel and unsecured instrument rates in the bottom panel. The dashed vertical lines indicate quarter-end months.

Figure B.3 plots the inter-dealer rate R_t, proxied by the GCF repo rate, which likewise increases at quarter-end.

Figure B.3: Repo Rates across Markets



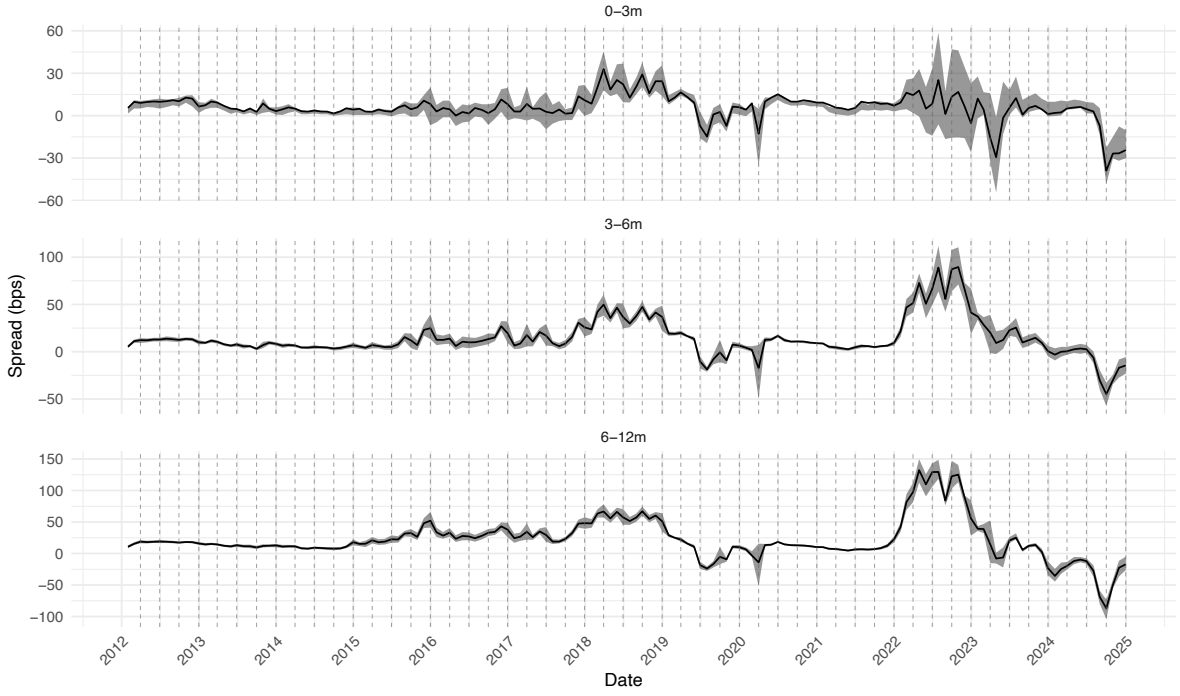
Notes: This figure plots repo rate spread across markets. The red line shows the market-value-weighted repo rate spread reported in Figure B.2, while the other line shows the volume-weighted GCF repo rate in the FICC GCF Repo market, which represents the repo rate in the interdealer market. The dashed vertical lines indicate quarter-end months.

B.2 Robustness and Heterogeneity of the Results in Section 3

B.2.1 Time Series of the Cross-Sectional Spread Distribution

Figure B.4 plots the time series of the cross-sectional security-spread distribution, partitioned into the three remaining-maturity buckets. The solid line is the daily value-weighted average and the shaded region the interquartile range. Two features are apparent. First, baseline spreads increase monotonically in remaining maturity: 0–3 month spreads hover near zero, 3–6 month spreads sit modestly higher, and 6–12 month spreads are highest, reflecting the liquidity and term premia of bills, as shorter-maturity issues are more liquid and carry a lower term premium. Second, cross-sectional dispersion widens noticeably after the 2016 MMF reform takes effect and again after the tightening cycle begins in 2022, with sizable positive and negative excursions at all three tenors.

Figure B.4: Daily Cross-Sectional Average Security Spread by Remaining-Maturity Bucket



Notes: The figure plots the spread, defined in equation (1), based on daily observations. The sample spans January 2012 to December 2024.

B.2.2 Regime Shifts: Basel III Phase-In and the 2016 MMF Reform

Table B.1 reports the estimates under different regimes. In the pre-2015 sub-period (Panel A) the quarter-end coefficient is essentially zero at the 3–6 and 6–12 month buckets, with only a small but statistically significant 1.2-basis-point effect at 0–3 months. One caveat is that this period also coincides with the zero-lower-bound regime (see Appendix B.2.10). Once Basel III is in force but before the MMF reform (Panel B), the coefficient becomes markedly more negative and traces the familiar maturity gradient: 5.5 basis points at 0–3 months, 5.1 at 3–6 months, and 3.5 at 6–12 months, with the longest tenor imprecisely estimated. After the October 2016 MMF reform (Panel C), the short-end effect is larger still.

Table B.1: Quarter-End Effect on T-Bill Spreads, by Sub-Period

Dependent Variable:	Spread (pp)		
Remaining maturity:	0–3 m	3–6 m	6–12 m
Panel A. Pre-Basel III (2012–2014)			
QEnd_t	-0.0117^{**}	-0.0053	0.0053
	(0.0041)	(0.0046)	(0.0070)
Observations	9,731	9,603	5,752
R^2	0.417	0.794	0.761
Within R^2	0.019	0.304	0.340
Panel B. Basel III, pre-reform (2015–Oct 2016)			
QEnd_t	-0.0548^{*}	-0.0507^{**}	-0.0350
	(0.0277)	(0.0200)	(0.0242)
Observations	5,799	5,749	3,442
R^2	0.274	0.443	0.521
Within R^2	0.072	0.310	0.315
Panel C. Post-MMF reform (Oct 2016–2024)			
QEnd_t	-0.0659^{***}	-0.0458^{*}	-0.0519
	(0.0239)	(0.0234)	(0.0338)
Observations	44,937	33,390	15,705
R^2	0.256	0.746	0.830
Within R^2	0.084	0.028	0.010

Notes: Estimates of equation (2) on Treasury bills only, split into three sub-periods. All specifications include Issuance × Year-Quarter fixed effects, security controls, and market-level controls. All else as in Table 1.

The portfolio side mirrors this regime shift: Tables B.11 and B.12 in Appendix B.3.4 show that MMF withdrawal magnitudes intensify sharply after 2015, concentrated in unsecured bank debt, while repo magnitudes are essentially unchanged.

B.2.3 Month-End Comparison-Group Robustness

A second concern is that the comparison group—all non-quarter-end trading days—may be too broad: short-term Treasury yields vary over the quarter with market liquidity, fiscal news, and the auction cycle, and fixed effects alone may not absorb this variation cleanly enough to provide a counterfactual for the quarter-end day. We therefore re-estimate the bills-only specification of Table 1 over the full 2012–2024 window with the comparison group restricted to month-end trading days, so that quarter-ends are benchmarked against days on which dealers and investors face similar end-of-month balance-sheet pressures (Table B.2).

Table B.2 reports this month-end specification. The point estimates remain negative and

significant at the shorter maturities and are slightly larger than in the baseline—6.9 basis points at 0–3 months in the bills-only panel and 10.0 basis points in the pooled panel. The front-end yield drop is therefore not an artifact of the broad non-quarter-end comparison group.

Table B.2: Quarter-End Effect on T-Bills: Month-End Comparison Group

Dependent Variable:	Spread (pp)					
Sample:	Panel A. T-bills only			Panel B. Notes + Bonds + Bills		
Remaining maturity:	0–3 m	3–6 m	6–12 m	0–3 m	3–6 m	6–12 m
QEnd_t	-0.0689^{***}	-0.0446^{**}	-0.0298	-0.1003^{***}	-0.0501^{***}	-0.0305
	(0.0230)	(0.0188)	(0.0245)	(0.0198)	(0.0138)	(0.0198)
Issuance × Year-Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Security controls	Yes	Yes	Yes	Yes	Yes	Yes
Market-level controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,910	2,310	1,223	4,563	4,165	5,095
R^2	0.310	0.784	0.838	0.274	0.771	0.859

Notes: Estimates of equation (2) restricting the comparison group to month-end trading days, over the full daily sample from January 2012 through December 2024. Panel A restricts the sample to Treasury bills only; Panel B uses the broader Notes + Bonds + Bills universe. All else as in Table 1.

B.2.4 Heterogeneity by Calendar Quarter

We next re-estimate the bills-only baseline separately by calendar quarter. Table B.3 shows that the effect is largest in the fourth quarter (the calendar year-end), consistent with intensified balance-sheet window dressing at year-end; only the third- and fourth-quarter estimates are statistically significant.

Table B.3: Heterogeneity by Calendar Quarter, T-bills only

Dependent Variable:	Spread (pp)				
Sample:	All Quarters	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
QEnd_t	-0.0449^{***}	-0.0255	-0.0342	-0.0482^{*}	-0.0677^{***}
	(0.0108)	(0.0158)	(0.0221)	(0.0252)	(0.0210)
Issuance × Year-Quarter × Bucket FE	Yes	Yes	Yes	Yes	Yes
Security controls	Yes	Yes	Yes	Yes	Yes
Market-level controls	Yes	Yes	Yes	Yes	Yes
Observations	134,108	32,400	33,783	34,320	33,605
R^2	0.601	0.560	0.637	0.656	0.545

Notes: Estimates of equation (2) on the Treasury-bills-only sample, split by calendar quarter. "All Quarters" pools the three maturity buckets with bucket-level fixed effects. All else as in Table 1.

B.2.5 FOMC Meeting Robustness

To rule out an influence of FOMC meetings, we re-estimate the baseline specifications after dropping all FOMC meeting days from the panel. Within the 2012–2024 window, only one FOMC meeting coincides with a quarter-end (March 31, 2020). Table B.4 reports the estimates, which are essentially unchanged relative to Table 1 across all three buckets and both samples, suggesting that the results are not driven by the timing of monetary policy.

Table B.4: Quarter-End Effect Excluding FOMC Days

Dependent Variable:	Spread (pp)					
Sample:	Panel A. T-bills only			Panel B. Notes + Bonds + Bills		
Remaining maturity:	0–3 m	3–6 m	6–12 m	0–3 m	3–6 m	6–12 m
QEnd_t	-0.0558^{***}	-0.0357^{**}	-0.0343	-0.0909^{***}	-0.0409^{***}	-0.0299^{*}
	(0.0175)	(0.0159)	(0.0216)	(0.0182)	(0.0115)	(0.0171)
Issuance × Year-Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Security controls	Yes	Yes	Yes	Yes	Yes	Yes
Market-level controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	58,488	47,155	24,130	96,296	85,324	102,855
R^2	0.244	0.753	0.828	0.187	0.747	0.846
Within R^2	0.066	0.039	0.022	0.017	0.034	0.028

Notes: Estimates of equation (2) after dropping all observations on FOMC meeting dates. Within 2012–2024, only one FOMC meeting (March 31, 2020) lands on a quarter-end day, so this restriction effectively removes the 2020Q1 quarter-end day from the panel. All else as in Table 1.

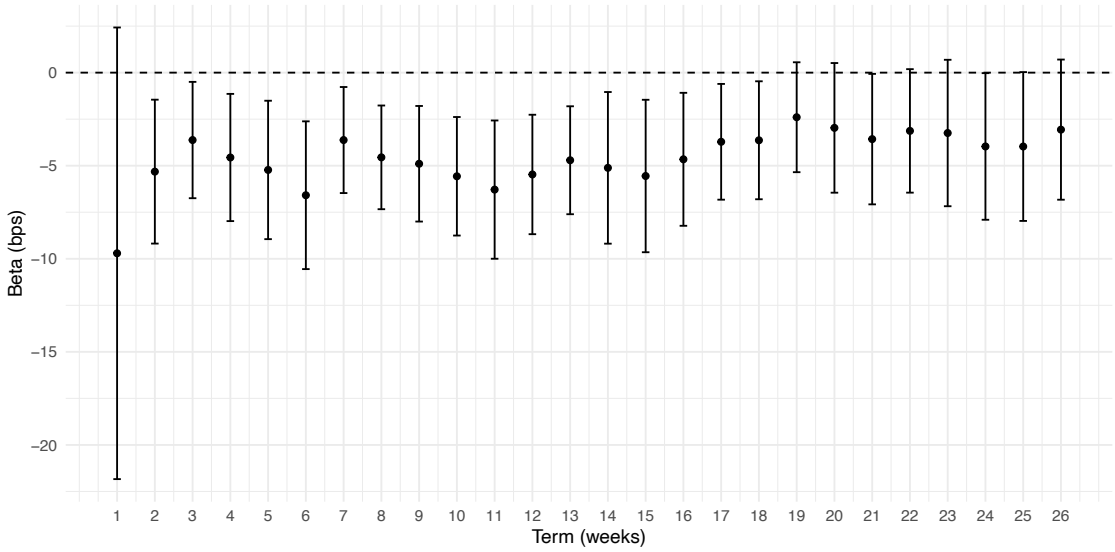
B.2.6 Leave-One-Quarter-Out Analysis

We also conduct a leave-one-quarter-out analysis, re-estimating the bills-only baseline specification while recursively excluding one calendar year-quarter at a time. The results remain stable across all iterations. For the 0–3 month bucket, the most influential quarter is 2024Q3, whose exclusion reduces the coefficient's absolute magnitude by 0.94 basis points relative to the full-sample estimate of −5.65 bps; for the 3–6 month bucket the most influential quarter is 2022Q4 (0.93-bps reduction in absolute magnitude relative to a −3.59-bps full-sample estimate); for the 6–12 month bucket the same quarter is the most influential, with a 1.04-bps reduction (relative to a −3.44-bps estimate). None of these exclusions affect the statistical significance of the baseline estimates, indicating that the results reflect a persistent pattern rather than a small number of influential quarters.

B.2.7 Index-Level Regression

As an additional robustness check, we examine the quarter-end pattern using monthly index-level Treasury bill yields across maturities from 1 to 26 weeks. For each maturity, we regress the monthly change in the Treasury yield spread on a quarter-end indicator. Figure B.5 plots the estimated quarter-end coefficients by maturity. The point estimates are negative throughout the maturity range, although their precision varies, providing further evidence of the quarter-end yield decline documented in the security-level analysis.

Figure B.5: Quarter-End Effects on Treasury Bill Spreads across Maturities



Notes: This figure reports estimated quarter-end coefficients from separate regressions of monthly changes in index-level Treasury bill yield spreads on a quarter-end indicator, for maturities ranging from 1 to 26 weeks. Spreads are measured relative to the lower bound of the federal funds target range. Points denote coefficient estimates, expressed in basis points, and error bars denote 95% confidence intervals.

B.2.8 Decomposition by Security Type and Original Tenor

Table B.5 restricts the sample to rolled-down notes and bonds across the three remaining-maturity buckets. The 0–3 month coefficient grows in magnitude to −15.4 basis points. The market for rolled-down notes and bonds is less liquid, producing a larger quarter-end yield drop, which helps account for the larger pooled estimate in Panel B of Table 1.

Table B.5: Quarter-End Effect by Security Type

Dependent Variable:	Spread (pp)		
Sample:	Notes + Bonds		
Remaining maturity:	0–3 m	3–6 m	6–12 m
QEnd_t	-0.1539^{***}	-0.0480^{***}	-0.0288
	(0.0366)	(0.0171)	(0.0217)
Issuance × Year-Quarter FE	Yes	Yes	Yes
Security controls	Yes	Yes	Yes
Market-level controls	Yes	Yes	Yes
Observations	39,059	39,434	81,362
R^2	0.162	0.740	0.849

Notes: Estimates of equation (2). The sample retains only rolled-down Treasury notes and bonds (no bills). All else as in Table 1.

Table B.6 pushes this further, splitting the rolled-down notes and bonds with 0–3 month remaining maturity by the security's original auction tenor: the quarter-end drop is uniformly significant and is generally larger in absolute magnitude for longer original tenors, from −15.8 basis points for 2–3 year notes to −22.2 basis points for 20–30 year bonds. Both patterns are consistent with an MMF-demand channel whose price impact is smaller for liquid short-term securities and larger for longer-tenor bonds that trade in thinner markets.

Table B.6: Quarter-End Effect by Original Tenor of Notes/Bonds

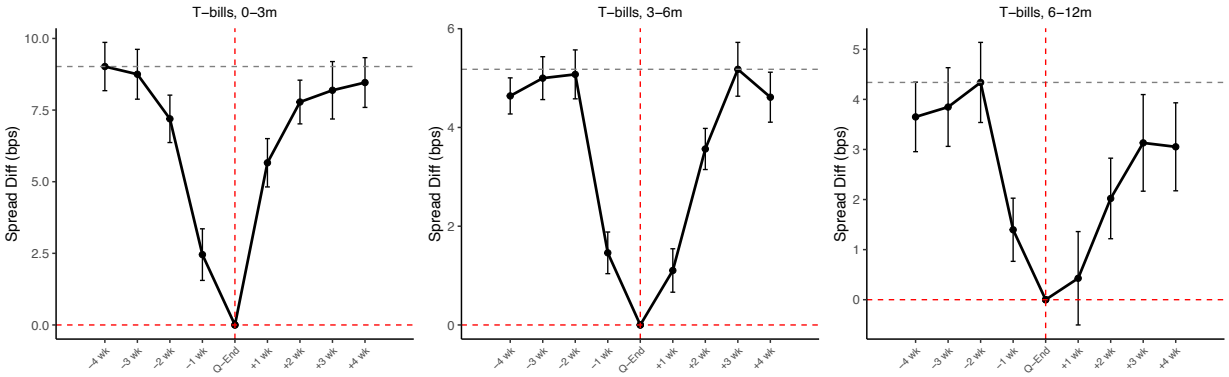
Dependent Variable:	Spread (pp)			
Sample:	Notes + Bonds with 0–3 m residual maturity			
Original tenor:	2–3 y note	5 y note	7–10 y note	20–30 y bond
QEnd_t	-0.1577^{***}	-0.1558^{***}	-0.1691^{***}	-0.2215^{***}
	(0.0403)	(0.0331)	(0.0511)	(0.0813)
Issuance × Year-Quarter FE	Yes	Yes	Yes	Yes
Security controls	Yes	Yes	Yes	Yes
Market-level controls	Yes	Yes	Yes	Yes
Observations	18,568	9,209	9,775	1,507
R^2	0.159	0.169	0.172	0.256

Notes: Estimates of equation (2) on rolled-down Treasury notes and bonds with 0–3 months remaining maturity, split by the security's original auction tenor. All else as in Table 1.

B.2.9 Event-Window Plots by Maturity Bucket and Placebo Tests

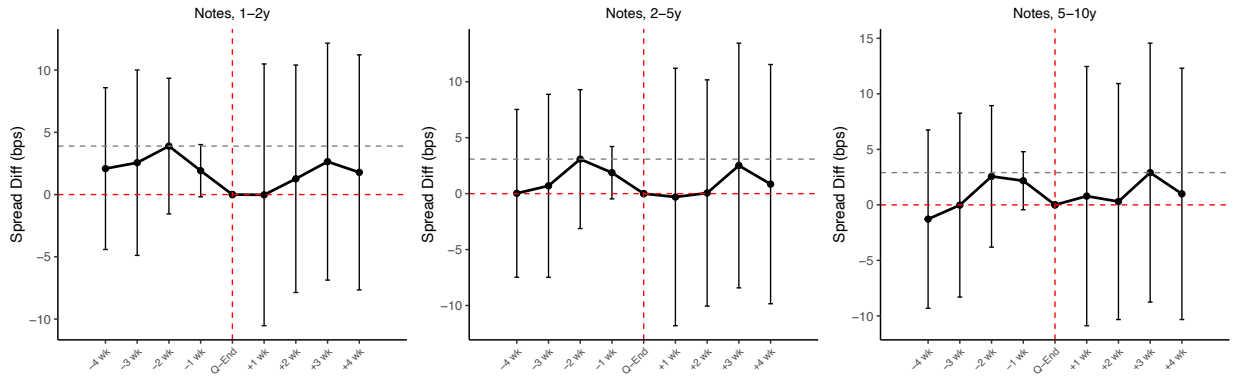
Figure B.6 reports the event-window coefficients for all three remaining-maturity buckets of Treasury bills. The V-shape is present in each bucket but flattens and loses significance as remaining maturity rises. Figures B.7 and B.8 report placebo event studies for Treasury notes and bonds with more than one year remaining maturity. Neither placebo shows a comparable quarter-end V-shape, reinforcing that the effect is concentrated in the short-maturity Treasury securities held by MMFs.

Figure B.6: Event-Study around Quarter-Ends for T-bills



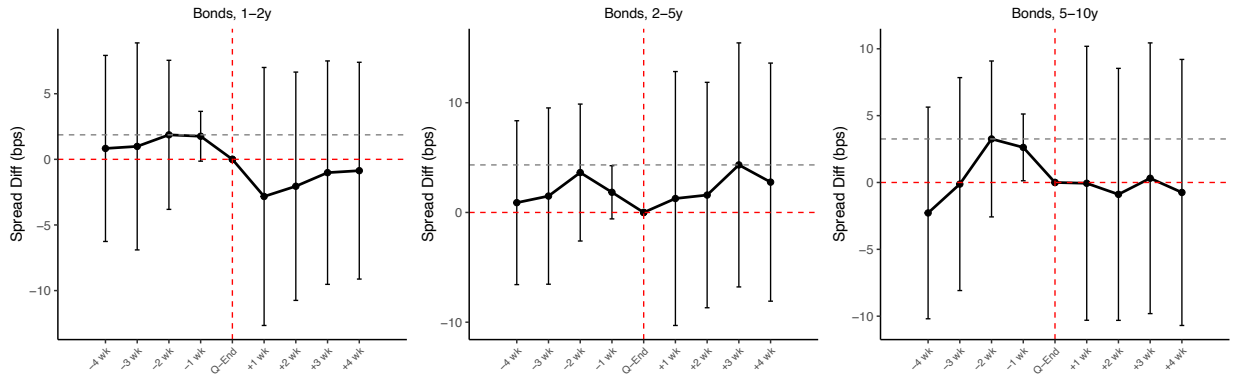
Notes: Event-window coefficients $-\hat{\beta}_k$ from equation (3) for $k \in \{-4, \ldots, +4\}$ weeks, estimated on Treasury bills and shown by remaining-maturity bucket (0–3, 3–6, and 6–12 months). The quarter-end week is normalized to zero; error bars are 95% confidence intervals. The sample spans January 2012 through December 2024.

Figure B.7: Placebo Event-Study: Treasury Notes only, >1y



Notes: Same event-time regression as Figure 1, estimated on Treasury notes with more than one year remaining maturity (bucketed 1–2y / 2–5y / 5–10y). The quarter-end week is normalized to zero; error bars are 95% confidence intervals.

Figure B.8: Placebo Event-Study: Treasury Bonds only, >1y



Notes: Same event-time regression as Figure 1, estimated on Treasury bonds with more than one year remaining maturity (bucketed 1–2y / 2–5y / 5–10y, the same buckets as for notes; bonds in these buckets are seasoned issues approaching maturity). The quarter-end week is normalized to zero; error bars are 95% confidence intervals.

B.2.10 Zero-Lower-Bound Regime Heterogeneity

The pooled estimates in Table 1 average over very different monetary-policy environments, and the measured quarter-end effect need not be constant across them. When the federal funds target range sits at its zero lower bound (ZLB), the policy floor compresses

the front-end spread on which MMF demand operates: bill yields are pinned close to zero and have limited room to fall further, so a given amount of quarter-end demand pressure should translate into a smaller observed yield response than in periods when the short rate is well above zero. If so, the reduced-form estimate that pools ZLB and non-ZLB quarters would understate the off-ZLB price impact. We test this by re-estimating the bills-only specification of equation (2) separately by policy-rate regime.

We classify a trading day as ZLB if the lower bound of the federal funds target range is exactly zero. This selects two episodes within the 2012–2024 sample: the post-crisis ZLB, from the start of the sample in January 2012 through 15 December 2015, and the pandemic ZLB, from 16 March 2020 through 16 March 2022. We keep the two episodes separate, because they differ along dimensions other than the policy rate—among them the scale of the Federal Reserve's balance sheet and reverse-repo facility, the level of bill supply, and the post-2015 strengthening of the window-dressing incentive documented in Appendix B.2.2—and label all remaining days non-ZLB. T-bill trading days split 990, 504, and 1,759 across the GFC-ZLB, COVID-ZLB, and non-ZLB regimes; Figure B.9 plots the policy floor with the two ZLB windows shaded.

Table B.7 re-estimates the quarter-end coefficient within each regime. The front-end yield drop is a non-ZLB phenomenon: in non-ZLB quarters the coefficient is 8.3, 6.5, and 7.6 basis points (in absolute value) across the three maturity buckets and significant throughout, close to the pooled baseline in Table 1. During the GFC ZLB the effect is largely absent, surviving only at the 0–3 month bucket and at a much smaller 1.3 basis points; during the COVID ZLB it is statistically indistinguishable from zero at every maturity.

Table B.8 confirms this pattern in a single pooled regression that interacts the quarter-end indicator with the three regime dummies, which allows the regime differences to be tested directly. Both ZLB coefficients are significantly smaller in absolute value than the non-ZLB coefficient: the GFC-minus-non-ZLB difference is 7.3, 5.7, and 7.1 basis points and the COVID-minus-non-ZLB difference 8.6, 7.6, and 10.1 basis points, significant in every bucket. The two ZLB coefficients are not significantly different from one another, so we read the attenuation as a common feature of the zero-rate environment rather than as specific to either episode.

Two qualifications apply to the COVID results. The pandemic ZLB contains only eight quarter-ends, so its coefficients are estimated imprecisely, and the failure to detect an effect may reflect limited statistical power rather than the absence of an effect. Relatedly, the small residual effect at the 0–3 month bucket during the GFC ZLB, absent during the COVID ZLB, should not be read as evidence that the attenuation was more complete under the pandemic, since the two ZLB coefficients cannot be distinguished. With these caveats, the regime split is consistent with the truncation mechanism above: where the spread can move freely the quarter-end yield drop is several basis points, and where the policy floor binds the measured response collapses. It also implies that the pooled estimates in Table 1, which include the two ZLB episodes, are if anything conservative measures of the off-ZLB *reduced-form* quarter-end effect.

Table B.7: Quarter-End Effect on T-Bill Spreads by Policy-Rate Regime

Dependent Variable:	Spread (pp)								
Regime:	ZLB–GFC			ZLB–COVID			Non-ZLB		
Remaining maturity:	0–3 m	3–6 m	6–12 m	0–3 m	3–6 m	6–12 m	0–3 m	3–6 m	6–12 m
QEnd_t	-0.0132^{***}	-0.0099	0.0023	0.0012	-0.0023	-0.0027	-0.0833^{***}	-0.0645^{**}	-0.0760^{**}
	(0.0040)	(0.0063)	(0.0078)	(0.0094)	(0.0138)	(0.0245)	(0.0284)	(0.0275)	(0.0350)
Issuance × Year-Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Security controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Market-level controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	12,854	12,675	7,594	12,960	9,936	3,852	34,653	26,131	13,453
R^2	0.404	0.543	0.670	0.467	0.729	0.809	0.269	0.746	0.832
Within R^2	0.019	0.164	0.194	0.121	0.034	0.066	0.101	0.041	0.016

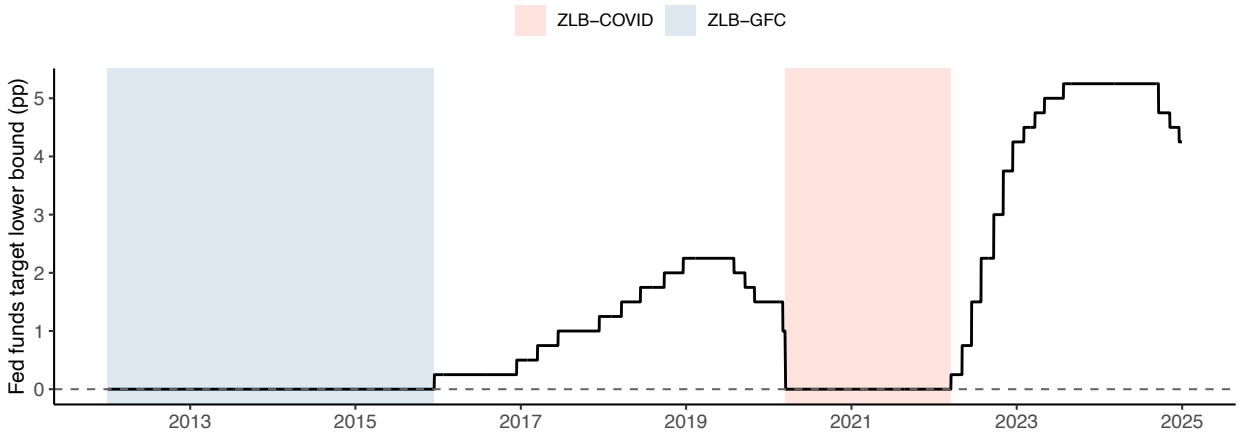
Notes: OLS estimates of equation (2) on Treasury bills only, estimated separately within each policy-rate regime. A trading day is ZLB if the lower bound of the federal funds target range equals zero (ZLB–GFC: January 2012–15 December 2015; ZLB–COVID: 16 March 2020–16 March 2022); all other days are non-ZLB. Security-level controls are duration and the bid–ask spread; market-level controls are the volume of newly auctioned Treasury bills and the face amount of maturing Treasury debt. Standard errors, clustered at the year-quarter level, are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table B.8: Quarter-End Effect by Policy-Rate Regime: Pooled Interaction

Dependent Variable:	Spread (pp)		
Remaining maturity:	0–3 m	3–6 m	6–12 m
$\text{QEnd}_t \times$ Non-ZLB	-0.0887^{***}	-0.0649^{**}	-0.0691^{*}
	(0.0273)	(0.0267)	(0.0358)
$\text{QEnd}_t \times$ ZLB–GFC	-0.0157^{***}	-0.0080	0.0021
	(0.0046)	(0.0056)	(0.0074)
$\text{QEnd}_t \times$ ZLB–COVID	-0.0028	0.0115	0.0316
	(0.0137)	(0.0154)	(0.0298)
Difference tests ($\delta = \beta_{\text{ZLB}} - \beta_{\text{Non-ZLB}}$):			
ZLB–GFC − Non-ZLB	0.0730^{***}	0.0569^{**}	0.0712^{*}
	(0.0268)	(0.0273)	(0.0368)
ZLB–COVID − Non-ZLB	0.0859^{***}	0.0764^{**}	0.1007^{**}
	(0.0286)	(0.0308)	(0.0474)
Issuance × Year-Quarter FE	Yes	Yes	Yes
Security controls	Yes	Yes	Yes
Market-level controls	Yes	Yes	Yes
Observations	60,467	48,742	24,899
R^2	0.245	0.752	0.827
Within R^2	0.066	0.038	0.023

Notes: OLS estimates of equation (2) on Treasury bills, pooling all regimes and interacting QEnd_t with indicators for the non-ZLB, ZLB–GFC, and ZLB–COVID regimes; the regime level effects are absorbed by the year-quarter fixed effects. The difference-test rows report $\delta = \beta_{\text{ZLB}} - \beta_{\text{Non-ZLB}}$, with standard errors computed from the estimated coefficient covariance matrix; a positive δ indicates a weaker (less negative) quarter-end yield drop at the ZLB. Controls, fixed effects, clustering, and significance stars are as in Table B.7.

Figure B.9: Federal Funds Target Lower Bound and ZLB Regimes



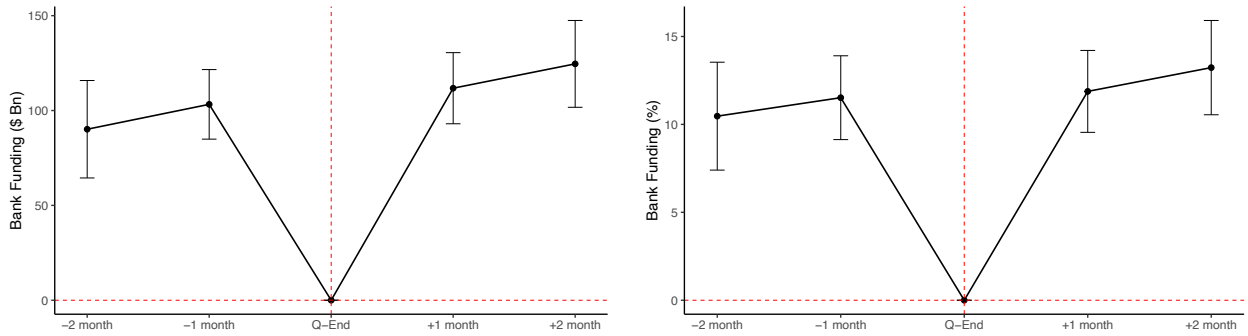
Notes: Daily lower bound of the federal funds target range, January 2012 through December 2024. Shaded regions mark the two zero-lower-bound episodes used in Tables B.7 and B.8: the post-crisis ZLB (through 15 December 2015) and the pandemic ZLB (16 March 2020–16 March 2022).

B.3 Robustness and Heterogeneity of the Results in Section 4

B.3.1 Aggregate Bank-Funding Event Study

Figure B.10 presents the event-study counterpart to the aggregate sawtooth series, averaging the change in MMF bank-funding holdings in the months around each quarter-end; the contraction is concentrated at the quarter-end ($t = 0$) and reverses the following month.

Figure B.10: Event Study of Aggregate Bank Funding



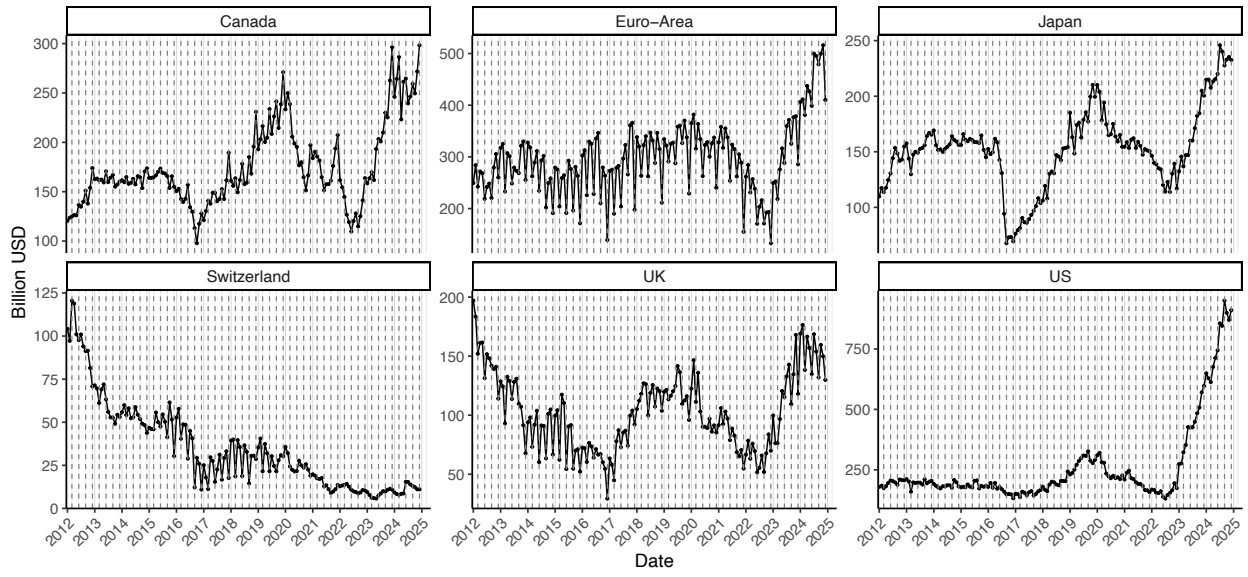
Δ Quantity ($ Bn) Δ Quantity (%)

Notes: The figure plots aggregate MMF holdings of bank funding (repo plus unsecured) in event time, over a window of ±2 months around the quarter-end ($t = 0$, normalized to zero). The left panel reports the change in holdings in USD billions and the right panel the percentage change. The sample spans 2012–2024, and the error bars denote 95% confidence intervals.

B.3.2 Decomposition by Issuer Region and Instrument

This appendix collects the decompositions of the aggregate quarter-end contraction summarized in Section 4.1. Figure B.11 decomposes MMF holdings of bank debt by the headquarters region of the issuing bank. The quarter-end pattern is concentrated in non-U.S. banks—especially those headquartered in the euro area, the UK, and Switzerland—which compute their leverage ratio from a quarter-end snapshot, whereas U.S. banks use a daily quarterly average and show no recurring quarter-end contraction. Canadian banks follow fiscal quarter-ends one month after the calendar quarter-ends, so their dip does not align with the dashed lines.

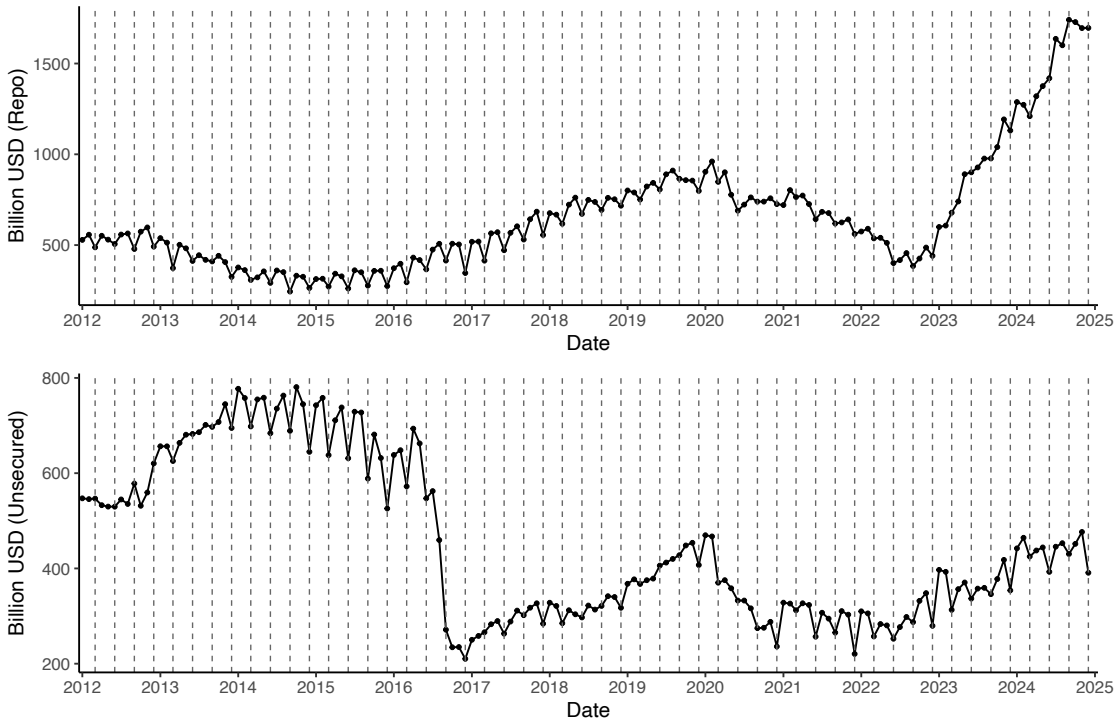
Figure B.11: Aggregate MMF Bank-Funding Holdings by Issuer Region around Quarter-Ends



Notes: The figure plots aggregate MMF holdings of bank debt by the headquarters region of the issuing bank, irrespective of security type. Dashed vertical lines mark standard calendar quarter-end dates. Canadian banks report on fiscal quarter-ends that fall one month after the calendar quarter-ends, so their quarter-end reduction does not align with the dashed lines. The sample spans 2012–2024.

Figure B.12 decomposes aggregate holdings into secured (repo) and unsecured (CP, CD, and time deposits) instruments. The recurring quarter-end contraction is visible in both panels, indicating that the window-dressing pattern operates through both funding channels. The two series differ in trend: unsecured holdings drop sharply around late 2016 and remain structurally lower thereafter, reflecting the 2016 MMF reform, whose principal provisions took effect in October 2016, when prime funds—the main holders of unsecured bank debt—lost roughly $1 trillion of assets to government funds, which by mandate cannot hold unsecured bank debt. Repo holdings, by contrast, continue to grow, since government funds can hold repo backed by Treasury or agency collateral. The 2016 level shift is therefore a regulatory shift rather than evidence against the window-dressing channel: the quarter-end seasonality is preserved both before and after 2016.

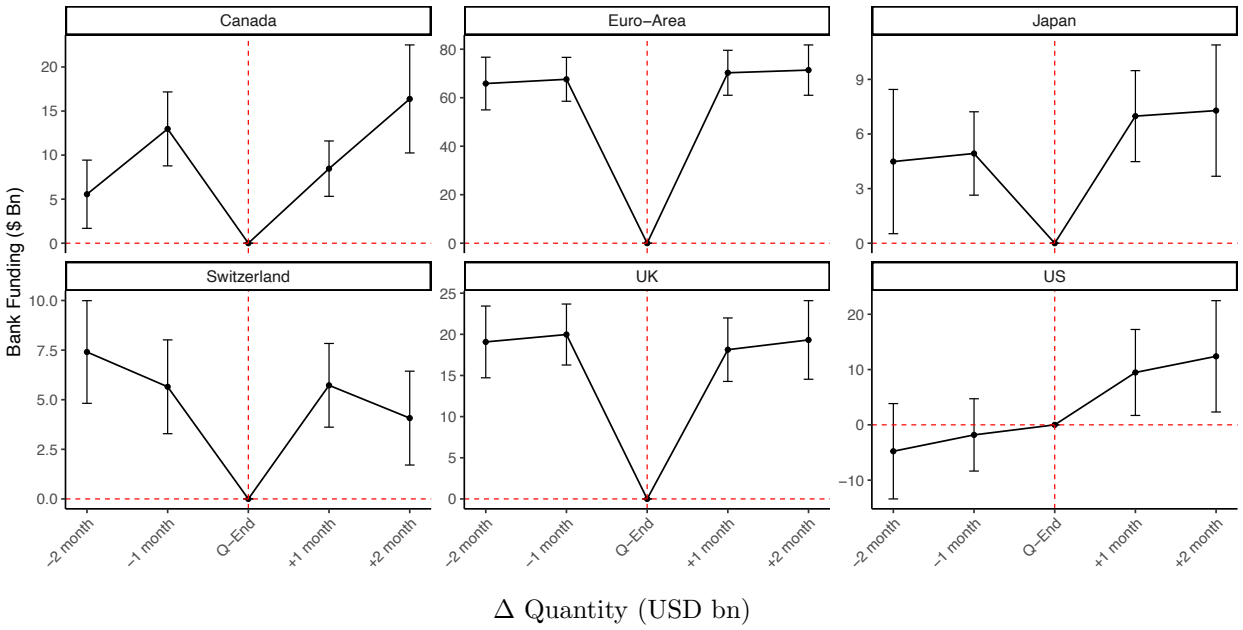
Figure B.12: MMF Holdings of Unsecured Bank Debt and Repo around Quarter-Ends



Notes: The figure plots aggregate MMF holdings of bank debt by security type, irrespective of issuer region. Dashed vertical lines mark standard calendar quarter-end dates. The top panel plots repurchase agreements (repo) of all collateral types and the bottom panel plots unsecured instruments (commercial paper, certificates of deposit, and time deposits). The sample spans 2012–2024.

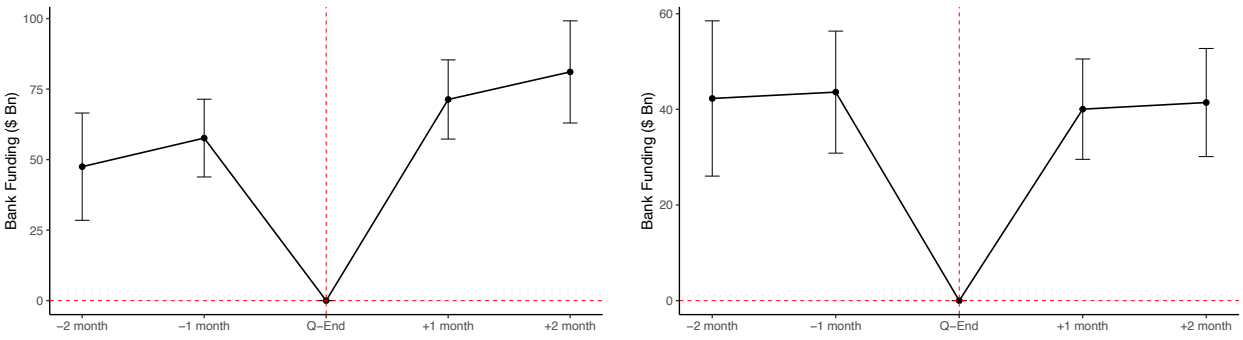
Figures B.13–B.14 report the event-window companions to Figure B.10, in dollar-change terms, by issuer region and separately for repo and unsecured instruments. In each case holdings fall sharply in the quarter-end month and rebound the following month, with the decline concentrated in euro-area, UK, and Swiss banks and spread across both repo and unsecured instruments.

Figure B.13: Event Window: Bank Funding by Issuer Region



Notes: The figure plots event-time averages of changes in MMF bank-funding holdings (in USD billions) by issuer region around each region's quarter-end. Canadian banks' quarter-ends fall in January, April, July, and October, one month after calendar quarter-ends, accounting for their distinct pattern. For all other regions, event time is anchored to quarter-ends in March, June, September, and December. The sample spans 2012–2024, and error bars denote 95% confidence intervals.

Figure B.14: Event Window: Bank Repo and Unsecured Debt



Notes: The figure plots event-time averages of MMF holdings against bank counterparties around quarter-ends. The left panel reports repo holdings and the right panel unsecured holdings (commercial paper, certificates of deposit, and time deposits), each as the change in holdings in USD billions. The sample spans 2012–2024, and error bars denote 95% confidence intervals.

B.3.3 Bank–Fund-Level Evidence on the Intensive and Extensive Margins

We examine the extensive margin at the bank–fund level, repeating the analysis of Table 3 on both sides of the fund–bank network: whether funds cut the number of banks they fund at quarter-end, and whether banks lose funding relationships as the number of funds lending to them falls. Table B.9 reports results consistent with the main extensive-margin findings: quarter-end months bring a statistically significant reduction in both the number of banks per fund and the number of funds per bank, across repo, unsecured debt, and the pooled specification.

Table B.9: Magnitude of Window Dressing: Extensive Margin

Dependent Variable:	Banks per Fund			Funds per Bank		
Asset Class:	Repo	Unsecured Debt	Both	Repo	Unsecured Debt	Both
QEnd_t	-0.1679^{***}	-0.7799^{***}	-0.5322^{***}	-1.216^{***}	-1.128^{***}	-1.553^{***}
	(0.0181)	(0.0469)	(0.0258)	(0.1828)	(0.0991)	(0.1169)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fund × Year-Quarter FE	Yes	Yes	Yes			
Bank × Year-Quarter FE				Yes	Yes	Yes
Observations	21,909	11,173	23,215	4,056	8,722	9,288
R^2	0.971	0.968	0.985	0.981	0.982	0.990
Within R^2	0.024	0.052	0.040	0.046	0.049	0.060

Notes: Columns (1)–(3) aggregate the data to the fund-year-month level and regress the number of unique banks with which a fund holds a positive position in the indicated asset class at month-end t on the quarter-end indicator, including fund × year-quarter fixed effects. Columns (4)–(6) aggregate to the bank-year-month level and regress the number of unique funds with a positive position in a given bank at month-end t on the quarter-end indicator, including bank × year-quarter fixed effects. Standard errors, clustered by fund × year-quarter in columns (1)–(3) and by bank × year-quarter in columns (4)–(6), are reported in parentheses, and ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

For the intensive margin, we estimate the following specification, in the same spirit as equation (5):

$$\text{Value}_{i,j,t} = \beta\,\text{QEnd}_t + \gamma' X_{i,j,t} + \alpha_{i,q(t)} + \mu_{j,q(t)} + \varepsilon_{i,j,t}, \tag{B.1}$$

where i indexes a money market fund, j indexes a bank issuer, $X_{i,j,t}$ denotes the set of controls, and $\alpha_{i,q(t)}$ and $\mu_{j,q(t)}$ are Fund × Year-Quarter and Issuer × Year-Quarter fixed effects, respectively. Because the specification conditions on continuing fund–bank relation-

ships and absorbs both fund-side and issuer-side quarterly shocks, β captures the within-pair contraction in exposures at quarter-end.

Table B.10 reports the results. The coefficient on QEnd_t is negative and statistically significant for both book-value and market-value measures across all asset classes. The estimated contraction is approximately \$0.032 billion per fund–bank pair for repo, \$0.016 billion per pair for unsecured debt, and \$0.026 billion per pair when both asset classes are pooled. The similarity between book-value and market-value estimates indicates that the contraction is not a valuation artifact. Moreover, the inclusion of issuer-side fixed effects helps rule out bank-specific quarterly shocks as the driver of the result.

Table B.10: Magnitude of Window Dressing: Intensive Margin Within Fund–Bank Pairs

Dependent Variables:	Book Value	Market Value	Book Value	Market Value	Book Value	Market Value
Asset Class:	Repo		Unsecured Debt		Both	
QEnd_t	-0.0324^{***}	-0.0324^{***}	-0.0163^{***}	-0.0163^{***}	-0.0260^{***}	-0.0259^{***}
	(0.0028)	(0.0028)	(0.0015)	(0.0015)	(0.0016)	(0.0016)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fund × Year-Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Issuer × Year-Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	158,717	158,717	207,952	207,952	338,132	338,132
R^2	0.472	0.472	0.588	0.588	0.446	0.446
Within R^2	0.0014	0.0014	0.0037	0.0037	0.0036	0.0036

Notes: The table reports fund–bank-pair OLS estimates of equation (B.1), where i indexes a fund, j indexes a bank issuer, and t indexes a month-end. The dependent variable is the book or market value of the pair's holdings in the indicated asset class (in USD billions). Fund × year-quarter fixed effects absorb fund-level quarterly shocks and issuer × year-quarter fixed effects absorb issuer-level quarterly shocks, so β measures the within-pair, within-quarter contraction at quarter-end. Standard errors, clustered by fund × year-quarter, are reported in parentheses, and ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

B.3.4 Window-Dressing Magnitude: Pre vs. Post Basel III

Appendix B.2.2 shows that the front-end yield drop is essentially absent before 2015 and emerges sharply once non-U.S. banks become subject to point-in-time leverage-ratio disclosure under Basel III. If our channel is correct, we should expect a parallel pattern on the MMF portfolio side: the magnitude of MMF withdrawal from bank funding at quarter-end should also be larger after 2015. We test this directly by re-estimating Tables 4 and B.10 separately on pre-2015 and post-2015 subsamples, focusing on market-value outcomes for

parsimony.[20]

Table B.11 reports the fund-level total-exposure split. The contraction in repo holdings is comparable across regimes ($0.39 billion per fund-month before 2015 vs. $0.35 billion afterwards), but the contraction in unsecured holdings widens dramatically post-2015: from $0.14 billion before 2015 to $0.85 billion after, a roughly six-fold increase. When the two asset classes are pooled, the post-2015 fund-level contraction ($0.70 billion) exceeds the pre-2015 contraction ($0.46 billion) by about 50%. The pattern is consistent with foreign banks shifting more aggressively out of unsecured funding once their balance-sheet snapshot rule begins binding, and with U.S. MMFs absorbing the corresponding withdrawal.

Table B.11: Fund-Level Total Exposure: Pre vs. Post Basel III

Dependent Variable:	Market Value (USD billions)					
Sample period:	Pre-Basel III (<2015)			Post-Basel III (≥2015)		
Asset class:	Repo	Unsec.	Both	Repo	Unsec.	Both
QEnd_t	-0.3929^{***}	-0.1442^{***}	-0.4633^{***}	-0.3514^{***}	-0.8502^{***}	-0.6955^{***}
	(0.0270)	(0.0413)	(0.0326)	(0.0315)	(0.0569)	(0.0382)
Fund × Year-Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,218	4,222	6,571	16,095	6,950	16,674
R^2	0.973	0.994	0.995	0.988	0.989	0.990
Within R^2	0.070	0.006	0.065	0.016	0.076	0.042

Notes: The table re-estimates equation (5) on the pre- and post-2015 subsamples, with the market value of fund-level holdings (in USD billions) as the dependent variable, reported separately for repo, unsecured debt, and the two pooled. All specifications include fund × year-quarter fixed effects. Standard errors, clustered by fund × year-quarter, are reported in parentheses, and ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table B.12 applies the same pre/post-2015 split to the extensive-margin regressions of Table 3. Panel A reports the year-month specification, in which the dependent variable is the count of unique money funds holding the indicated asset class at month-end. The post-2015 estimates are statistically significant across all three asset classes (−1.40, −0.98, and −1.26 funds for repo, unsecured, and both pooled, respectively), whereas the pre-2015 estimates

[20]Security-level controls (duration, ratings, repo haircut) are dropped here because the cleaned MMF panel records weighted-average-maturity and credit-rating fields only from 2016 onward; including them would mechanically empty the pre-Basel III sample. The QEnd coefficient is identified by within-fund (and within-pair) remaining variation after the fixed effects, so it remains cleanly interpretable without those controls.

are noisier and only marginally significant for repo. Panel B reports the year-month-by-fund specification, in which the dependent variable is the number of unique bank counterparties of a given fund at month-end. The contrast across regimes is sharpest for unsecured debt, where the post-2015 estimate of −1.16 banks per fund is roughly six times the pre-2015 estimate of −0.20; the repo estimate is essentially unchanged across regimes, and the pooled estimate is about 2.5 times larger after 2015.

Table B.12: Extensive Margin: Pre vs. Post Basel III

Sample period:	Pre-Basel III (<2015)			Post-Basel III (≥2015)		
Asset class:	Repo	Unsec.	Both	Repo	Unsec.	Both
Panel A. Active Funds per Month (Year-Month panel)						
QEnd_t	−3.625*	−1.417	−2.833	−1.400***	−0.975**	−1.263***
	(1.962)	(0.814)	(1.798)	(0.423)	(0.366)	(0.398)
Observations	36	36	36	120	120	120
R^2	0.753	0.803	0.714	0.981	0.996	0.992
Within R^2	0.224	0.209	0.186	0.136	0.107	0.150
Panel B. Banks per Active Fund (Year-Month × Fund panel)						
QEnd_t	−0.1695***	−0.2038***	−0.2698***	−0.1628***	−1.164***	−0.6395***
	(0.0331)	(0.0616)	(0.0437)	(0.0205)	(0.0640)	(0.0307)
Fund × Year-Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,218	4,222	6,571	16,095	6,950	16,674
R^2	0.956	0.977	0.985	0.975	0.965	0.985
Within R^2	0.008	0.005	0.011	0.007	0.086	0.049

Notes: The table re-estimates the extensive-margin specifications of Table 3 and Table B.9 separately on the pre- and post-2015 subsamples. Panel A aggregates the data to a year-month panel, with the count of unique funds holding the indicated asset class as the dependent variable. Panel B aggregates to a fund-year-month panel, with the count of unique bank counterparties of fund i as the dependent variable and fund × year-quarter fixed effects. Standard errors, clustered by fund × year-quarter, are reported in parentheses, and ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table B.13 reports the intensive-margin contraction across the two regimes at two levels of aggregation: the fund level (Panel A) and within continuing fund–bank pairs (Panel B). At both levels, the contraction in unsecured exposures strengthens markedly after 2015.

Table B.13: Intensive Margin, Fund-Level and Within Fund–Bank Pairs: Pre vs. Post Basel III

Dependent Variable:	Market Value (USD billions)					
Sample period:	Pre-Basel III (<2015)			Post-Basel III (≥2015)		
Asset class:	Repo	Unsec.	Both	Repo	Unsec.	Both
Panel A. Fund-Level Total Exposure (Fund × Year-Quarter FE)						
QEnd_t	-0.4077^{***}	-0.1407^{***}	-0.4633^{***}	-0.3443^{***}	-0.8335^{***}	-0.6955^{***}
	(0.0278)	(0.0416)	(0.0326)	(0.0318)	(0.0568)	(0.0382)
Observations	6,223	4,219	6,581	16,135	6,963	16,745
R^2	0.970	0.994	0.995	0.988	0.989	0.990
Within R^2	0.0697	0.0061	0.0648	0.0148	0.0731	0.0418
Panel B. Within Fund–Bank Pairs (Fund × YQ FE + Issuer × YQ FE)						
QEnd_t	-0.0428^{***}	-0.0033^{*}	-0.0195^{***}	-0.0299^{***}	-0.0210^{***}	-0.0298^{***}
	(0.0032)	(0.0018)	(0.0017)	(0.0032)	(0.0020)	(0.0020)
Observations	44,202	78,885	112,905	135,602	147,416	262,728
R^2	0.469	0.639	0.581	0.464	0.633	0.490
Within R^2	0.0022	0.0000	0.0005	0.0002	0.0006	0.0003

Notes: The table re-estimates the intensive-margin (market-value) specifications separately on the pre- and post-2015 subsamples, with the market value of holdings (in USD billions) as the dependent variable. Panel A is the fund-level specification of equation (5) (fund × year-quarter fixed effects); Panel B is the fund–bank-pair specification of equation (B.1), which additionally absorbs issuer × year-quarter fixed effects. Security controls are omitted for comparability with the pre-2015 sample. Standard errors, clustered by fund × year-quarter, are reported in parentheses, and ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Together, Tables B.11–B.13 show that the timing and composition of MMF withdrawal from non-U.S. bank funding line up with the post-2015 strengthening of the regulatory window-dressing incentive documented in Section 3 and Appendix B.2.2.

B.3.5 Same-Fund Substitution by Sub-Period and by Fund Type

Table B.14 re-estimates the same-fund reallocation of Table 6 (in dollar units) separately on the 2012–2014, 2015–Oct 2016, and Oct 2016–2024 sub-periods. The Treasury loading on withdrawn bank funding stays positive and significant throughout, easing from about \$0.18 per dollar before 2015 to roughly \$0.10 afterward, so the substitution into Treasuries survives in every regime. The RRP loading, by contrast, rises sharply once the Federal Reserve's overnight reverse repo facility expands—from \$0.34 per dollar in 2012–2014 to \$0.77 in 2015–Oct 2016 and \$0.59 thereafter—so the RRP becomes the dominant destination

for withdrawn funding in the later regimes.

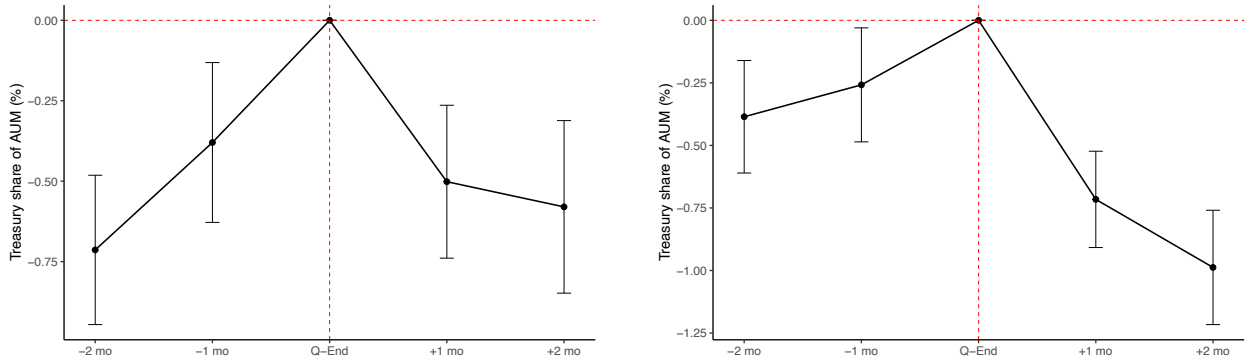
Table B.14: Same-Fund Reallocation into Treasuries and RRP, by Sub-Period

Dependent Variable:	ΔTreasury ($bn)			ΔRRP ($bn)		
Sub-period:	2012–2014	2015–Oct 2016	Oct 2016–2024	2012–2014	2015–Oct 2016	Oct 2016–2024
Model:	(1)	(2)	(3)	(4)	(5)	(6)
$-\Delta$ BankFunding	0.180***	0.103**	0.104***	0.344***	0.772***	0.589***
	(0.042)	(0.048)	(0.032)	(0.054)	(0.097)	(0.050)
NetFlow (control)	0.170***	0.123***	0.117***	0.248***	0.511***	0.524***
	(0.034)	(0.035)	(0.025)	(0.059)	(0.051)	(0.047)
Fund × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,364	1,245	4,571	2,364	1,245	4,571
Within R^2	0.096	0.087	0.057	0.154	0.527	0.432

Notes: The table re-estimates equation (6) (in dollar units) separately on the 2012–2014, 2015–Oct 2016, and Oct 2016–2024 subsamples, using quarter-end months. The dependent variables are the within-fund changes in Treasury and RRP holdings (in USD billions); $-\Delta$ BankFunding is the negative of the change in the fund's bank-funding holdings, and the fund's net flow enters as a control. All specifications include fund × year fixed effects. Standard errors, clustered by fund × year, are reported in parentheses, and ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Figure B.15 reproduces the fund-level Treasury-share event window of main-text Figure 3 separately for government (panel a) and prime (panel b) MMFs. In both groups the Treasury/AUM share peaks at the quarter-end and is lower in the surrounding months.

Figure B.15: Treasury Share Event Window by Fund Type



(a) Government MMFs (b) Prime MMFs

Notes: The figure repeats the fund-level Treasury-share event window of Figure 3, restricted to government MMFs (panel a) and prime MMFs (panel b). The specification includes fund × event-quarter fixed effects. The sample spans 2012–2024, and error bars denote 95% confidence intervals.

B.4 Derivations and Robustness for Section 5

B.4.1 Proof of Orthogonality under the Strong Orthogonality Assumption

Proof.

$$
\begin{aligned}
&\mathbb{E}\Big[\epsilon_{t,\tau} \cdot WDIT^{QE}_{t,\tau} \;\Big|\; \alpha_t, \gamma_{\tau,q(t)}, X_{t,\tau}\Big] \\
&= \mathbb{E}\left[\epsilon_{t,\tau} \sum_{i,j} \mathrm{QEnd}_t \theta^{Bill}_{i,\tau,t-1} \theta^{Bank}_{i,j,t-1} \Delta Debt_{j,t} \;\middle|\; \alpha_t, \gamma_{\tau,q(t)}, X_{t,\tau}\right] && \Big\}\text{Substitute the definition} \\
&= \sum_{i,j} \mathbb{E}\Big[\epsilon_{t,\tau} \mathrm{QEnd}_t \theta^{Bill}_{i,\tau,t-1} \theta^{Bank}_{i,j,t-1} \Delta Debt_{j,t} \;\Big|\; \alpha_t, \gamma_{\tau,q(t)}, X_{t,\tau}\Big] && \Big\}\text{Linearity} \\
&= \sum_{i,j} \mathbb{E}\Big[\mathrm{QEnd}_t \Delta Debt_{j,t}\, \mathbb{E}_{\tau,j,t}\Big[\epsilon_{t,\tau} \theta^{Bill}_{i,\tau,t-1} \theta^{Bank}_{i,j,t-1} \;\Big|\; \alpha_t, \gamma_{\tau,q(t)}, X_{t,\tau}\Big] \;\Big|\; \alpha_t, \gamma_{\tau,q(t)}, X_{t,\tau}\Big] && \Big\}\text{Law of iterated expectations} \\
&= \sum_{i,j} \mathbb{E}\Big[\mathrm{QEnd}_t \Delta Debt_{j,t} \cdot 0 \;\Big|\; \alpha_t, \gamma_{\tau,q(t)}, X_{t,\tau}\Big] && \Big\}\text{By Assumption} \\
&= 0
\end{aligned}
$$

□

B.4.2 Bank-side Characteristics Coverage

Table B.15: Exposure coverage of bank-characteristic measures

Characteristic	Banks	Cells	Mean	Median	P10	$\geq 50\%$
Log CDS spread	28	1,795	80.9%	83.0%	60.1%	95.5%
Tier 1 capital ratio (pp)	27	1,795	78.9%	81.2%	56.3%	94.3%
Short-term borrowing ratio (pp)	27	1,795	79.0%	81.3%	56.7%	94.5%
ROA (pp)	27	1,795	78.2%	80.3%	56.3%	94.0%

Notes: Banks reports the number of distinct banks with available data. Cells reports the number of maturity–month cells with an observed characteristic. Coverage is the share of each cell's bank exposure represented by banks with available data. Mean, Median, and P10 report its mean, median, and tenth percentile. The final column reports the percentage of cells with coverage of at least 50%. Coverage statistics use the 1,795 cells with positive bank exposure. The original regression sample contains 1,840 cells, but 45 lack usable lagged bank exposure. The regressions retain these cells by replacing missing characteristics with observed-cell means and including missing-value indicators. Filled values do not count as observed coverage, and the dependent variable is unchanged.

B.4.3 Baseline Regression with Additional Controls

Table B.16: Baseline WDIT coefficient with controls for bank and fund characteristics

	(1) Baseline	(2) + Bank	(3) + Fund	(4) + Both
$WDIT^{QE}$	4.4757***	4.6225***	4.5984***	4.7664***
	(1.7245)	(1.7586)	(1.7255)	(1.7231)
95% confidence interval	[1.089, 7.862]	[1.169, 8.076]	[1.210, 7.987]	[1.382, 8.150]
Observations	1,840	1,840	1,840	1,840
Quarter × tenor clusters	623	623	623	623
WDIT coefficient change	Reference	+3.3%	+2.7%	+6.5%

Notes: Every column retains the same specification as the original baseline. Bank controls are log CDS, Tier 1 capital ratio, short-term borrowing / total funding, ROA, and their four coverage fractions. Fund controls are cash, seven-day yield, WAM, daily liquidity, log minimum investment, log AUM, net flow and government-fund share. All four columns are estimated on the same 1,840 baseline cells and 623 year-quarter-by-maturity-bucket clusters as Table 7; the controls enter in levels. Bank characteristics are mapped to the cell by equation (14) and fund characteristics by equation (13). Missing cell characteristics are filled with their observed-cell means and enter with missing-value indicators, so no column loses an observation. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

B.4.4 Rotemberg Weights: Algebra and Full Results

$$\hat{\alpha}_{j,t} = \frac{\underbrace{\text{QEnd}_t \Delta Debt_{j,t}}_{shift} \sum_\tau [\underbrace{\sum_i \theta^{Bill}_{i,\tau,t-1}\theta^{Bank}_{i,j,t-1}}_{share}] \tilde{WDIT}^{QE}_{t,\tau}}{\sum_{t,\tau} \tilde{WDIT}^{QE}_{t,\tau}{}^2}$$

$$\hat{\beta}_{j,t} = \frac{\sum_\tau \underbrace{\sum_i \theta^{Bill}_{i,\tau,t-1}\theta^{Bank}_{i,j,t-1}}_{share} \tilde{y}_{t,\tau}}{\sum_\tau \underbrace{\sum_i \theta^{Bill}_{i,\tau,t-1}\theta^{Bank}_{i,j,t-1}}_{share} \tilde{WDIT}^{QE}_{t,\tau}}$$

where tildes denote variables residualized with respect to the other controls and fixed effects in equation (9).

Table B.17 reports the full summary of the Rotemberg decomposition discussed in Section 5.

Table B.17: Summary of Rotemberg weights

Panel A: Negative and positive issuer-month weights

Sign	Sum	Mean (bp)	Share
Negative	-0.108	-1.152	0.089
Positive	1.108	7.263	0.911

Panel B: Correlations across issuer-month components

	$\hat{\alpha}_{jt}$	$WDIT_{jt}$	$\hat{\beta}_{jt}$	F-stat	$\text{Var}_i(\theta_{ijt})$
$\hat{\alpha}_{jt}$	1.000				
$WDIT_{jt}$	-0.833	1.000			
$\hat{\beta}_{jt}$	-0.015	0.019	1.000		
F-stat	0.306	-0.303	-0.018	1.000	
$\text{Var}_i(\theta_{ijt})$	-0.117	0.159	-0.014	-0.135	1.000

Panel C: Variation across calendar years

Year	Sum $\hat{\alpha}$	Mean $\hat{\alpha}$ (bp)
2012	0.022	1.320
2013	0.042	2.132
2014	0.068	3.575
2015	0.063	3.288
2016	0.209	11.302
2017	0.203	11.266
2018	0.125	6.581
2019	0.064	3.089
2020	0.114	5.683
2021	0.059	3.017
2022	0.010	0.538
2023	0.009	0.518
2024	0.012	0.612

Panel D: Top five positive issuer-level Rotemberg weights

Issuer	$\hat{\alpha}_j$	$WDIT_{jt}$	$\hat{\beta}_j$	95% CI	$\hat{\beta}$ contribution	F-stat
Credit Agricole	0.239	-33.151	6.084	(2.00, 10.30)	1.455	577.645
BNP Paribas	0.139	-16.157	0.303	(-3.60, 4.20)	0.042	253.140
Societe Generale	0.117	-23.524	3.656	(0.70, 6.70)	0.426	361.414
Barclays PLC	0.099	-21.305	6.587	(1.00, 12.20)	0.649	343.167
Credit Suisse	0.059	-7.094	2.951	(-1.30, 7.40)	0.174	328.898

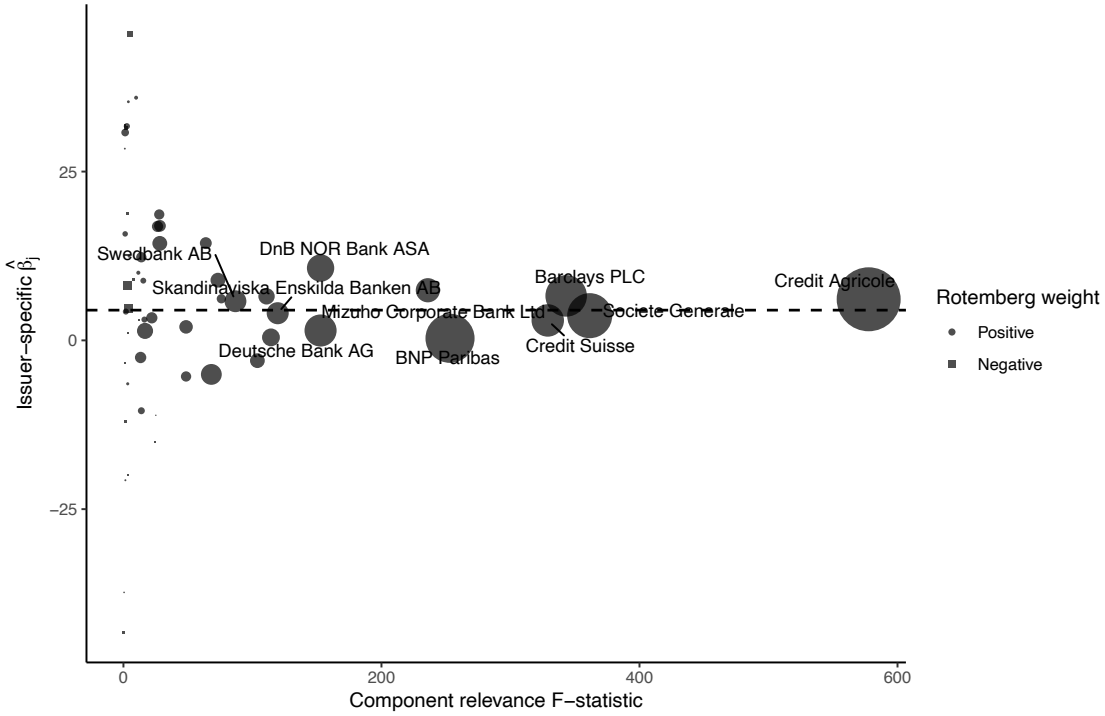
Panel E: Component estimates for positive and negative weights

Sign	$\hat{\alpha}$-weighted sum	Share of overall $\hat{\beta}$	Mean $\hat{\beta}_{jt}$
Negative	-1.357	-0.303	110.501
Positive	5.833	1.303	47.818

Notes: Panels A–C and E use issuer-by-quarter-end-month components. Mean Rotemberg weights in Panels A and C are multiplied by 10,000 and reported in basis points of weight. Panel A's Share is the group's share of total absolute Rotemberg-weight mass. In Panel B, the F-statistic is the squared clustered t statistic from the auxiliary relevance regression of aggregate WDIT on the issuer-month WDIT component, using the equation (9) controls and fixed effects. It is not a substantive first stage for equation (9). $\text{Var}_i(\theta_{ijt})$ is the sample variance across active funds of $\theta_{ijt} = Value_{ij,t-1}/Value_{j,t-1}$; active funds with no exposure to issuer j are assigned zero. Panel D first aggregates across months within issuer and reports the five largest positive issuer weights. Its confidence set inverts clustered tests of a zero coefficient on the issuer-specific component in a regression of $\Delta spread - bWDIT^{QE}$ on that component, the bid–ask control, report-month fixed effects, and tenor-by-quarter fixed effects. The search grid is $[-20, 20]$ with step 0.100. The aggregate equation (9) estimate is $\hat{\beta} = 4.476$. In Panel E, the $\hat{\alpha}$-weighted sums are the identification-relevant quantities and reconcile to $\hat{\beta} = 4.476$. The unweighted Mean $\hat{\beta}_{jt}$ in the last column is reported only for completeness. An individual issuer-month estimate $\hat{\beta}_{jt}$ becomes very large in either sign whenever that component's residualized exposure to the aggregate instrument is close to zero, so this raw mean is driven by a handful of such issuer-months rather than being informative.

Figure B.16 visualizes the relationship among bank-specific Rotemberg weights, component-relevance F-statistics, and just-identified estimates.

Figure B.16: Rotemberg Weights, F-statistic and $\hat{\beta}_j$



Notes: This figure illustrates the relationship between the Rotemberg weight, the F-statistic, and the bank-specific estimator $\hat{\beta}_j$. Marker size is proportional to the magnitude of the corresponding $\hat{\alpha}_j$. Squares denote negative Rotemberg weights, while circles denote positive Rotemberg weights. The dashed horizontal line indicates the benchmark market-level estimate. The ten banks with the largest Rotemberg weights are labeled correspondingly.

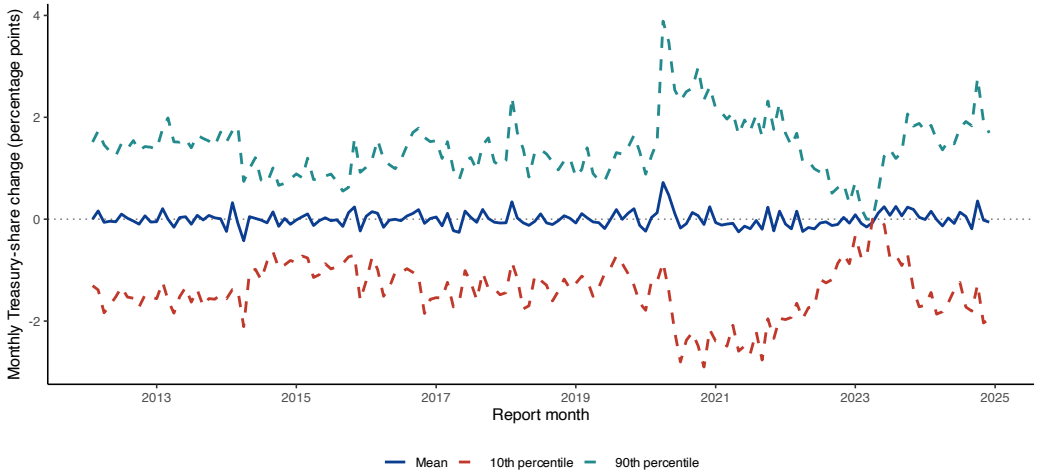
Table B.18: Relationship between Shares and Fund Characteristics

	(1) Credit Agricole	(2) BNP Paribas	(3) Societe Generale	(4) Barclays PLC	(5) Credit Suisse	(6) Aggregate WDIT
Cash / AUM (pp)	0.0106	0.0026	0.0261	0.0139	0.0056	0.0040
	(0.0282)	(0.0280)	(0.0280)	(0.0267)	(0.0199)	(0.0029)
Seven-day gross yield (pp)	-0.0205	-0.0649	-0.0234	-0.0856	-0.0356	0.0067
	(0.1904)	(0.1778)	(0.1748)	(0.1582)	(0.1406)	(0.0232)
Government-fund share (pp)	-0.0052	-0.0042	-0.0023	-0.0065	-0.0109***	0.0007
	(0.0050)	(0.0053)	(0.0051)	(0.0054)	(0.0041)	(0.0005)
WAM (days)	-0.0024	-0.0081	-0.0103	-0.0172	-0.0056	0.0005
	(0.0124)	(0.0135)	(0.0132)	(0.0127)	(0.0098)	(0.0013)
Daily liquid assets / AUM (pp)	-0.0031	-0.0042	0.0016	0.0049	-0.0004	-0.0005
	(0.0098)	(0.0094)	(0.0088)	(0.0083)	(0.0069)	(0.0009)
Log(min investment threshold)	-0.0155	-0.0161	-0.0231	-0.0476*	-0.0087	0.0022
	(0.0267)	(0.0281)	(0.0269)	(0.0255)	(0.0206)	(0.0026)
Log (AUM)	0.0473	0.1186	0.0768	0.0640	0.1108*	0.0022
	(0.0847)	(0.0839)	(0.0772)	(0.0783)	(0.0633)	(0.0077)
Net flow / AUM (pp)	0.0150	0.0024	0.0038	0.0138	0.0082	-0.0014
	(0.0158)	(0.0135)	(0.0140)	(0.0140)	(0.0112)	(0.0015)
Joint characteristic test (p)	0.910	0.884	0.832	0.363	0.207	0.604
Report month	Yes	Yes	Yes	Yes	Yes	Yes
YearQuarter × Maturity bucket	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,840	1,840	1,840	1,840	1,840	1,840
R^2	0.76905	0.74845	0.76842	0.75071	0.75585	0.77261
Within R^2	0.00430	0.00526	0.00856	0.00875	0.01339	0.00764

Notes: Columns (1)–(5) report regressions of the pre-quarter-end bank–tenor exposure share on fund characteristics, for the five banks with the largest Rotemberg weights; column (6) uses aggregate $WDIT_{t,\tau}^{QE}$ as the dependent variable. The unit of observation is the report-month-by-maturity-bucket cell, on the exact 1,840 keys of the baseline estimation sample in Table 7. All columns include report-month and year-quarter-by-maturity-bucket fixed effects, and standard errors are clustered by year-quarter-by-maturity bucket, as in Table 7. All fund characteristics are measured and mapped to the tenor level following equation (13). Cash is the total cash holdings of the fund. Seven-day gross yield is the annualized seven-day gross yield offered to shareholders. WAM is the weighted-average maturity of the fund's overall portfolio. Daily liquid assets are defined according to SEC Rule 2a-7. Minimum investment threshold is the minimum initial investment required by the fund. Net flow is the monthly net capital flow into the fund. The joint characteristic test is a Wald test that their coefficients are jointly zero in that column. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

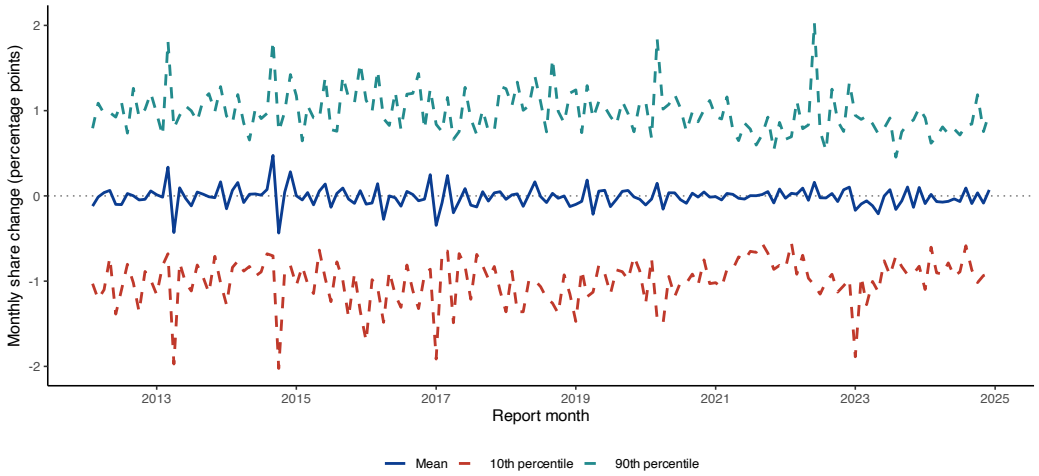
B.4.5 $\Delta\theta^{Bill}_{i,\tau,t}$ and $\Delta\theta^{Bank}_{i,j,t}$ over Time

Figure B.17: Monthly Changes in Bill Shares



Notes: This figure plots monthly changes in bill shares, $\Delta\theta^{Bill}_{i,\tau,t}$, over time.

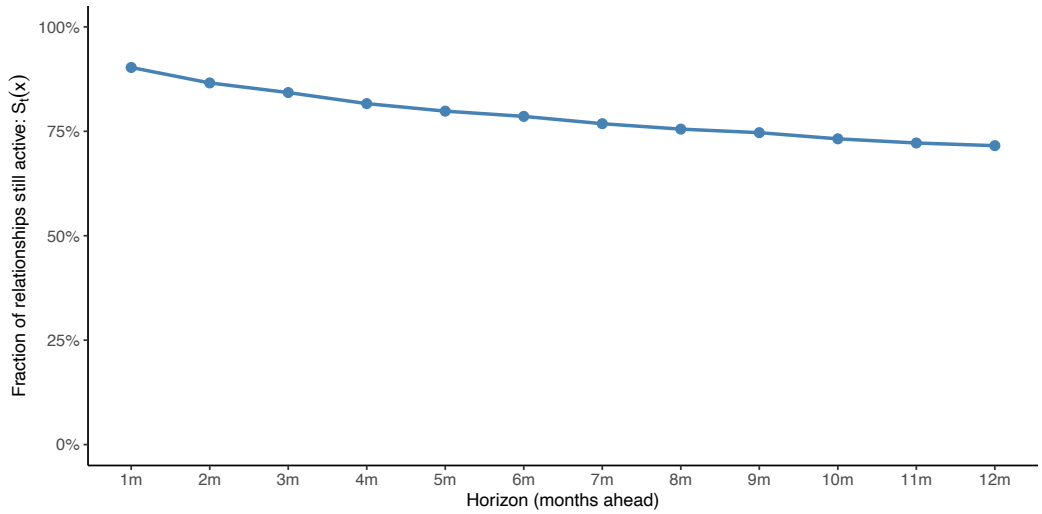
Figure B.18: Monthly Changes in Bank Shares



Notes: This figure plots monthly changes in bank shares, $\Delta\theta^{Bank}_{i,j,t}$, over time.

B.4.6 Persistence of Fund–Bank Relationships

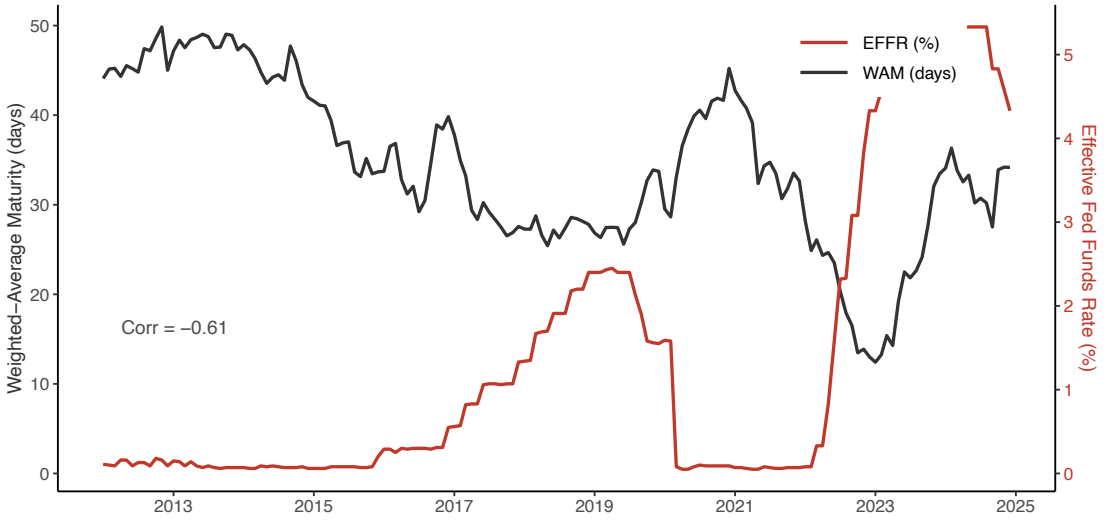
Figure B.19: Bank–Fund Relationship: Extensive Margin



Notes: This figure plots the retention rate of fund–bank relationships. It is computed as follows: for each date t, form the set of unique fund–bank pairs $\mathcal{L}_t$. For horizon x, compute the survival rate $S_t(x) = \frac{|\mathcal{L}_t \cap \mathcal{L}_{t+x}|}{|\mathcal{L}_t|}, \quad \forall t \in T$. Repeat across horizons x to trace the persistence profile of relationships.

B.4.7 MMF Portfolio Duration and the Policy Rate

Figure B.20: MMF Portfolio Duration and Interest Rate



Notes: This figure plots the portfolio maturity of our MMF sample and the effective federal funds rate.

B.4.8 The Overnight Reverse Repurchase (ON RRP) Facility

The Federal Reserve's overnight reverse repurchase agreement (ON RRP) facility allows eligible counterparties to place cash with the Fed overnight at a fixed rate, which puts a floor under short-term money-market rates. The Fed launched it on a test basis in September 2013. In contrast to its standing repo operations with primary dealers, the ON RRP admits a wider set of counterparties that includes money market funds, government-sponsored enterprises, and banks, with money market funds eligible from the start. The amount each counterparty can place is capped on a daily, per-counterparty basis. That cap was initially $30 billion (set in September 2014, alongside an aggregate limit of $300 billion), was raised to $80 billion in March 2021, and again to $160 billion in September 2021.

The size of this cap matters for our setting because it governs how readily a fund can substitute toward the Fed rather than toward Treasury bills. When the facility is small or its cap binds tightly, as in the early years of our sample, a fund that pulls back from bank funding at quarter-end has little to reallocate into besides Treasury bills, so the price effect on bills is larger. Once the cap is relaxed, the ON RRP becomes an elastic place to park

cash; it absorbs a larger share of the quarter-end reallocation and weakens the effect on Treasury prices.

B.4.9 Further Share Lags

Table B.19: Lag Robustness: $WDIT^{QE}$ Coefficient across Share-Lag Combinations

	Treasury weight lag k_2						
Fund–bank lag k_1	-1	-2	-3	-4	-5	-6	Mean
-1	4.606***	3.734**	2.374*	4.634***	4.120***	2.400*	6.222***
	(1.632)	(1.530)	(1.421)	(1.629)	(1.184)	(1.442)	(1.806)
-2	4.625***	3.858**	2.791*	5.083***	4.532***	2.818*	6.681***
	(1.658)	(1.559)	(1.463)	(1.764)	(1.258)	(1.517)	(1.871)
-3	5.457***	4.044**	2.798*	5.752***	5.221***	3.519**	7.476***
	(1.763)	(1.690)	(1.607)	(1.720)	(1.346)	(1.649)	(1.989)
-4	5.152***	4.166***	2.874*	5.517***	4.782***	3.244**	7.044***
	(1.772)	(1.593)	(1.472)	(1.853)	(1.301)	(1.607)	(1.945)
-5	5.490***	4.451***	3.155**	5.722***	4.879***	3.299**	7.281***
	(1.842)	(1.622)	(1.508)	(1.995)	(1.347)	(1.639)	(2.029)
-6	7.049***	5.712***	4.303***	7.115***	5.643***	4.071**	9.110***
	(2.015)	(1.666)	(1.626)	(2.059)	(1.401)	(1.782)	(2.169)
Mean	6.303***	5.174***	3.634**	6.620***	5.208***	3.653**	8.120***
	(1.977)	(1.737)	(1.671)	(2.045)	(1.426)	(1.734)	(2.121)

Notes: Each cell reports the coefficient on $WDIT^{QE}$ using the indicated fund–bank exposure shares and Treasury tenor weights. Negative labels denote exact calendar-month lags. Mean denotes the equal-weight average of the preceding 12 monthly shares, computed separately for each fund–bank or fund–tenor pair and requiring all 12 shares to be available. Absent Treasury holdings are treated as zero only in months with valid reported fund AUM. Missing contributions are omitted; an all-missing sum is zero. The original estimation sample (N = 1,840 throughout) and ΔDebt shocks are retained. All specifications include the same controls as the baseline specification. Standard errors, clustered by year-quarter-by-maturity bucket, are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

B.4.10 Weaker Identification Assumption and Estimates

To relax Assumption 1, we construct an alternative version of $WDIT^{QE}_{t,\tau}$ that removes fund-bank specific relationship exposure. Define

$$WDIT^{BankAvg}_{t,\tau} \equiv \text{QEnd}_t \sum_{i,j} \theta^{Bill}_{i,\tau,t-1} \bar{\theta}^{Bank}_{j,t-1} \Delta Debt_{j,t},$$

where $\bar{\theta}^{Bank}_{j,t-1} \equiv \frac{1}{N}\sum_i \theta^{Bank}_{i,j,t-1}$ denotes the simple average fund-bank relationship exposure across the sample, i.e., the MMF sector-level bank funding weight. Consistency now requires only the weaker assumption:

Assumption 2. *(Weak Share Exogeneity) For any Treasury tenor τ in each period t, the lagged tenor exposure $\theta^{Bill}_{i,\tau,t-1}$ is orthogonal to the unobserved tenor-level price shock $\epsilon_{t,\tau}$:*

$$\mathbb{E}_{\tau,t}\left[\epsilon_{t,\tau}\theta^{Bill}_{i,\tau,t-1}\middle|\ \alpha_t,\ \gamma_{\tau,q(t)},\ X_{t,\tau}\ \right] = 0, \quad \forall i,\tau,t.$$

Compared with the baseline variable, $WDIT^{BankAvg}_{t,\tau}$ requires a weaker identifying assumption because it removes the cross-sectional variation in fund-bank allocations. By replacing the bilateral exposure with an average measure, the identifying variation is driven solely by predetermined investment allocation choices rather than potentially endogenous fund-bank decisions.

We re-estimate the baseline specification using $WDIT^{BankAvg}$. The estimates are reported in Table B.20 below, alongside the baseline and the tenor-level variant introduced next.

We next replace the fund-specific tenor share with a market-level tenor share. Replacing fund-specific Treasury-tenor allocations with a common market-level allocation collapses the

two-layer network into a one-layer exposure design. Specifically, define

$$\begin{aligned} WDIT_{t,\tau}^{BillAvg} &\equiv \text{QEnd}_t \frac{100}{Size_{t-1,\tau}} \sum_{i,j} \bar{\theta}_{\tau,t-1}^{Bill} \theta_{i,j,t-1}^{Bank} \Delta Debt_{j,t} \\ &= \text{QEnd}_t \frac{100\bar{\theta}_{\tau,t-1}^{Bill}}{Size_{t-1,\tau}} \sum_{j} \Delta Debt_{j,t} \left(\sum_{i} \theta_{i,j,t-1}^{Bank} \right) \\ &= \text{QEnd}_t \frac{100\bar{\theta}_{\tau,t-1}^{Bill}}{Size_{t-1,\tau}} \sum_{j} \Delta Debt_{j,t}. \end{aligned}$$

The final equality follows from the definition of bank-funding shares, which sum to one across the fund universe:

$$\sum_{i} \theta_{i,j,t-1}^{Bank} = 1.$$

Let

$$D_t \equiv \sum_{j} \Delta Debt_{j,t}, \qquad s_{\tau,t-1} \equiv \frac{100\bar{\theta}_{\tau,t-1}^{Bill}}{Size_{t-1,\tau}}.$$

The regressor can therefore be written as

$$WDIT_{t,\tau}^{BillAvg} = \text{QEnd}_t \, s_{\tau,t-1} D_t.$$

Thus, the remaining regressor combines a common aggregate bank-funding shift with a predetermined tenor-specific exposure. Cross-tenor variation reflects the market-level Treasury allocation relative to each tenor's outstanding supply, while bilateral bank–fund relationships drop out. Under the share exogeneity framework, identification rests on the exogeneity of these normalized Treasury-tenor exposures conditional on the included controls and fixed effects. Table B.20 reports both weaker-assumption variants next to the baseline, estimated on the same sample.

Table B.20: Window-Dressing Spillovers under the Weaker Identification Assumptions

Dependent Variable:	$\Delta spread$		
	Baseline	Market-level bank shares	Market-level bill shares
Model:	(1)	(2)	(3)
$WDIT^{QE}$	4.4757***		
	(1.7245)		
$WDIT^{BankAvg}$		4.8432**	
		(2.2303)	
$WDIT^{BillAvg}$			3.4922**
			(1.4393)
Controls	Yes	Yes	Yes
YearQuarter × Maturity bucket	Yes	Yes	Yes
Report month	Yes	Yes	Yes
Observations	1,840	1,840	1,840
R^2	0.83075	0.83068	0.83124
Within R^2	0.00613	0.00575	0.00903

Notes: OLS estimates of equation (9) on the full sample. Column (1) repeats the baseline specification of Table 7, column (2) replaces the bilateral fund–bank exposure $\theta^{Bank}_{i,j,t-1}$ with its cross-fund average, and column (3) replaces the fund-specific tenor share $\theta^{Bill}_{i,\tau,t-1}$ with a market-level tenor share. The dependent variable is the month-end change in the spread to the federal-funds lower bound, in basis points. The same short-maturity Treasury universe—nominal bills, notes, and bonds with under one year to maturity—is used to construct each regressor and $\Delta spread_{t,\tau}$, and securities within 15 days of maturity are excluded. All three columns are estimated on the same observations and include report-month and year-quarter-by-maturity-bucket fixed effects and control for the bid–ask spread. Standard errors, clustered by year-quarter-by-maturity bucket, are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

B.4.11 Leave-One-Out Bank Shift

The bank-level shift $\Delta Debt_{j,t}$ in equation (10) is the full change in bank j's MMF funding. An alternative construction nets out fund i's own contribution to that change,

$$B^{LOO}_{i,t} = \sum_{j} \theta^{Bank}_{i,j,t-1} \left(\Delta Debt_{j,t} - \Delta Value_{i,j,t} \right),$$

Table B.21 shows the choice is immaterial empirically. The two regressors have a correlation of 0.998 on quarter-end cells, the full-sample point estimate differs by 9.4% of the baseline, and the difference between them has a standard deviation only 11% as large as

the regressor itself, so the two cannot be told apart statistically: entered jointly, neither is significant and the baseline standard error roughly doubles.

Table B.21: Baseline and Leave-One-Out Bank Shift

Dependent Variable:	$\Delta spread$		
	Full Sample	2012–2014	2015–2024
Specification:	(1)	(2)	(3)
Panel A: baseline, full bank shift (as in the paper)			
$WDIT^{QE}$	4.4757***	6.4342***	4.1874**
	(1.7245)	(2.2323)	(1.9462)
Panel B: leave-one-out bank shift			
$WDIT^{QE,LOO}$	4.8951**	6.8703***	4.5652**
	(1.9186)	(2.3489)	(2.1997)
Controls	Yes	Yes	Yes
YearQuarter × Maturity bucket	Yes	Yes	Yes
Report month	Yes	Yes	Yes
Observations	1,840	432	1,408

Notes: Both panels use identical estimation observations, controls and fixed effects; the only difference is whether fund i's own change in its holdings of bank j is netted out of the bank-level shift in equation (10). On quarter-end cells the two regressors have a correlation of 0.998; the standard deviation of their difference is 0.111 times that of the baseline regressor. Entered jointly in the full sample, the baseline regressor gives 1.58 (3.35) and the difference term −27.42 (28.14), neither significant, with the baseline standard error rising from 1.72 to 3.35. Standard errors, clustered by year-quarter-by-maturity bucket, are in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

B.4.12 Prior-Month Placebo

Table B.22: Prior-Month Placebo: Full Sample and Subsamples

Dependent variable:	$\Delta spread_{t-1,\tau}$		
	Full sample	2012–2014	2015–2024
	(1)	(2)	(3)
$WDIT_{t,\tau}$	-0.6136	-1.2609	-0.5089
	(1.6763)	(2.2858)	(1.9085)
Bid–ask spread control	Yes	Yes	Yes
Year-quarter × maturity bucket FE	Yes	Yes	Yes
Report-month FE	Yes	Yes	Yes
Observations	1,804	432	1,372
R^2	0.83325	0.57070	0.83605
Within R^2	0.00330	0.01613	0.00295

Notes: This table reports OLS regressions of the previous month's spread change, $\Delta spread_{t-1,\tau}$, measured in basis points, on current trading pressure $WDIT_{t,\tau}$. All columns use the baseline trading-pressure measure and quarter-end indicator. Subsamples are defined by the outcome month, $t-1$. Observations require exact calendar-month matches for the same maturity bucket within the baseline estimation sample; the matched outcome sample ends in November 2024. The bid–ask spread control, report-month fixed effects, and year-quarter-by-maturity-bucket fixed effects are aligned with the outcome month. Standard errors, clustered at the outcome-year-quarter-by-maturity-bucket level, are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

B.4.13 Additional Results Incorporating the Reversal Period

Table B.23: Results Incorporating the Post-Quarter-End Reversal Period

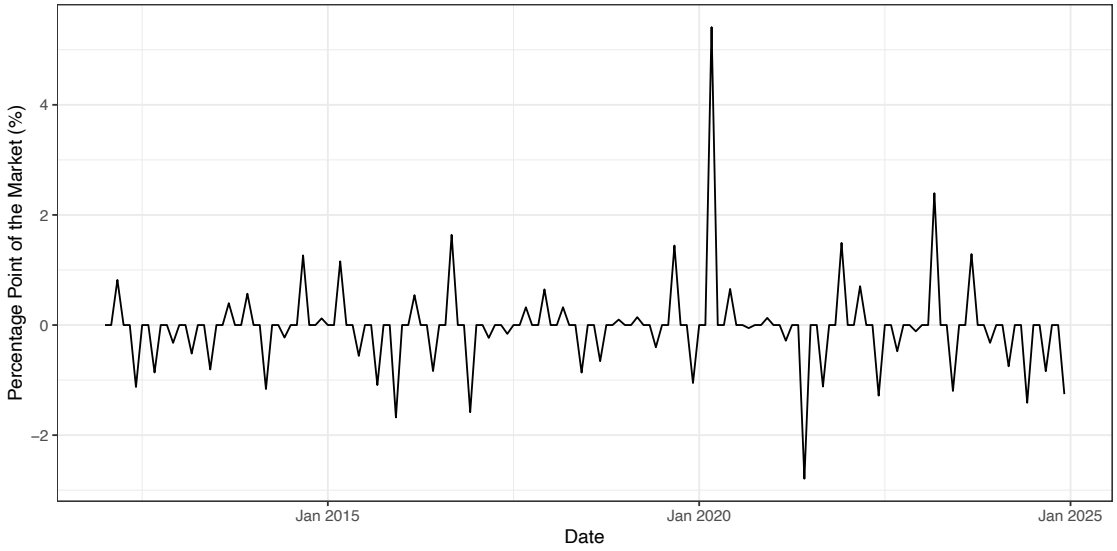
Dependent variable:	Δspread		
Model:	Quarter-end (1)	Month after quarter-end (2)	Quarter-end and following month (3)
Variables			
$WDIT$	4.4757***	6.1145***	3.9874***
	(1.7245)	(2.1610)	(1.1317)
Controls	Yes	Yes	Yes
Sample	Full sample	Full sample	Full sample
Fixed effects			
Year-quarter × maturity bucket	Yes	Yes	Yes
Report month	Yes	Yes	Yes
Observations	1,840	1,840	1,840
R^2	0.83072	0.83109	0.83127
Within R^2	0.00600	0.00817	0.00920

Notes: This table reports OLS estimates of equation (9). The dependent variable is the month-end change in the spread relative to the lower bound of the federal funds target range, measured in basis points. The same universe of short-maturity Treasury securities—nominal bills, notes, and bonds with less than one year to maturity—is used to construct both $WDIT^{QE}_{t,\tau}$ and $\Delta\text{spread}_{t,\tau}$. Securities within 15 days of maturity are excluded. All specifications include report-month and year-quarter-by-maturity-bucket fixed effects and control for the bid–ask spread. Standard errors, clustered at the year-quarter-by-maturity-bucket level, are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

B.4.14 $WDIT^{QE}_{t,\tau}$ versus $\Delta Bill^{QE}_{t,\tau}$

Figure B.21 plots the time series of the market-level difference between $WDIT^{QE}_{t,\tau}$ and $\Delta Bill^{QE}_{t,\tau}$. The figure shows that $WDIT^{QE}_{t,\tau}$ is systematically smaller in magnitude than the observed change in Treasury bill holdings, $\Delta Bill^{QE}_{t,\tau}$.

Figure B.21: Market-Level Difference between $WDIT^{QE}_{t,\tau}$ and $\Delta Bill^{QE}_{t,\tau}$



Notes: This figure plots the market-level difference between $WDIT^{QE}_{t,\tau}$ and $\Delta Bill^{QE}_{t,\tau}$.

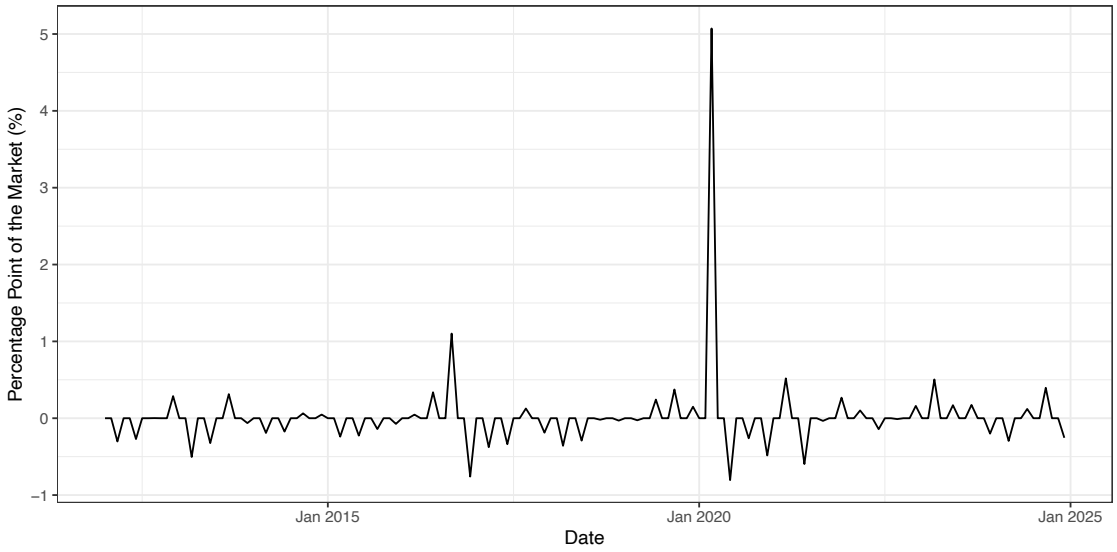
This gap is expected by construction. The measure $WDIT^{QE}_{t,\tau}$ captures only the reallocation of funds induced by quarter-end reductions in bank borrowing from MMFs. It does not capture changes in Treasury demand arising from the allocation of contemporaneous fund inflows. As documented in Section 4, MMFs are more likely to direct new fund inflows into Treasury securities at quarter-end than during other periods. Because these flow-driven portfolio adjustments are excluded from the construction of $WDIT^{QE}_{t,\tau}$, the measure captures only a subset of the total quarter-end increase in Treasury bill demand, leading it to be systematically smaller than $\Delta Bill^{QE}_{t,\tau}$. To recover the importance of this omitted flow channel, we estimate the following regression:

$$\Delta Bill^{QE}_{i,\tau,t} = \beta \times \text{NetFlow}_{i,t} + \alpha_t + \gamma_\tau + \zeta_i + \epsilon_{i,t,\tau}$$

We then use the fitted values from this regression to isolate the component of Treasury bill purchases attributable to fund inflows and remove it from the aggregate Treasury demand measure. Figure B.22 plots the difference between $WDIT^{QE}_{t,\tau}$ and the resulting adjusted

measure, $\Delta\widehat{Bill}_{t,\tau}^{QE}$. The discrepancy becomes substantially smaller.[21]

Figure B.22: Market-Level Difference between $WDIT_{t,\tau}^{QE}$ and $\Delta\widehat{Bill}_{t,\tau}^{QE}$



Notes: This figure plots the market-level difference between $WDIT_{t,\tau}^{QE}$ and $\Delta\widehat{Bill}_{t,\tau}^{QE}$, where $\Delta\widehat{Bill}_{t,\tau}^{QE}$ is constructed by removing the estimated component associated with fund net inflows.

Finally, we compare realized holdings changes with predicted reallocation. For this exercise, we construct predicted reallocation without interacting it with the quarter-end indicator, allowing the instrument to vary in both quarter-end and non-quarter-end months. We then use this measure to instrument realized holdings changes in the spread regression, retaining the controls and fixed effects in equation (9). All three specifications use the same 1,840 observations. Column (1) of Table B.24 reports a first-stage coefficient of 1.317, indicating that a one-percentage-point change in predicted reallocation is associated with a 1.32-percentage-point change in realized bill holdings. The point estimate exceeds one, consistent with the possibility of additional flow-driven reallocation beyond that captured by our proxy, although the difference from one is not statistically significant. Column (2) reports an IV multiplier of 3.107, smaller than but of the same order of magnitude as the baseline estimate of 4.5. In contrast, the OLS regression of spread changes on realized holdings changes in column (3)

[21]The main exception is March 2020. During the COVID-19 "dash-for-cash" episode, MMF portfolio reallocations were unusually large and abrupt, making them difficult to capture using lagged portfolio weights.

yields a small and statistically insignificant coefficient of -0.144. This contrast is consistent with, rather than conclusive evidence of, the endogeneity concerns discussed in Section 5: realized portfolio changes may reflect responses to yields and other shocks that jointly affect prices and holdings, obscuring the price impact of demand-driven reallocation.

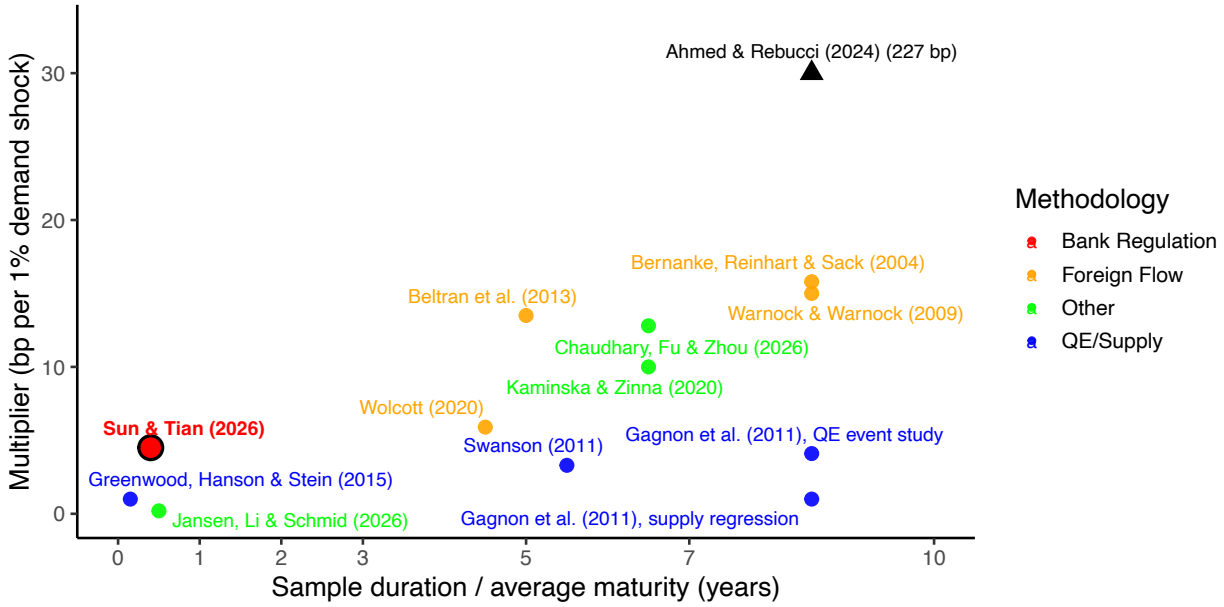
Table B.24: Predicted Reallocation, Realized Holdings Changes, and Treasury Spreads

Dependent Variables:	$\Delta Bill^{QE}$	$\Delta Spread$	
	(1)	(2)	(3)
$WDIT^{QE}$	1.317***		
	(0.490)		
$\Delta Bill^{QE}$		3.107**	-0.144
		(1.460)	(0.145)
Controls	Yes	Yes	Yes
YearQuarter × Maturity bucket	Yes	Yes	Yes
Report month	Yes	Yes	Yes
Observations	1,840	1,840	1,840
R^2	0.375	0.696	0.830
Within R^2	0.011	-0.783	0.004
Wald (1st stage)		7.227	

Notes: This table reports regressions at the maturity-bucket–month level. Column (1) presents the first-stage OLS regression of realized changes in money market funds' Treasury holdings, $\Delta Bill^{QE}_{t,\tau}$, on predicted window-dressing-induced reallocation, $WDIT^{QE}_{t,\tau}$. Column (2) reports two-stage least-squares estimates of Treasury spread changes on realized holdings changes, instrumenting $\Delta Bill^{QE}_{t,\tau}$ with $WDIT^{QE}_{t,\tau}$. Column (3) reports OLS estimates of the relationship between spread changes and realized holdings changes, without instrumenting the latter. Holdings changes and predicted reallocation are expressed in percentage points of the outstanding market size of the corresponding maturity bucket. $\Delta Spread_{t,\tau}$ is the month-end change in the Treasury yield spread relative to the lower bound of the federal funds target range, measured in basis points. All specifications control for the bid–ask spread and include report-month and year-quarter-by-maturity-bucket fixed effects. Standard errors, clustered at the year-quarter-by-maturity-bucket level, are reported in parentheses. "Wald (1st stage)" reports the cluster-robust Wald statistic testing the null that the coefficient on the excluded instrument is zero in the first stage. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Overall, this exercise suggests that $WDIT^{QE}_{t,\tau}$ captures an important component of the total quarter-end demand shock, while potentially understating its magnitude by excluding additional flow-driven reallocation. To the extent that omitted demand reinforces the variation captured by our measure, conditional on the controls and fixed effects, the baseline estimate $\hat{\beta}$ may overstate the market multiplier. The smaller IV point estimate is consistent with this interpretation, although the evidence does not establish a tight upper bound.

Figure B.23: Estimated Treasury Yield Multiplier in the Literature



Notes: This figure compares estimated price multipliers across studies. The horizontal axis reports the average duration or maturity of the securities studied, measured in years. The vertical axis reports the estimated yield response, in basis points, to a demand shock equal to 1% of the outstanding market value of the corresponding asset class. Estimates are rescaled to a common unit whenever necessary. Colors indicate the identification strategy employed by each study. The red marker highlights our estimate. To preserve readability, the estimate from Ahmed & Rebucci (2024) is shown separately as an annotated outlier.

Table B.25: Remaining Days to Final Maturity by Instrument

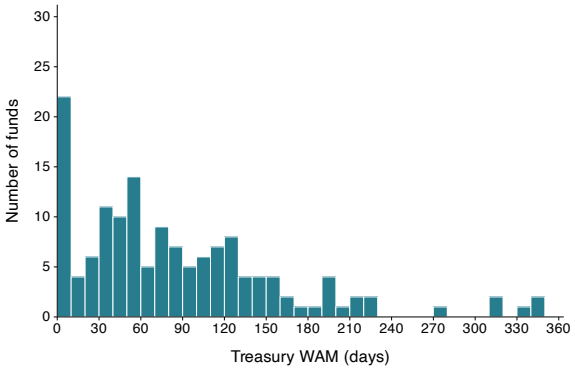
Instrument	N	Mean	SD	P25	Median	P75	VW mean
Treasury Repo	193,359	23.6	476.6	1.0	3.0	8.0	19.4
Agency Repo	164,300	26.3	605.9	1.0	3.0	7.0	32.2
Other Repo	97,397	60.5	611.6	2.0	6.0	38.0	88.0
Financial CP	338,187	87.1	81.3	27.0	67.0	126.0	84.8
Non-Financial CP	52,366	49.8	190.7	8.0	28.0	65.0	48.5
Other CP	61,196	53.7	57.4	13.0	35.0	73.0	65.0
ABCP	233,548	63.4	62.6	16.0	45.0	87.0	64.2
CD/TD	485,105	88.3	130.2	18.0	66.0	130.0	71.9

Notes: The sample comprises prime and government money market fund holdings in Form N-MFP, 2012 to 2024, across all counterparties. Maturity is the number of calendar days from month-end to final maturity. N counts security–fund–month observations with non-missing, nonnegative maturity. SD is the sample standard deviation; P25 and P75 are the 25th and 75th percentiles. VW mean is weighted by positive security value; all other statistics are unweighted. Repo denotes repurchase agreements, classified by collateral type (Treasury, agency, or other). CP denotes commercial paper; Financial and Non-Financial CP denote commercial paper issued by financial and non-financial firms, respectively, while Other CP denotes commercial paper whose issuer type cannot be identified. ABCP denotes asset-backed commercial paper. CD/TD denotes certificates of deposit and time deposits.

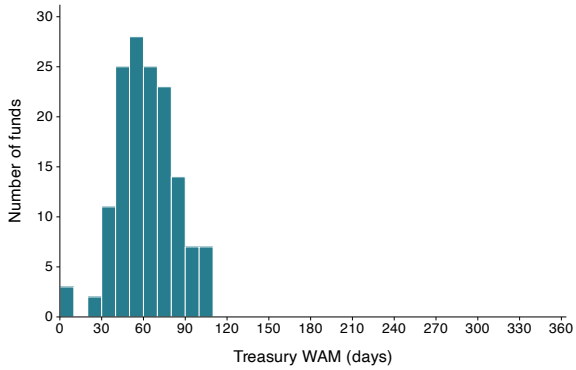
B.5 Additional Results for Section 6

B.5.1 Heterogeneous Duration Targets

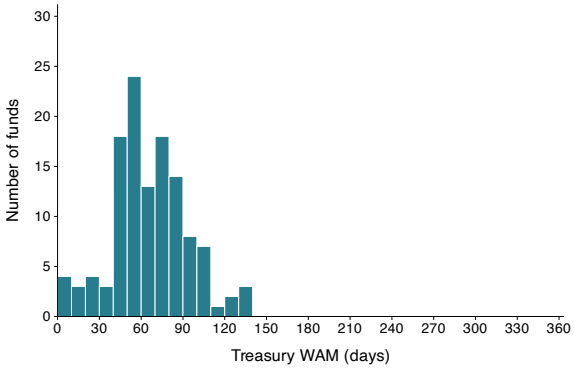
Figure B.24: Cross-Fund Heterogeneity in Treasury Portfolio Maturity



(a) January 2015



(b) January 2021



(c) December 2024

Notes: This figure shows the cross-sectional distribution of Treasury weighted-average maturity (WAM) among prime and government money market funds. Each observation represents one fund in the indicated month. Treasury WAM is the market-value-weighted average maturity of a fund's direct Treasury holdings, measured in days using the regulatory WAM convention, which incorporates interest-rate-reset dates where applicable. Fund-level WAM is winsorized at the 1st and 99th percentiles separately within each month. Histograms use 10-day bins and common axis scales; each fund receives equal weight. The monthly samples are formed separately and contain 145, 145, and 122 funds in panels (a), (b), and (c), respectively.

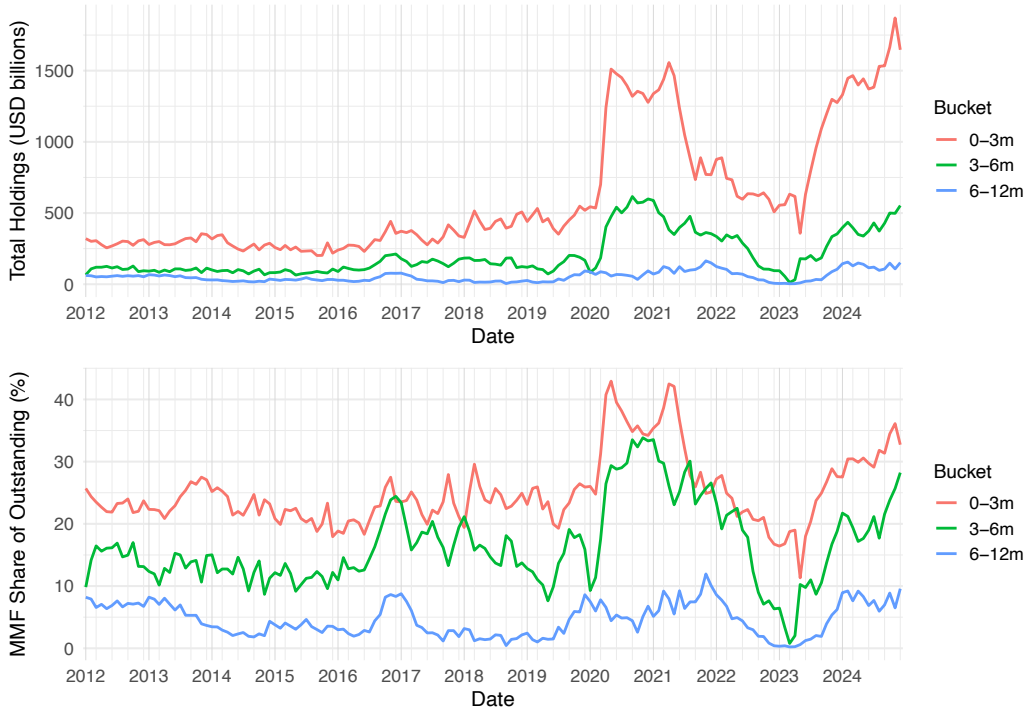
B.6 Additional Empirical Results

B.6.1 MMF Holdings of Treasury Securities

Money-fund Treasury demand is short-dated and large relative to supply: Figure B.25 shows MMF holdings concentrated in short maturities, amounting to roughly a third of

marketable short-term Treasuries outstanding.

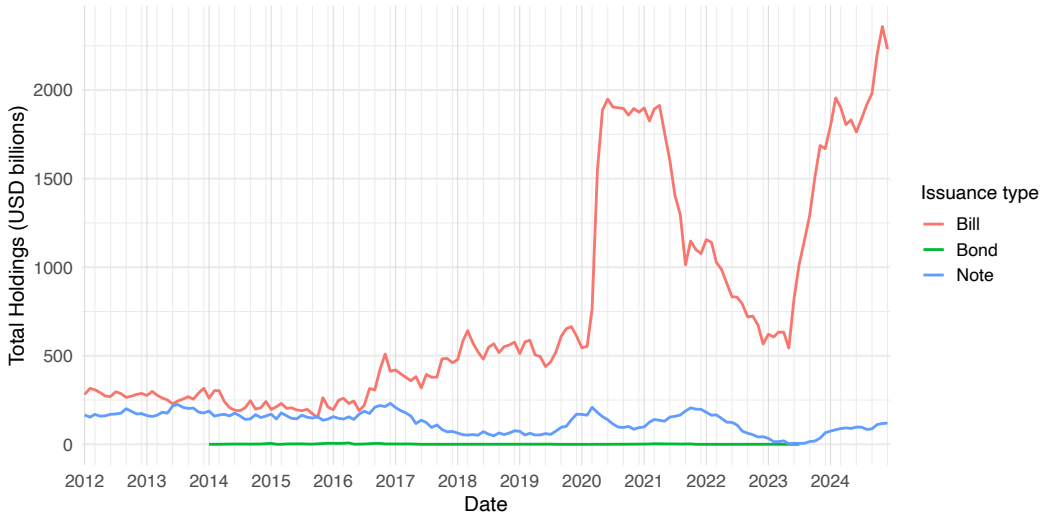
Figure B.25: MMF Holdings of Short-Term Treasury Securities



Notes: The top panel presents the dollar value of MMF holdings by remaining-maturity bucket since 2012, while the bottom panel reports these holdings as a percentage of total outstanding in the corresponding remaining-maturity bucket.

Figure B.26 further shows this demand falls predominantly on Treasury bills:

Figure B.26: MMF Holdings of Short-Term Treasury by Treasury Issuance Type

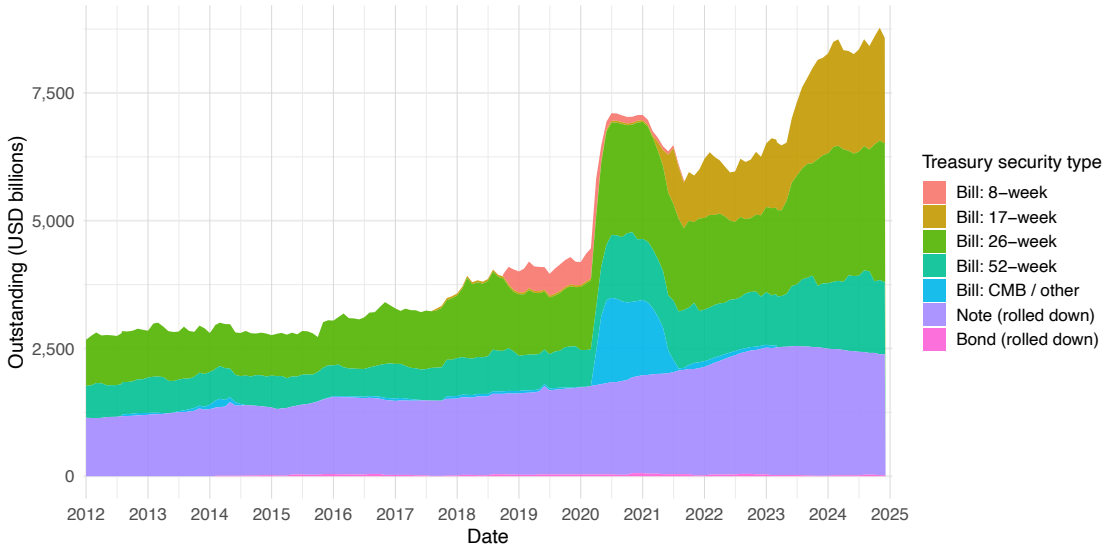


Notes: The plot presents the dollar value of MMF holdings by Treasury security type.

B.6.2 Composition of the Short-Term Treasury Market

Figure B.27 plots the composition of the short-maturity Treasury universe over the sample period. Treasury bills dominate the stack at all dates, with the regular auction tenors at 4-, 8-, 13-, 17-, 26-, and 52-week maturities accounting for the bulk of outstanding face value. Cash-management bills, which the Treasury issues at irregular intervals to bridge short-term cash-flow imbalances, have grown notably since 2020 in response to pandemic-era financing needs and recurring debt-ceiling episodes. The 17-week bill is a 2022 addition to the regular auction calendar. Originally longer-maturity Treasury notes and bonds that have rolled down to within one year of maturity also contribute to the short-end universe but at substantially smaller volumes than bills. This breakdown is useful background for the empirical analysis in Sections 3 and 4, where Treasury bills are the central object of interest because they are MMFs' dominant Treasury holding.

Figure B.27: Short-Term Treasury Outstanding by Treasury Security Type



Notes: This figure plots the market value of outstanding U.S. Treasury securities with one year or less to maturity from January 2012 to December 2024, broken down by Treasury security type. Bills are bucketed by their original auction tenor (4-, 8-, 13-, 17-, 26-, and 52-week, with cash-management bills grouped as "CMB / other"). Notes and bonds that have rolled down to within one year of maturity are shown as separate categories at the top of the stack.

B.6.3 Treasury Auctions and Reopenings

This subsection summarizes the institutional features of the U.S. Treasury auction process that are relevant for the empirical analysis. Two features matter for our dataset construction: the regular auction calendar of bills determines when new short-end supply enters the market, and the practice of *reopening* an existing security creates multiple distinct issue dates that share the same maturity date and coupon. Both features inform how we map issue and maturity records in the CRSP daily file and how we construct the auction-week issuance control in equation (2).

Regular bill auction calendar. The U.S. Treasury auctions Treasury bills on a regular schedule across six tenor buckets: 4-week, 8-week, 13-week, 17-week, 26-week, and 52-week.[22]

[22]The 17-week bill was introduced in October 2022 to fill the gap between the 13-week and 26-week tenors and is now part of the standing auction calendar. Earlier in our sample, only 4-, 8-, 13-, 26-, and 52-week tenors are auctioned regularly.

The 4-, 8-, 13-, and 26-week bills are auctioned weekly. The 17-week bill has also been auctioned weekly since its introduction in 2022, while the 52-week bill is auctioned every four weeks. Auction announcements typically occur a few days before the auction date, and securities are issued and delivered shortly after the auction clears. Each auction announcement specifies the security type, the offering amount, the auction date, the issue date, the maturity date, and the closing times for non-competitive and competitive bidding.

Cash-management bills (CMBs). In addition to the regular bills, the Treasury issues cash-management bills at irregular intervals to bridge short-term cash-flow imbalances around quarterly tax-receipt dates, debt-ceiling episodes, and unexpected fiscal needs. CMBs share the bill structure—they are zero-coupon, sold at a discount, and pay face value at maturity—but their auction frequency, offering size, and original tenor are determined ad hoc by the Treasury. CMB issuance has expanded notably since 2020 in response to pandemic-era financing needs (see Figure B.27). In our empirical analysis we treat CMBs as part of the bill universe because they are economically equivalent to regular bills for the purposes of MMF demand.

Coupon securities and reopenings. Treasury notes (originally 2-, 3-, 5-, 7-, and 10-year tenors) and Treasury bonds (originally 20- and 30-year tenors) are auctioned monthly or quarterly depending on the tenor, with most tenors involving an initial offering followed by one or more *reopenings* of the same security in subsequent months. A reopening is an additional offering of an outstanding issue: the reopened security has the same CUSIP, maturity date, and coupon rate as the original, but a different issue date and a different auction-determined price.[23] Reopenings imply that the same CRSP issue can have multiple recorded *issuance dates* but a single *maturity date*, which we account for when constructing the auction-issuance control variable for the price regressions.

Bidding mechanism. Treasury auctions are single-price (uniform-price) auctions: all successful bidders receive the same rate, yield, or discount margin set by the highest accepted

[23]For a non-reopening note auction, the coupon rate is set during the auction itself at the highest 1/8-percent increment that does not produce a price greater than 100 percent of principal. Reopened notes inherit the original coupon, with only the price determined in the new auction.

bid. Non-competitive bids—submitted by smaller investors and capped at $10 million per auction—are filled at the auction-clearing rate before competitive bids are awarded. Competitive bids submitted by institutional investors specify a maximum rate, yield, or discount margin and are awarded from lowest to highest until the offering size is filled, with each bidder limited to 35% of the offering. Pre-auction trading occurs in the *when-issued* market, which serves as a price-discovery mechanism between announcement and auction. Although when-issued trading is informative about expected auction prices, it does not enter our empirical specifications because our identifying variation comes from quarter-end demand pressure rather than pre-auction expectations.

C Data Preparation and Cleaning

We obtain money market fund (MMF) data from Form N-MFP filings available through the U.S. Securities and Exchange Commission (SEC) EDGAR database. Each submission contains information at multiple levels:

- *SUBMISSION*: This dataset identifies each N-MFP filing and related metadata, with each row corresponding to a unique fund-level submission. We remove duplicate filings, as funds may amend and resubmit reports. We also exclude funds that are liquidating or merging into other funds.

- *SERIESLEVELINFO*: This dataset contains fund-level characteristics for each submission. We extract the *seriesid*, which uniquely identifies a fund across filings. Key variables include the *percentage of liquid assets*, *fund type*, *weighted average maturity (WAM)*, *weighted average life (WAL)*, *cash holdings*, *net asset value (NAV)*, and *seven-day gross yield*. Notably, since late 2024, the reporting frequency for variables such as *percentage of liquid assets* and *seven-day gross yield* has changed to daily. To ensure consistency over time, we retain only month-end observations throughout the sample period.

- *CLASSLEVELINFO*: This dataset provides share-class-level information, where each *seriesid* may correspond to multiple *classid*s. We extract subscription and redemption flows to measure investor inflows and outflows, as well as the *seven-day net yield* to capture after-fee returns. We also use the *minimum investment threshold* as a proxy for investor type (institutional vs. retail). Similar to the fund-level data, subscriptions and redemptions are reported at daily frequency after late 2024; we again aggregate to month-end values for consistency.

- *ADVISER*: This dataset identifies each fund's investment adviser, which we use to aggregate holdings to the adviser (i.e., fund family) level. Raw adviser strings, however, are inconsistent along several dimensions: the same adviser may appear under slightly different legal names across filings, advisers may rebrand (for example, Wells Fargo Funds Management becomes Allspring Funds Management in 2021), and some series

are linked to multiple adviser strings over their reporting history, typically reflecting corporate actions rather than a genuine change of adviser. To construct a consistent adviser identifier, we first collapse the submission-level data to a series–adviser panel and identify series that report more than one adviser string over time. For each such case, we manually verify whether the multiple names correspond to a single underlying adviser or to a true change of adviser, using SEC filings and public announcements, and we assign a single consolidated adviser name to each series. We then aggregate to the adviser level, keeping only advisers with a non-trivial presence in prime or government fund segments. The final adviser panel associates each *seriesid* with a unique parent adviser entity and is used throughout the analysis whenever fund-family-level aggregation is required.

- *SCHPORTFOLIOSECURITIES*: This dataset reports each individual security held by a fund at the reporting date and is the workhorse of our holdings-based analysis. Each row corresponds to a unique fund–security–date observation. From the raw filings we retain the security identifier (CUSIP and, when available, LEI), the issuer name, the investment category (U.S. Treasury debt, U.S. Government agency debt, certificates of deposit, financial and non-financial commercial paper, repurchase agreements, variable-rate demand notes, asset-backed commercial paper, and municipal debt), the final maturity date and the maturity date used for WAM and WAL computation, the principal amount, the amortized cost, the market value, the coupon or yield, and a flag indicating whether the rate is fixed or floating. We drop securities with non-positive market value or missing issuer identifiers, and we harmonize the investment-category labels across the SEC schema revisions of 2016 and 2023. All monetary values are expressed in nominal U.S. dollars, and we compute the tenor of each security in weeks to final maturity. For repurchase agreements we additionally retain the counterparty (repo seller) identity, which is later reconciled with the corresponding records in *COLLATERALISSUERS*.

- *COLLATERALISSUERS*: For each repurchase agreement reported in *SCHPORTFOLIOSECURITIES*, this dataset contains one row per piece of collateral pledged by the

repo counterparty. Each row records the collateral issuer name and category (U.S. Treasury, Agency MBS, corporate bond, equity, and so on), the coupon and maturity of the underlying collateral, and the principal and market value pledged. Because a single repo position may be backed by many pieces of collateral, we aggregate the data to the fund–repo–date level by summing market values across rows and computing the share of each collateral category. We classify a repo as "safe-asset-backed" if its collateral consists solely of U.S. Treasury or Agency securities. The aggregated collateral information is then merged back into *SCHPORTFOLIOSECURITIES* via the repo identifier, so that every repo position in the holdings panel carries both a counterparty identity and a full decomposition of its collateral.

After cleaning each of these tables individually, we merge the submission, series, share-class, portfolio, and collateral files into a single long panel indexed by seriesid–classid–CUSIP–date. We then drop residual duplicates that arise when the same security is reported more than once within a filing, retaining the record with the largest reported market value.

Crane Data The N-MFP filings do not contain standardized issuer identifiers (such as LEIs or parent–subsidiary linkages), and fields such as principal amounts and credit ratings are unevenly populated over time. To fill these gaps we supplement the N-MFP data with proprietary data from Crane Data, a commercial provider that tracks U.S. MMFs at the share-class, fund, and security level. We aggregate the raw Crane files to a monthly, fund-level panel and use them as the primary auxiliary source for issuer identity, security-level principal, and short-term credit ratings.

Sample Restriction Our analysis focuses on prime and government MMFs, which together account for the vast majority of U.S. MMF assets and are the funds most directly exposed to bank debt and short-term Treasury securities. We therefore restrict both the N-MFP and Crane panels to prime and government funds using a consistent fund-type classification.

Issuer Identity The issuer names reported in N-MFP filings are inconsistent across time and across funds. For example, the same bank may appear as "JPM," "JPMorgan," "JP

Morgan Chase & Co.," or "JPMorgan Chase Bank, N.A." across different filings. To harmonize these strings, we exploit the fact that Crane maintains a cleaned issuer taxonomy with parent–subsidiary linkages. We first construct a crosswalk between raw Crane issuer strings and a unified "issuer identity" at the parent-entity level. We then fuzzy-match the N-MFP issuer strings to this crosswalk, allowing for common misspellings, abbreviations, and punctuation differences, and manually verify ambiguous cases. The resulting issuer identity is assigned to each unsecured holding (commercial paper, certificates of deposit, time deposits, and other bank debt) and to the counterparty of each repurchase agreement. This harmonization is essential for constructing fund–bank exposure measures and the Bartik instrument described in Section 5.

Principal A first-order data issue is that, following the 2016 MMF reform, whose amendments to Form N-MFP took effect on April 14, 2016, many MMFs ceased to report security-level principal amounts to the SEC, and the principal field becomes systematically missing for a large share of holdings beginning in early 2016. This break in coverage makes it impossible to measure changes in security-level holdings consistently over time using the N-MFP data alone. Crane Data, however, reconstructs principal at the security level by collecting the underlying information directly from each fund's website and shareholder portal, and reports principal on a continuous basis throughout our sample period. We therefore build a security-level principal crosswalk from Crane and merge it back to the cleaned N-MFP holdings panel using the combination of issuer identity, investment category, coupon, and maturity date as the matching key. For the residual set of holdings without a Crane match, we back out the principal from the reported market value, coupon, and time to maturity using a standard zero-coupon or coupon-bond pricing formula, depending on the security type.

Credit Ratings Short-term credit ratings are reported unevenly across issuers and time in the N-MFP filings and are not mandatory fields after 2023. We therefore rely on Crane, which consolidates short-term ratings from Moody's, S&P, and Fitch at the issuer level. We construct an issuer-level rating crosswalk and attach the relevant short-term rating to each N-MFP holding using issuer identity and the filing date, retaining the most recent rating

available prior to the reporting date. For issuers rated by multiple agencies, we preserve each agency's rating separately and, where necessary, map the alphanumeric scales to a common ordinal index.

The resulting dataset is a monthly, fund–security-level panel covering prime and government MMFs from 2012 to 2024, in which each holding carries a harmonized issuer identity, a cleaned principal amount, and issuer-level short-term credit ratings. This panel serves as the basis for the portfolio-allocation and window-dressing analyses presented in the main text.